

MCS-062

Introduction to Data Science

Block

3

DATA VISUALIZATION AND INTERPRETATION

Unit 8 Data Visualisation and Interpretation

Unit 9 Excel for Descriptive Data Analysis

Unit 10 Excel for Inferential Data Analysis

PROGRAMME DESIGN COMMITTEE

Prof. D. K. Lobiyal, JNU, New Delhi
Prof. Dinesh Kumar Vishwakarma, DTU, New Delhi

Prof. Sandeep Singh Rawat, SOCIS, IGNOU
Prof. P. V. Suresh, SOCIS, IGNOU
Prof. V.V. Subrahmanyam, SOCIS, IGNOU
Dr. Akshay Kumar, Associate Professor, SOCIS, IGNOU
Dr. M.P. Mishra, Associate Professor, SOCIS, IGNOU
Dr. Sudhansh Sharma, Assistant Professor, SOCIS, IGNOU

COURSE DESIGN COMMITTEE

Prof. D. K. Lobiyal, JNU, New Delhi
Prof. Dinesh Kumar Vishwakarma, DTU, New Delhi
Prof. T. V. Vijay Kumar, JNU, New Delhi
Prof. D. P. Vidyarthi, JNU, New Delhi
Prof. Vivek Kumar Singh, University of Delhi, Delhi
Prof. A. K. Mohapatra, IGDTUW, New Delhi
Mr. Kapil Pandey, HCL Technologies Limited, Delhi – NCR
Prof. Manish Trivedi, SOS (Statistics), IGNOU
Dr. Rajesh Kaliraman, Assistant Professor, SOS(Statistics), IGNOU
Dr. Prabhat Kumar Sangal, Assistant Professor, SOS (Statistics), IGNOU

Prof. Sandeep Singh Rawat, SOCIS, IGNOU
Prof. P.V. Suresh, SOCIS, IGNOU
Prof. V.V. Subrahmanyam, SOCIS, IGNOU
Prof. Divakar Yadav, SOCIS, IGNOU
Dr. Akshay Kumar, Associate Professor, SOCIS IGNOU
Dr. M.P. Mishra, Associate Professor, SOCIS IGNOU
Dr. Sudhansh Sharma, Assistant Professor, SOCIS IGNOU

SOCIS FACULTY

Prof. Sandeep Singh Rawat, Director, SOCIS, IGNOU
Prof. V.V. Subrahmanyam, SOCIS, IGNOU
Prof. Akshay Kumar, SOCIS, IGNOU
Dr. Sudhansh Sharma, Associate Professor, SOCIS, IGNOU

Prof. P.V. Suresh, IGNOU, SOCIS, IGNOU
Prof. Divakar Yadav, SOCIS, IGNOU
Prof.M.P. Mishra, SOCIS, IGNOU

COURSE PREPARATION TEAM

Content Writer (Unit-8)

(Adopted from MCS 226: Unit-4)

Dr. Priya Gupta
Jawahar Lal Nehru University (JNU), New Delhi

Content Writer (Unit-9)

(Partially Adapted from BCSL 044)

Dr. Sudhansh Sharma & Dr. Akshay Kumar
Assistant Professor Associate Professor
School of Computer and Information Sciences, IGNOU

Content Writer (Unit-10)

Dr. Neetu Singh, Assistant Professor,
Department of Computer Applications
Jaypee Institute of Information Technology(JIIT), Noida, U.P.

Content Editor (Unit-8 & 9)

Prof. T.V. Vijay Kumar
Jawahar Lal Nehru University (JNU)
New Delhi

Content Editor (Unit-10)

Prof. Manish Trivedi
School of Sciences, IGNOU

Language Editor

Prof. Parmod Kumar
School of Humanities, IGNOU

COURSE COORDINATOR

Dr. Sudhansh Sharma,
Associate Professor
School of Computers and Information Sciences, IGNOU.

PRINT PRODUCTION

Sh. Sanjay Aggarwal
Assistant Registrar
MPDD, IGNOU, New Delhi

October, 2025

©Indira Gandhi National Open University, 2025

ISBN-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University.

Further information on the Indira Gandhi National Open University courses may be obtained from the University's office at Maidan Garhi, New Delhi-110068.

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by MPDD, IGNOU.

Laser Typesetter: Tamal Basu

BLOCK 3 DATA VISUALIZATION AND INTERPRETATION

Block-3 of the “Introduction to Data Science” course explores the vital role of data visualization and interpretation in the data analysis process. This block focuses on equipping learners with the tools and techniques necessary to communicate data-driven insights effectively—a competency that is increasingly in demand across industries and academic research disciplines.

Unit 8: Data Visualization introduces learners to various types of graphical representations such as histograms, box plots, scatter plots, heat maps, bubble charts, and distribution plots. This unit emphasizes the importance of visual storytelling in making complex data comprehensible and engaging. In an industry landscape where dashboards and real-time analytics are critical, the ability to design effective visualizations is indispensable. Whether in business analytics, journalism, or scientific communication, this unit supports the learner in becoming proficient at conveying insights that influence decisions.

Unit 9: Excel for Descriptive Data Analysis highlights the practical use of Microsoft Excel, a universally accessible and widely used tool in data handling and analysis. The unit covers data import, preprocessing, and descriptive statistics such as measures of central tendency and dispersion. By mastering Excel-based analysis, students acquire hands-on skills that are highly applicable in roles ranging from data entry and reporting to business analysis and operations management. The unit reinforces the foundational statistical understanding introduced in previous blocks using a tool familiar to most organizations.

Unit 10: Excel for Inferential Data Analysis extends the learner's capability to perform hypothesis testing and inferential statistics using Excel. Topics include t-tests, chi-square tests, F-tests, ANOVA, correlation, regression analysis, and time series forecasting. These techniques are essential in academic research as well as industry for conducting experiments, validating assumptions, and making data-informed predictions. Proficiency in Excel-based inferential analysis enhances employability by allowing learners to apply statistical reasoning in domains such as marketing analytics, quality control, finance, and human resources.

Together, the units in Block-3 bridge analytical thinking with visual and practical expression. By the end of this block, learners will be able to not only conduct detailed analyses but also present their findings in a clear, professional, and impactful manner. This block prepares students for real-world reporting and presentation scenarios, making them valuable contributors in academic research groups, corporate analytics teams, and beyond.



UNIT 8 DATA VISUALISATION AND INTERPRETATION

Structure

- 8.0 Introduction
- 8.1 Objectives
- 8.2 Different Types of Plots
- 8.3 Histograms
- 8.4 Box Plots
- 8.5 Scatter Plots
- 8.6 Heat Map
- 8.7 Bubble Chart
- 8.8 Bar Chart
- 8.9 Distribution Plot
- 8.10 Pair Plot
- 8.11 Line Graph
- 8.12 Pie Chart
- 8.13 Doughnut Chart
- 8.14 Area Chart
- 8.15 Summary
- 8.16 Answers
- 8.17 References

8.0 INTRODUCTION

The previous units of this course cover detail on different aspects of data analysis, including the basics of data science, basic statistical concepts related to data science and data pre-processing. This Unit explains the different types of plots for data visualisation and interpretation. This Unit covers the details of the charts/plots for data visualisation and further discusses their constructions and various use cases associated with various data visualisation plots. This Unit will help you to appreciate the real-world need for a workforce trained in visualisation techniques. It will help you to design, develop, and interpret visual representations of data. The Unit also defines the best practices associated with the construction of different types of plots.

8.1 OBJECTIVES

After going through this Unit, you will be able to:

- Explain the key characteristics of various types of plots for data visualisation;
- Explain how to design and create data visualisations;

- Summarise and present the data in meaningful ways;
- Define appropriate methods for collecting, analysing, and interpreting numerical information.

8.2 DIFFERENT TYPES OF PLOTS

As more and more data is available to us today, there are several varieties of charts and graphs than before. In reality, the amount of data we produce, acquire, copy, and use will nearly double by 2025. Therefore, Data visualisation is a crucial and powerful tool for organisations. One can benefit from graphs and charts in the following ways:

- Encouraging the group to act proactively.
- Showcasing progress toward the goal to the stakeholders
- Displaying the core values of a company or an organisation to the audience.

Moreover, data visualisation can bring heterogeneous teams together around new objectives and foster trust among the team members. Let us discuss various graphs and charts that can be utilised in various business communications.

8.3 HISTOGRAMS

A histogram visualises the distribution of data across distinct groups with continuous classes. It is represented by a set of rectangular bars with widths equal to the class intervals and areas proportional to frequencies in the respective classes. A histogram may hence be defined as a graphic of a grouped frequency distribution with continuous classes. It provides an estimate of the distribution of values, their extremes, and the presence of any gaps or out-of-the-ordinary numbers. They are useful in providing a basic understanding of the probability distribution.

Constructing a Histogram: To construct a histogram, the data is grouped into specific class intervals, or "bins" and plotted along the x-axis. These represent the range of the data. Then, the rectangles are constructed for each class with their bases along the intervals. The height of these rectangles is measured along the y-axis representing the frequency for each class interval. It is important to remember that in these representations, every rectangle is next to another because the base spans the spaces between class boundaries.

Use Cases: When it is necessary to illustrate or compare the distribution of specific numerical data across several ranges of intervals, histograms can be employed. They can aid in visualising the key meanings and patterns associated with a lot of data. They may help a business or organisation in the decision-making process. Some of the use cases of histograms include-

- Distribution of salaries in an organisation
- Distribution of height in one batch of students of a class, student performance on an exam,
- Distribution of different products sold in a year.

Best Practices

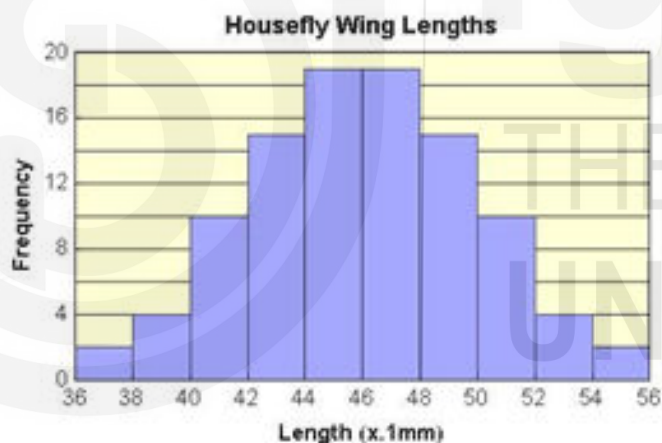
- **Analyse various data groups:** The best data groupings can be found by creating a variety of histograms.
- **Break down compartments using colour:** The same chart can display a second set of categories by colouring the bars that represent each category.

Types of Histogram

Normal distribution: In a normal distribution, the probability that points will occur on each side of the mean is the same.

Example: Consider the following bins, which show the frequency of length of wings of houseflies in 1/10 of a millimetre.

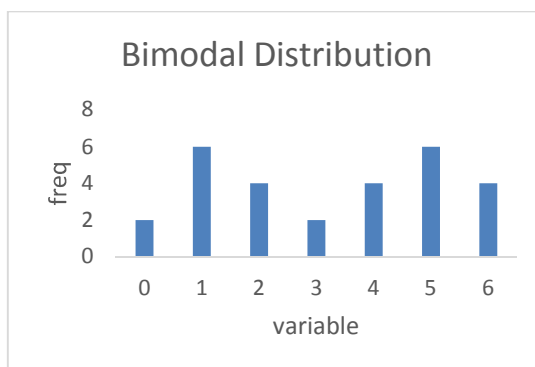
<i>Bin</i>	<i>Frequency</i>	<i>Bin</i>	<i>Frequency</i>
36-38	2	46-48	19
38-40	4	48-50	15
40-42	10	50-52	10
42-44	15	52-54	4
44-46	19	54-56	2



Bimodal Distribution: This distribution has two peaks. In the case of a bimodal distribution, the data must be segmented before being analysed as normal distributions in their own right.

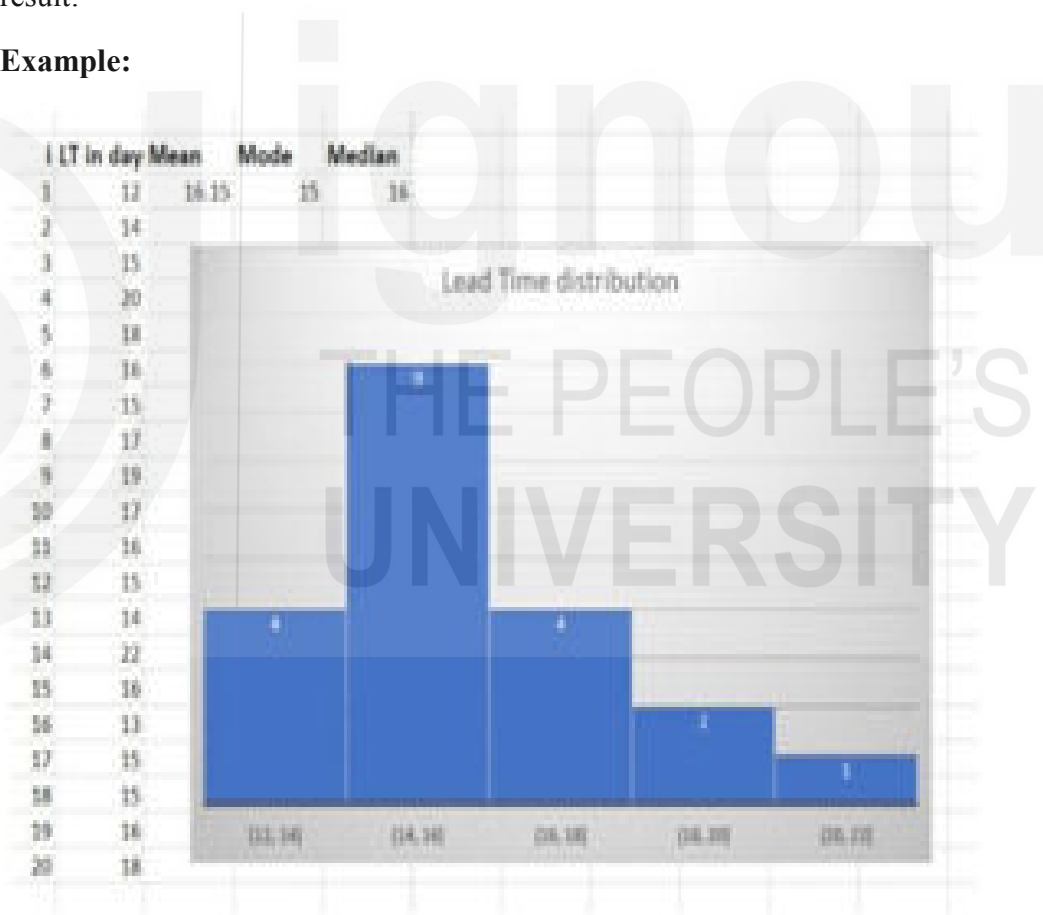
Example:

Variable	Frequency
0	2
1	6
2	4
3	2
4	4
5	6
6	4



Right-skewed distribution: A distribution that is skewed to the right is sometimes referred to as a positively skewed distribution. A right-skewed distribution is one that has a greater percentage of data values on the left and a lesser percentage on the right. Whenever the data have a range boundary on the left side of the histogram, a right-skewed distribution is frequently the result.

Example:



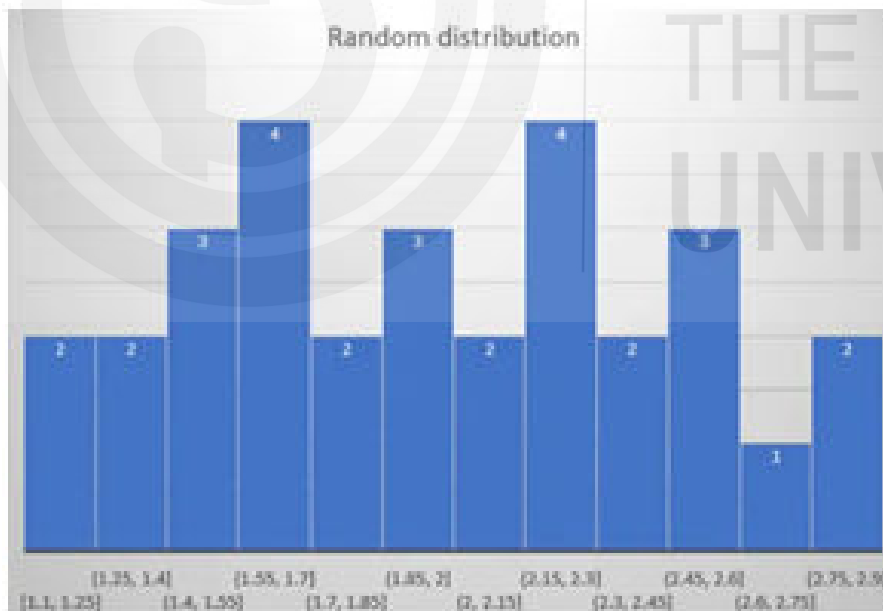
Left-skewed distribution: A distribution that is skewed to the left is sometimes referred to as a negatively skewed distribution. A left-skewed distribution will have a greater proportion of data values on the right side of the distribution and a lesser proportion of data values on the left. A right-skewed distribution commonly results when the data have a range limit on the right side of the histogram. An alternative name for this is a right-tailed distribution.

Example:

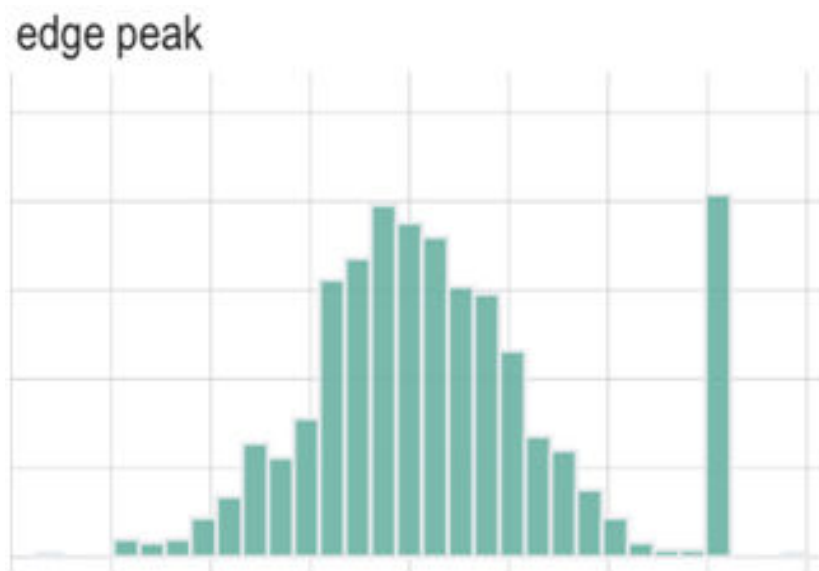


A random distribution: A random distribution is characterised by the absence of any clear pattern and the presence of several peaks. When constructing a histogram using a random distribution, it is possible that several distinct data attributes will be blended into one. In such cases, the data ought to be partitioned and investigated independently.

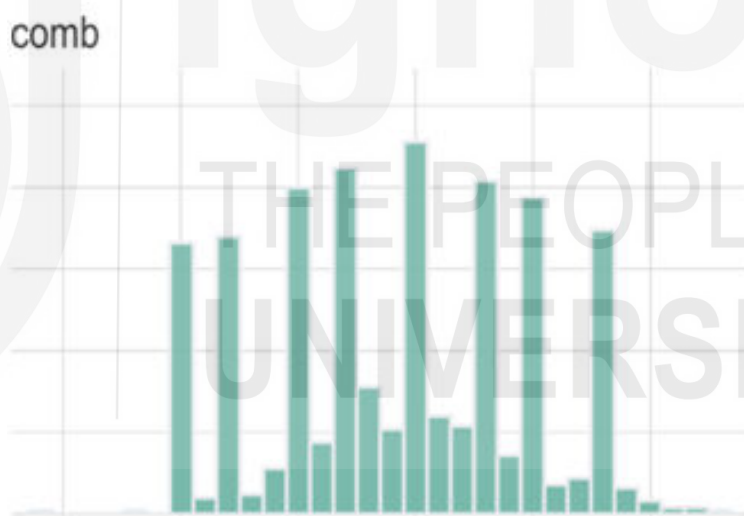
Example:



Edge Peak Distribution: When there is an additional peak at the edge of the distribution that does not belong there, this type of distribution is called an edge peak distribution. Unless you are very positive that your data set has the expected number of outliers, this almost always indicates that you have plotted (or collected) your data incorrectly (i.e. a few extreme views on a survey).



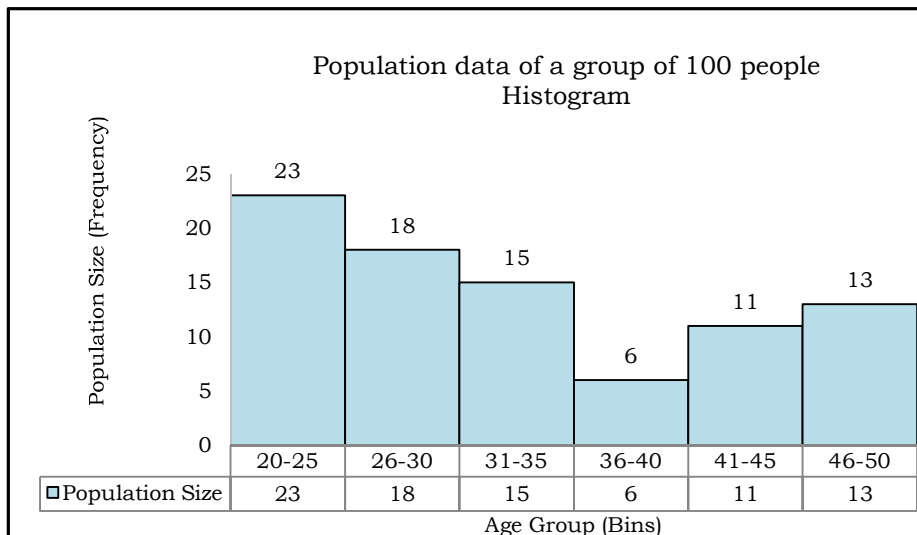
Comb Distribution: Because the distribution seems to resemble a comb, with alternating high and low peaks, this type of distribution is given the name "comb distribution." Rounding off an object might result in it having a comb-like form. For instance, if you are measuring the height of the water to the nearest 10 centimetres but your class width for the histogram is only 5 centimetres, you may end up with a comb-like appearance.



Example

Histogram for the population data of a group of 86 people:

Age Group (in years)	Population Size
20-25	23
26-30	18
31-35	15
36-40	6
41-45	11
46-50	13
TOTAL	86



Check Your Progress 1

1. What is the difference between a Bar Graph and a Histogram?

.....

.....

.....

2. Draw a Histogram for the following data:

Class Interval	Frequency
0 – 10	35
10 – 20	70
20 – 30	20
30 – 40	40
40 – 50	50

3. Why is a histogram used?

.....

.....

.....

4. What do histograms show?

.....

.....

.....

8.4 BOX PLOTS

Box-and-whisker plots, also known as box plots, are widely employed when displaying data distributions using the five essential summary statistics of minimum, first quartile, median, third quartile, and maximum. It is a visual depiction of data that aids in determining how widely distributed or how much the data values change. These box plots make it simple to compare the

distributions since it makes the centre, spread, and overall range understandable. They are utilised for data analysis wherein the graphical representations are used to determine the following:

1. Shape of Distribution
2. Central Value
3. Variability of Data

Constructing a Box plot: The two components of the graphic are described by their names: the box, which shows the median value of data along with the first and third quartiles (25 percentile and 75 percentile), and the whiskers, which shows the remaining data. The 3rd quartile's difference from the first quartile of data is called the interquartile range. The highest and minimum points in the data can also be displayed using the whiskers. The points beyond the 1.5 $\times$ interquartile range can be identified as suspected outliers.

Use Cases: A box plot is frequently used to demonstrate whether a distribution is skewed and whether the data set contains any potential outliers or odd observations. Box plots are also very useful for comparing or involving Big data sets. Examples of box plots include plotting the following:

- Gas efficiency of vehicles
- Time spent reading across readers

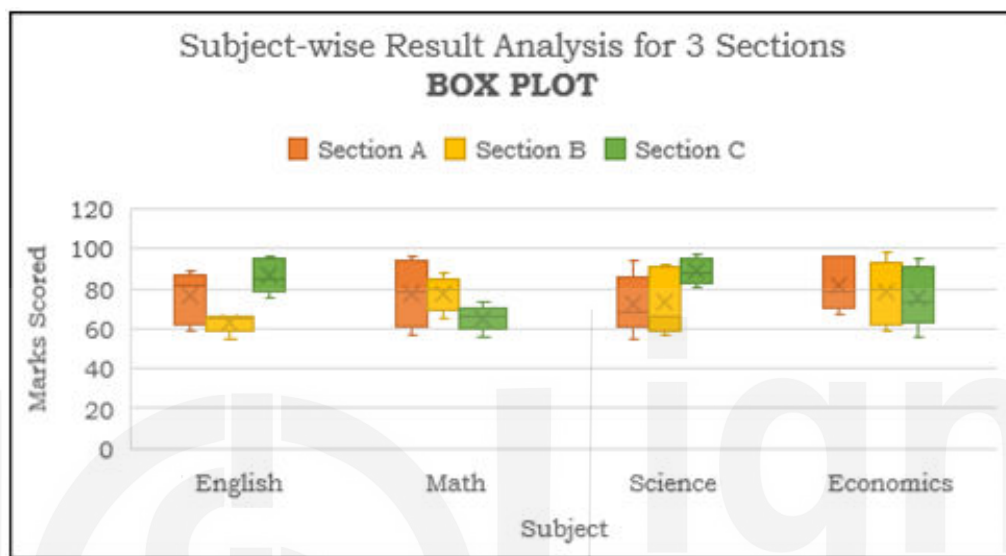
Best Practices

- **Cover the points within the box:** This aids the viewer in concentrating on the outliers.
- **Box plot comparisons between categorical dimensions:** Box plots are excellent for quickly comparing dataset distributions.

Example

Subject	Section A	Section B	Section C
English	59	65	82
Math	96	73	66
Science	78	57	81
Economics	96	79	73
English	65	55	94
Math	78	65	56
Science	68	61	85
Economics	96	98	56
English	85	63	85
Math	93	88	68
Science	94	66	94
Economics	67	59	86
English	82	66	96

Math	64	79	63
Science	55	90	97
Economics	73	89	95
English	89	66	75
Math	57	81	73
Science	67	92	88
Economics	78	65	69



The box plots clearly show that Section B has performed poorly in English, whereas Section C has performed poorly in Maths. Section A has mostly balanced performance, but the marks of the students are most dispersed.

Check Your Progress 2

- How to correctly interpret a box plot?

.....

.....

.....

- What are the most important parts of a box plot?

.....

.....

.....

- What are the uses of box plots?

.....

.....

- How do you describe the distribution of a box plot?

.....

.....

8.5 SCATTER PLOTS

A scatter plot is the most commonly used chart when observing the relationship between two quantitative variables. It works particularly well for quickly identifying possible correlations between different data points. The relationship between multiple variables can be efficiently studied using scatter plots, which show whether one variable is a good predictor of another or whether they normally fluctuate independently. Multiple distinct data points are displayed on a single graph in a scatter plot. Following that, the chart can be enhanced with analytics like trend lines or cluster analysis. It is especially useful for quickly identifying potential correlations between data points.

Constructing a Scatter Plot: Scatter plots are mathematical diagrams or plots that rely on Cartesian coordinates. In this type of graph, the categories being compared are represented by the circles on the chart (shown by the colour of the circles) and the numerical volume of the data (indicated by the circle size). One colour on the graph allows you to represent two values for two variables related to a data set, but two colours can also be used to include a third variable.

Use Cases: Scatter charts are great in scenarios where you want to display both distribution and the relationship between two variables.

- Display the relationship between time-on-platform (How Much Time Do People Spend on Social Media?) and churn (the number of people who stopped being customers during a set period of time).
- Display the relationship between salary and years spent at the company

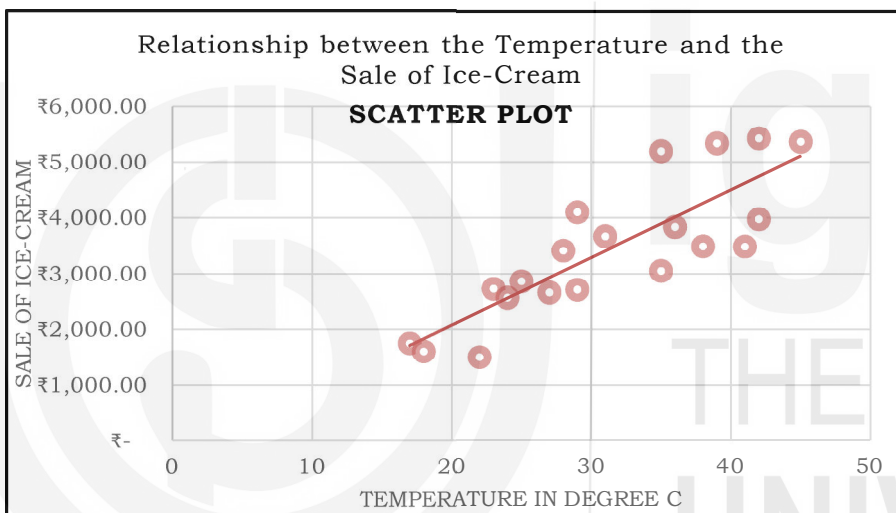
Best Practices

- **Analyse clusters to find segments:** Based on your chosen variables, cluster analysis divides the data points into discrete parts.
- **Employ highlight actions:** You can rapidly identify which points in your scatter plots share characteristics by adding a highlight action while keeping an eye on the rest of the dataset.
- **Mark customisation:** individual markings add a simple visual hint to your graph, which helps distinguish various point groups easily.

Example

Temperature (in deg C)	Sale of Ice-Cream
17	₹ 1,750.00
18	₹ 1,603.00
22	₹ 1,500.00
29	₹ 2,718.00
27	₹ 2,667.00
28	₹ 3,422.00
31	₹ 3,681.00

23	₹ 2,734.00
24	₹ 2,575.00
25	₹ 2,869.00
35	₹ 3,057.00
36	₹ 3,846.00
38	₹ 3,500.00
41	₹ 3,496.00
42	₹ 3,984.00
29	₹ 4,109.00
39	₹ 5,336.00
35	₹ 5,197.00
42	₹ 5,426.00
45	₹ 5,365.00



Please note that a linear trendline has been fitted to the scatter plot, indicating a positive change in sales of ice cream with an increase in temperature.

Check Your Progress 3

1. What are the characteristics of a scatter plot?

.....

.....

.....

2. What components make up a scatter plot?

.....

.....

.....

3. What is the purpose of a scatter plot?

.....

.....

4. What are the three types of correlations that can be inferred from scatter plots?

.....
.....
.....

8.6 HEAT MAP

Heatmaps are two-dimensional graphics that show data trends through colour shading. They are an example of a part-to-whole chart in which values are represented using colours. A basic heat map offers a quick visual representation of the data. A user can comprehend complex data sets with the help of more intricate heat maps. Heat maps can be presented in a variety of ways, but they all have one thing in common: **they all make use of colour to convey correlations between data values**. Heat maps are more frequently utilized to present a more comprehensive view of massive amounts of data. It is especially helpful because colours are simpler to understand and identify than plain numbers.

Heat maps are highly flexible and effective at highlighting trends. Heatmaps are naturally self-explanatory, in contrast to other data visualisations that require interpretation. The greater the quantity/volume, the deeper the colour (or the higher the value, the tighter the dispersion, etc.). Heat Maps dramatically improve the ability of existing data visualisations to convey important data insights quickly.

Use Cases: Heat Maps are primarily used to illustrate the enormous amounts of data contained inside a dataset and help guide users to the parts of data visualisations that matter the most.

- Average monthly temperatures across the years
- Departments with the highest amount of attrition over time.
- Traffic across a website or a product page.
- Population density/spread in a geographical location.

Best Practices

- **Select the proper colour scheme:** This chart relies heavily on colour; therefore, it is important to pick a colour scheme that complements the data.
- **Specify a legend:** As a related point, a heatmap must typically contain a legend describing how the colours correspond to numerical values.

Example

Region-wise monthly sale of a SKU (stock-keeping unit)

	MONTH											
ZONE	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
NORTH	75	84	61	95	77	82	74	92	58	90	54	83
SOUTH	50	67	89	61	91	77	80	72	82	78	58	63
EAST	62	50	83	95	83	89	72	96	96	81	86	82
WEST	69	73	59	73	57	61	58	60	97	55	81	92

The distribution of sales of a company is shown in the sample heatmap above, broken down by zone and spanning a 12-month period. Like in a typical data table, each cell displays a numeric count, but the count is also accompanied by a colour, with higher counts denoting deeper hues.

Check Your Progress 4

1. What type of input is needed for a heat map?

.....

.....

.....

2. What kind of information does a heat map display?

.....

.....

.....

3. What can be seen in a heatmap?

.....

.....

.....

8.7 BUBBLE CHART

Bubble diagrams are used to show the relationships between different variables. They are frequently used to represent data points in three dimensions, specifically when the bubble size, y-axis, and x-axis are all present. Bubble charts demonstrate relationships between data points using location and size. However, bubble charts have a restricted data size capability since too many bubbles can make the chart difficult to read.

Although technically not a separate type of visualisation, bubbles can be used to show the relationship between three or more measurements in scatter plots or maps by adding complexity. By altering the size and colour of circles, large amounts of data are presented concurrently in visually pleasing charts.

Constructing a Bubble Chart: For each observation of a pair of numerical variables (A, B), a bubble or disc is drawn and placed in a Cartesian coordinate system horizontally according to the value of variable A and vertically according to the value of variable B. The area of the bubble serves as a representation for a third numerical variable (C). Using various colours in bubbles, you may even add a fourth dataset (D: numerical or categorical).

By using location and proportions, bubble charts are frequently used to compare and illustrate the relationships between circles that have been classified. Bubble Charts' overall image can be utilised to look for patterns and relationships.

Use Cases: Usually, the positioning and ratios of the size of the bubbles/circles on this chart are used to compare and show correlations between variables. Additionally, it is utilised to spot trends and patterns in data.

- AdWords' analysis: CPC vs Conversions vs share of total conversions
- Relationship between life expectancy, GD per capita and population size

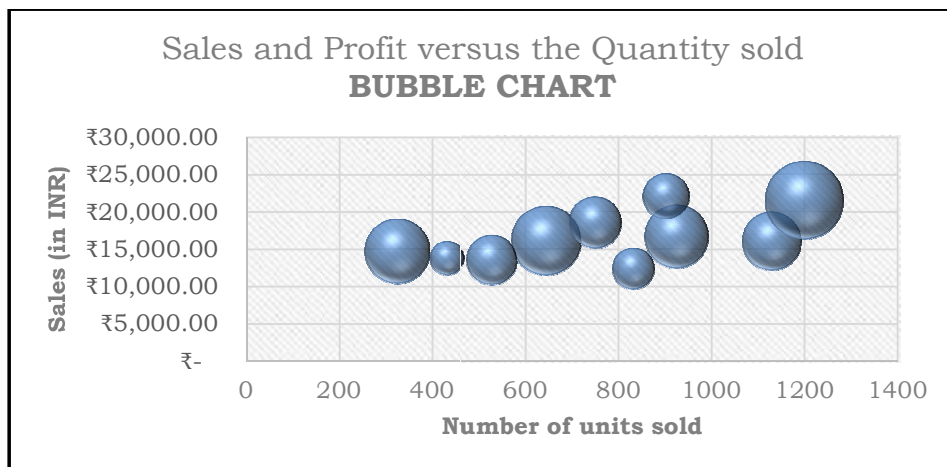
Best Practices:

- **Add colour:** A bubble chart can gain extra depth by using colour.
- Set bubble size in appropriate proportion.
- **Overlay bubbles on maps:** From bubbles, a viewer can immediately determine the relative concentration of data. These are used as an overlay to give the viewer a context for geographically-related data.

Example

Item Code	Units Sold	Sales (in Rs.)	Profit %
PC001	325	₹ 14,687.00	22%
PC002	1130	₹ 16,019.00	18%
PC003	645	₹ 16,100.00	25%
PC004	832	₹ 12,356.00	9%
PC005	1200	₹ 21,500.00	32%
PC006	925	₹ 16,669.00	21%
PC007	528	₹ 13,493.00	13%
PC008	750	₹ 18,534.00	14%
PC009	432	₹ 13,768.00	6%
PC0010	903	₹ 22,043.00	11%

he three variables in this example are sales, profits, and the number of units sold. Therefore, a bubble chart can display all three variables and their relationship simultaneously.



Check Your Progress 5

1. What is a bubble chart?

.....

.....

.....

2. What is a bubble chart used for?

.....

.....

.....

3. What is the difference between a scatter plot and a bubble chart?

.....

.....

.....

4. What is bubble size in a bubble chart?

.....

.....

.....

8.8 BAR CHART

A bar chart is a graphical depiction of numerical data that uses rectangles (or bars) with equal widths and varied heights. In the field of statistics, bar charts are one of the methods for handling data.

Constructing a Bar Chart: The x-axis corresponds to the horizontal line, and the y-axis corresponds to the vertical line. The y-axis represents frequency in this graph. Write the names of the data items whose values are to be noted along the horizontal x-axis.

Along the horizontal axis, choose the uniform width of bars and the uniform gap between the bars. Pick an appropriate scale to go along the y-axis that runs vertically so that you can figure out the height of the bars based on the

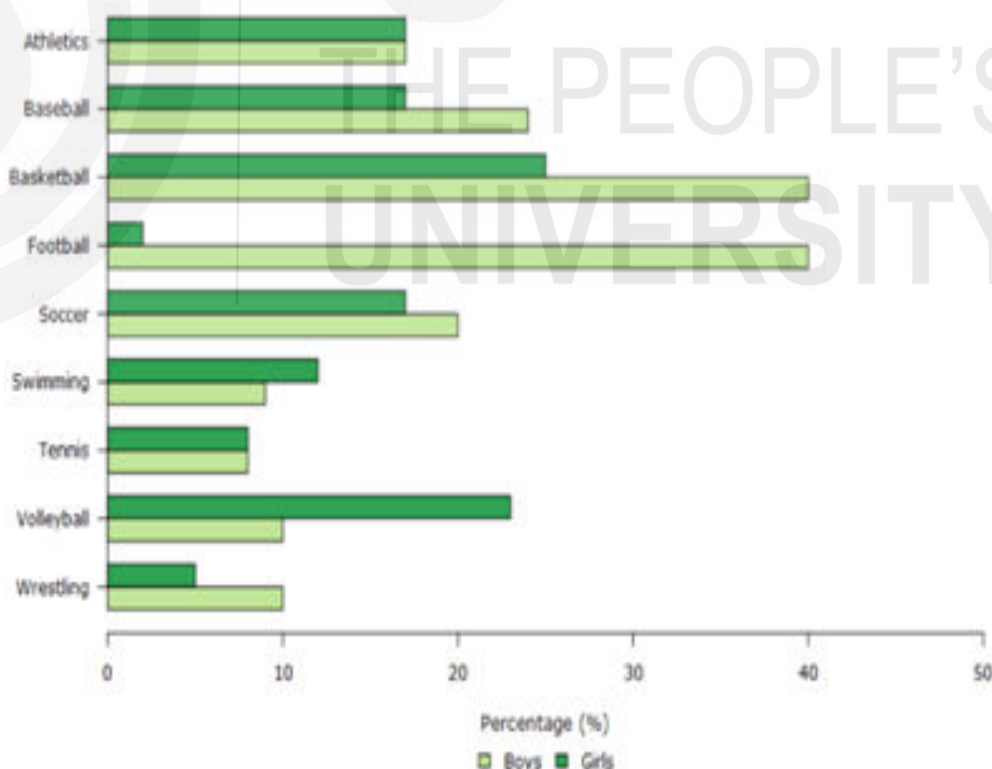
value of the variable to be displayed. Finally, draw the bars using that information.

Types of Bar Charts: Bar Charts are mainly classified into two types:

Horizontal Bar Charts: Horizontal bar charts are the type of graph that is used when the data being analysed is to be depicted on paper in the form of horizontal bars with their respective measures. When using a chart of this type, the categories of the data are indicated on the y-axis.

Example:

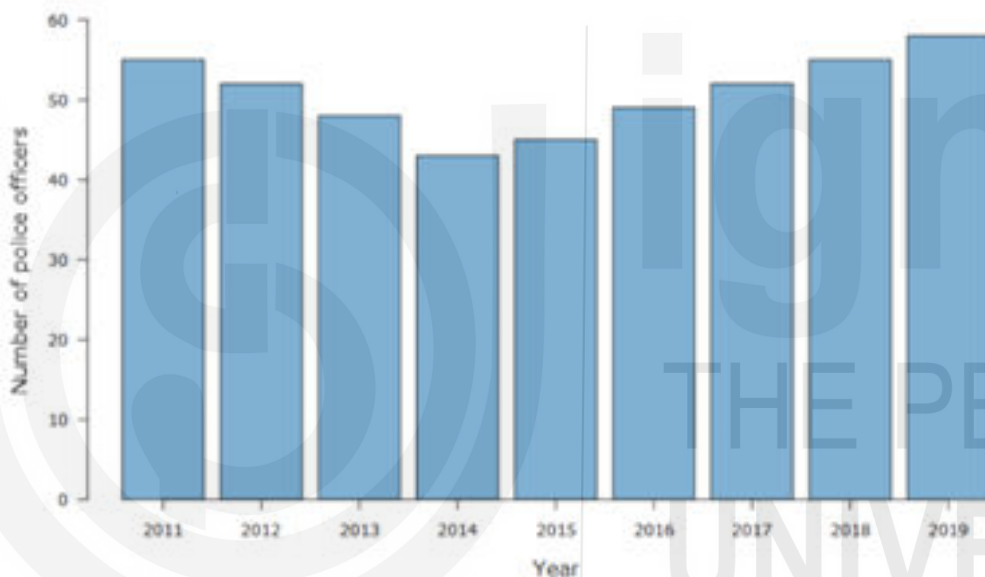
Sport	Percentage of body (%)	Percentage of girls (%)
Athletics	17	17
Baseball	24	17
Basketball	40	25
Football	40	2
Soccer	20	17
Swimming	9	12
Tennis	8	8
Volleyball	10	23
Wrestling	10	5



Vertical Bar Charts: A vertical bar chart displays vertical bars on graph (chart) paper. These rectangular bars in a vertical orientation represent the measurement of the data. These rectangular bars represent the quantities of the variables that are written along the x-axis.

Example:

Year	Number of Police Officers
2011	55
2012	52
2013	48
2014	43
2015	45
2016	49
2017	52
2018	55
2019	58



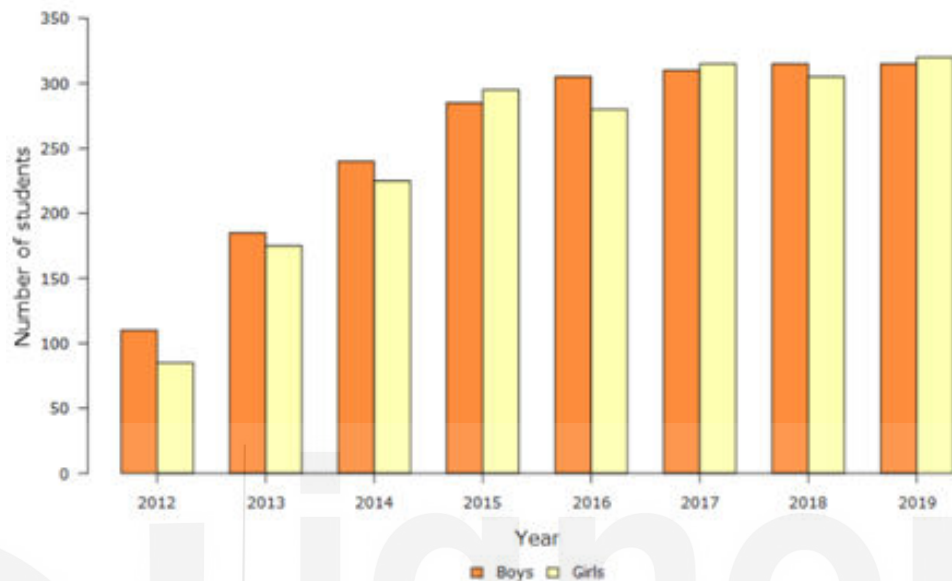
We can further divide bar charts into two basic categories:

Grouped Bar Charts: The grouped bar graph is also referred to as the clustered bar graph (graph). It is valuable for at least two separate types of data. The horizontal (or vertical) bars in this are categorised according to their position. For instance, if a bar chart is designed to show three groups, each of which has numerous values, then different colours will be used to indicate each group. When there is a close relationship between two sets of data, each group's colour coding will be the same.

Example:

Year	Number of boys	Number of girls
2012	110	85
2013	185	175
2014	240	225
2015	285	295

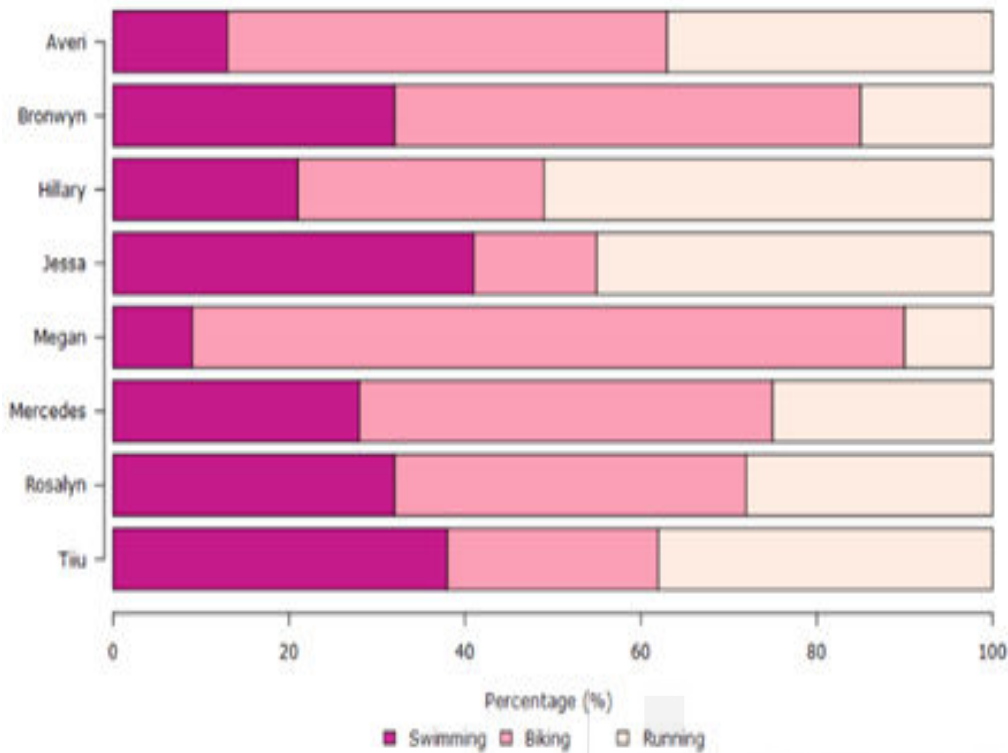
2016	305	280
2017	310	315
2018	315	305
2019	315	320



Stacked Bar Charts: The composite bar chart is also referred to as the stacked bar chart. It illustrates how the overall bar chart has been broken down into its component pieces. You should use bars of varying colours and clear labelling to determine the category of each item. As a result, in a chart with stacked bars, each parameter is represented by a single rectangular bar. Multiple segments, each of a different colour, are displayed within the same bar. Various components of each separate label are represented by the numerous segments of the bar. It is possible to draw it in either the vertical or the horizontal plane.

Example:

First Name	Percentage of time spent swimming (%)	Percentage of time spent cycling (%)	Percentage of time spent running (%)
Averi	13	50	37
Bronwyn	32	53	15
Hillary	21	28	51
Jessa	41	14	45
Megan	9	81	10
Mercedes	28	47	25
Rosalyn	32	40	28
Tilu	38	24	38



Use cases: Bar charts are typically employed to display quantitative data. The following is a list of some of the applications of the bar chart-

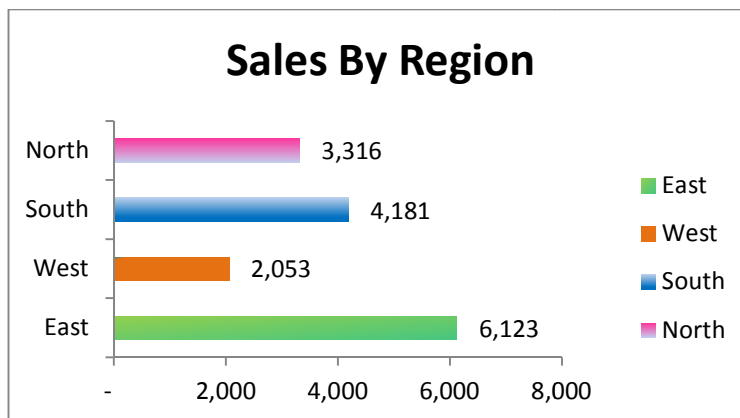
- In order to clearly illustrate the relationships between various variables, bar charts are typically utilised. When presented in a pictorial format, the parameters can be envisioned more quickly and easily by the user.
- Bar charts are the quickest and easiest way to display extensive amounts of data while saving time. It is utilised for studying trends over extended amounts of time.

Best Practices:

- Use a common zero-valued baseline
- Maintain rectangular forms for your bars
- Consider the ordering of category level and use colour wisely.

An Example of a Horizontal Bar Graph is given below:

Region	Sales
East	6,123
West	2,053
South	4,181
North	3,316



Check your progress 6:

1. When should we use a bar chart?

.....

.....

.....

2. What are the different types of bar charts?

.....

.....

.....

3. Draw a vertical bar chart.

.....

.....

.....

4. Draw a horizontal bar chart.

.....

.....

Use the following data to answer the questions 3 and 4:

Month	January	February	March
Number of visitors	150	300	250

8.9 DISTRIBUTION PLOT

Visually assessing the distribution of sample data, distribution charts do this by contrasting the actual distribution of the data with the theoretical values expected from a certain distribution. In addition to more traditional hypothesis tests, distribution plots can be used to establish whether the data from the sample follows a particular distribution. The distribution plot is useful for analysing the relationship between the range of a set of numerical data and its distribution. The values of the data are represented as points along an axis.

Constructing a Distribution Plot: In a distribution plot, you must utilise one or two dimensions together with one measure. You will get a single-line visualisation if you only use one dimension. If you use two dimensions, each value of the outer second dimension will produce a separate line.

Use Cases: The distribution of a data set shows the frequency of occurrence of each possible outcome of a repeatable event observed many times. For instance:

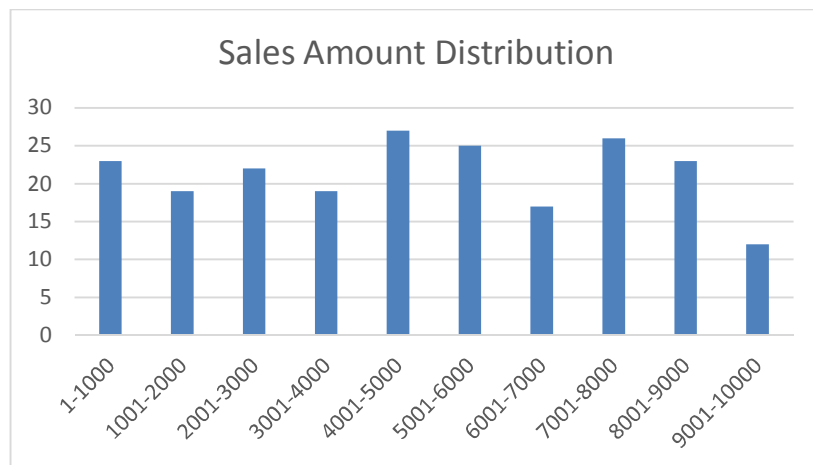
- Height of a population.
- Income distribution in an economy
- Test scores, listed by percentile.

Best Practices:

- It is advisable to have equal class widths.
- The class intervals should be mutually exclusive and non-overlapping.
- Open-ended classes at the lower and upper limits (e.g., <10, >100) should be avoided.

Example

Sales Amount	No. of Clients
1-1000	23
1001-2000	19
2001-3000	22
3001-4000	19
4001-5000	27
5001-6000	25
6001-7000	17
7001-8000	26
8001-9000	23
9001-10000	12
Grand Total	213



Check Your Progress 7:

1. What is a distribution plot?

.....

.....

.....

2. When should we use a distribution plot?

.....

.....

.....

3. What do distribution graphs show?

.....

.....

.....

8.10 PAIR PLOT

The pairs plot is an extension of the histogram and the scatter plot, which are both fundamental figures. The scatter plots on the upper and lower triangles show the relationship (or lack thereof) between the two variables. The histogram along the diagonal gives us the ability to see the distribution of a single variable, while the scatter plots on the upper and lower triangles show the relationship (or lack thereof) between two variables.

A pair plot can be utilised to gain an understanding of the optimum collection of characteristics to describe a relationship between two variables or to create clusters that are the most distinct from one another. Additionally, it is helpful to construct some straightforward classification models by drawing some straightforward lines or making linear separations in our data set.

Constructing a Pair Plot: If you have m attributes in your dataset, it creates a figure with $m \times m$ subplots. Each attribute's univariate histograms (distributions) make up the main-diagonal subplots. For a non-diagonal subplot, assume a position (i, j) . The dataset's samples are all plotted using a coordinate system with the characteristics i and j as the axes. In other words,

it projects the dataset on these two attributes only. This is particularly interesting to visually inspect how the samples are spread with respect to these two attributes ONLY. The "shape" of the spread can give you valuable insight into the relation between the two attributes.

Use Cases: A pairs plot allows us to see both distribution of single variables and relationships between two variables. It helps to identify the most distinct clusters or the optimum combination of attributes to describe the relationship between two variables. It can have the following two use cases:

- By creating some straightforward linear separations or basic lines in our data set, it helps to create some straightforward classification models.
- Analysing socio-economic data of a population.

Best Practices:

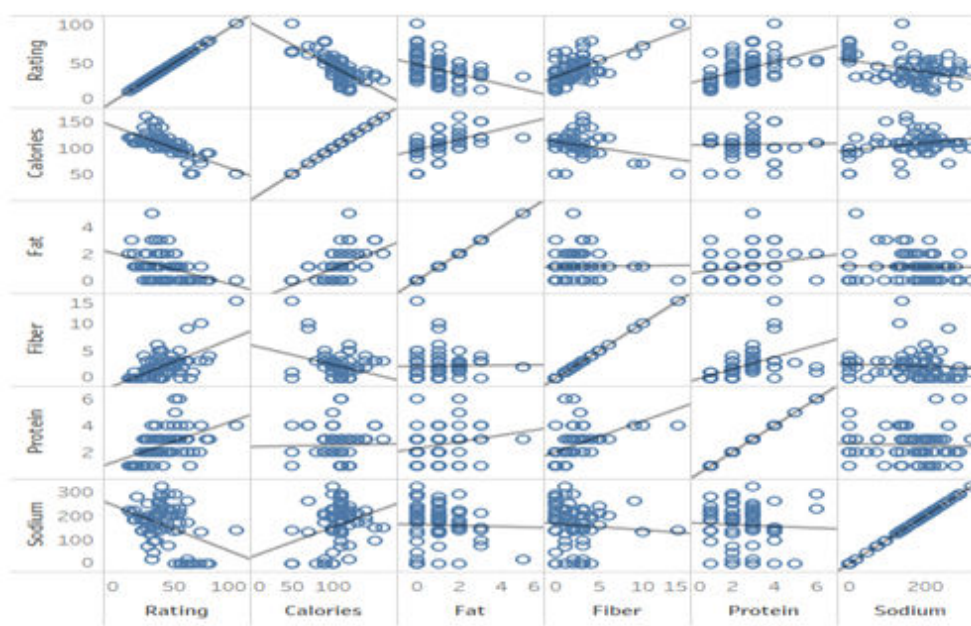
- Use a different colour palette.
- For each colour level, use a different marker.

Example:

Calories	Protein	Fat	Sodium	Fiber	Rating
70	4	1	130	10	68.40297
120	3	5	15	2	33.98368
70	4	1	260	9	59.42551
50	4	0	140	14	93.70491
110	2	2	180	1.5	29.50954
110	2	0	125	1	33.17409
130	3	2	210	2	37.03856
90	2	1	200	4	49.12025
90	3	0	210	5	53.31381
120	1	2	220	0	18.04285
110	6	2	290	2	50.765
120	1	3	210	0	19.82357
110	6	2	290	2	50.765
120	1	3	210	0	19.82357
110	3	2	140	2	40.40021
110	1	1	180	0	22.73645
110	2	0	280	0	41.44502
100	2	0	290	1	45.86332
110	1	0	90	1	35.78279
110	1	1	180	0	22.39651

110	3	3	140	4	40.44877
110	2	0	220	1	46.89564
100	2	1	140	2	36.1762
100	2	0	190	1	44.33086
110	2	1	125	1	32.20758
110	1	0	200	1	31.43597
100	3	0	0	3	58.34514
120	3	2	160	5	40.91705
120	3	0	240	5	41.01549
110	1	1	135	0	28.02577
100	2	0	45	0	35.25244
110	1	1	280	0	23.80404
100	3	1	140	3	52.0769
110	3	0	170	3	53.37101
120	3	3	75	3	45.81172
120	1	2	220	1	21.87129
110	3	1	250	1.5	31.07222
110	1	0	180	0	28.74241
110	2	1	170	1	36.52368
140	3	1	170	2	36.47151
110	2	1	260	0	39.24111
100	4	2	150	2	45.32807
110	2	1	180	0	26.73452
100	4	1	0	0	54.85092
150	4	3	95	3	37.13686
150	4	3	150	3	34.13977
160	3	2	150	3	30.31335
100	2	1	220	2	40.10597
120	2	1	190	0	29.92429
140	3	2	220	3	40.69232
90	3	0	170	3	59.64284
130	3	2	170	1.5	30.45084
120	3	1	200	6	37.84059
100	3	0	320	1	41.50354

50	1	0	0	0	60.75611
50	2	0	0	1	63.00565
100	4	1	135	2	49.51187
100	5	2	0	2.7	50.82839
120	3	1	210	5	39.2592
100	3	2	140	2.5	39.7034
90	2	0	0	2	55.33314
110	1	0	240	0	41.99893
110	2	0	290	0	40.56016
80	2	0	0	3	68.23589
90	3	0	0	4	74.47295
90	3	0	0	3	72.80179
110	2	1	70	1	31.23005
110	6	0	230	1	53.13132
90	2	0	15	3	59.36399
110	2	1	200	0	38.83975
140	3	1	190	4	28.59279
100	3	1	200	3	46.65884
110	2	1	250	0	39.10617
110	1	1	140	0	27.7533
100	3	1	230	3	49.78745
100	3	1	200	3	51.59219
110	2	1	200	1	36.18756



The pair plot can be interpreted as follows:

Along the boxes of the diagonal, the variable names are displayed.

A scatterplot of the correlation between each pairwise combination of factors is shown in each of the remaining boxes. For instance, a scatterplot of the values for rating and sodium can be seen in the matrix's box in the top right corner. A scatterplot of values for rating that is positively connected with rating and so forth may be seen in the box in the upper left corner. We can see the association between each pair of variables in our dataset from this single visualisation. For instance, calories and rating appear to have a negative link but protein and fat appear to be unrelated.

Check your progress 8:

1. Why is the pair plot used?

.....

.....

.....

2. How do you read a pair plot?

.....

.....

.....

3. What does a pair plot show?

.....

.....

.....

8.11 LINE GRAPH

A graph that depicts change over time by means of points and lines is known as a line graph, line chart, or line plot. It is a graph that shows a line connecting a lot of points or a line that shows how the points relate to one another. The graph is represented by the line or curve that connects successive data points to show quantitative data between two variables that are changing. The values of these two variables are compared along a vertical axis and a horizontal axis in linear graphs.

One of the most significant uses of line graphs is tracking changes over both short and extended time periods. It is also used to compare the changes that have taken place for diverse groups over the course of the same time period. It is strongly advised to use a line graph rather than a bar graph when working with data that only has slight fluctuations.

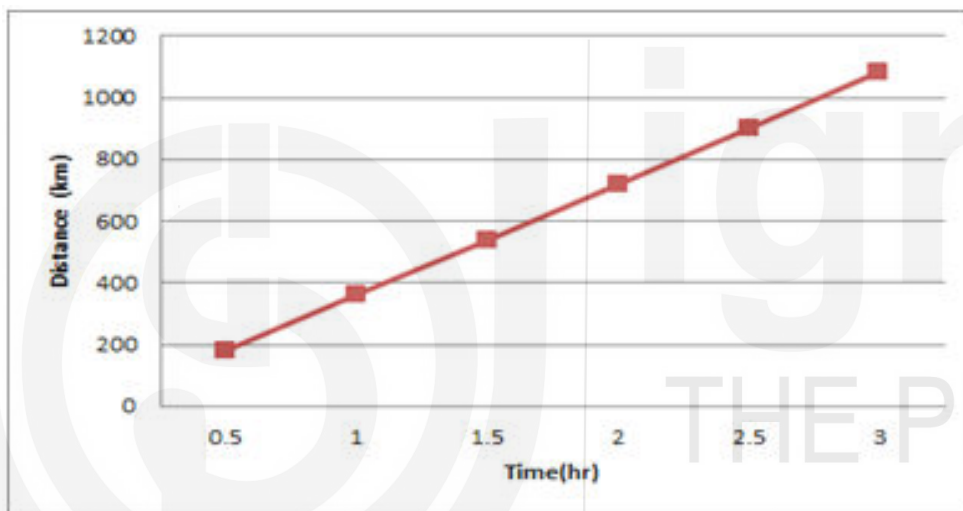
As an illustration, the finance department of a company would want to visualise how its current cash balance has changed over time. If so, they will plot the points over the horizontal and vertical axis using a line graph. It typically refers to the time period that the data span.

Following are the types of line graphs:

1. **Simple Line Graph:** Only a single line is plotted on the graph.

Example:

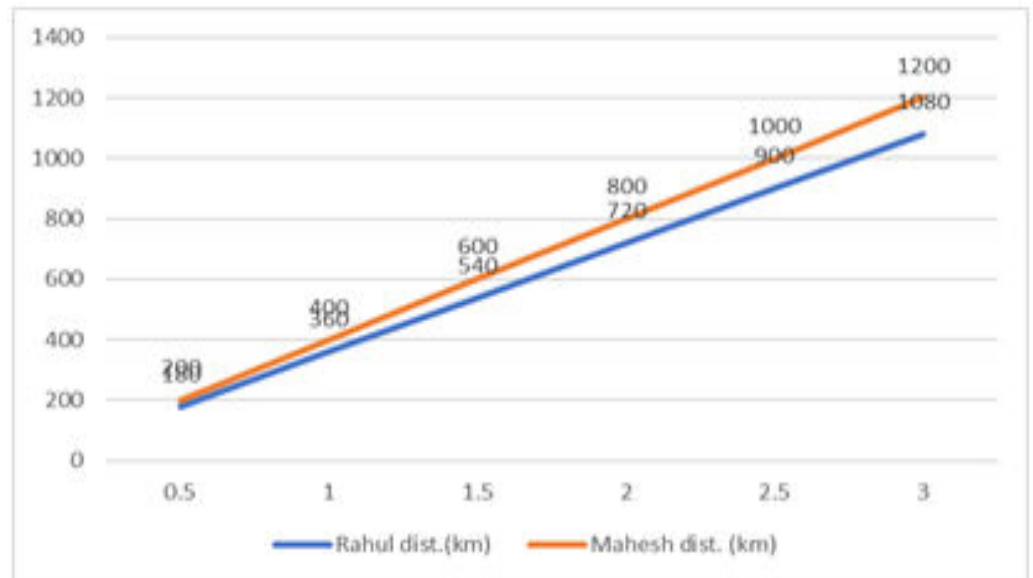
Time (hr)	Distance (km)
0.5	180
1	360
1.5	540
2	720
2.5	900
3	1080



2. **Multiple Line Graph:** The same set of axes is used to plot several lines. An excellent way to compare similar objects over the same time period is via a multiple-line graph.

Example:

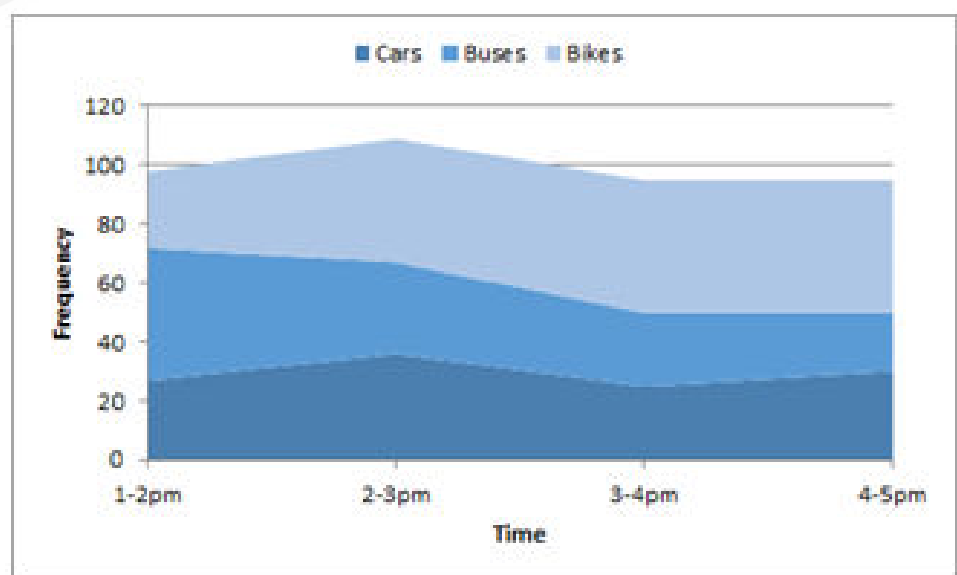
Time(hr)	Rahul dist.(km)	Mahesh dist. (km)
0.5	180	200
1	360	400
1.5	540	600
2	720	800
2.5	900	1000
3	1080	1200



3. **Compound Line Graph:** Whenever one piece of information may be broken down into two or more distinct pieces of data. A compound line graph is the name given to this particular kind of line graph. To illustrate each component that makes up the whole, lines are drawn. The line at the top displays the total, while the line below displays a portion of the total. The size of each component can be determined by the distance that separates every pair of lines.

Example:

Time	Cars	Buses	Bikes
1-2pm	37	45	42
2-3pm	44	34	26
3-4pm	23	39	27
4-5pm	29	41	48



Constructing a line graph: When we have finished creating the data tables, we will then use those tables to build the linear graphs. These graphs are constructed by plotting a succession of points, which are then connected together with straight lines to offer a straightforward method for analysing data gathered over a period of time. It provides a very good visual format of the outcome data that was gathered over the course of time.

Use cases: Tracking changes over both short and long time periods is an important application of line graphs. Additionally, it is utilised to compare changes over the same time period for various groups. Anytime there are little changes, using a line graph rather than a bar graph is always preferable.

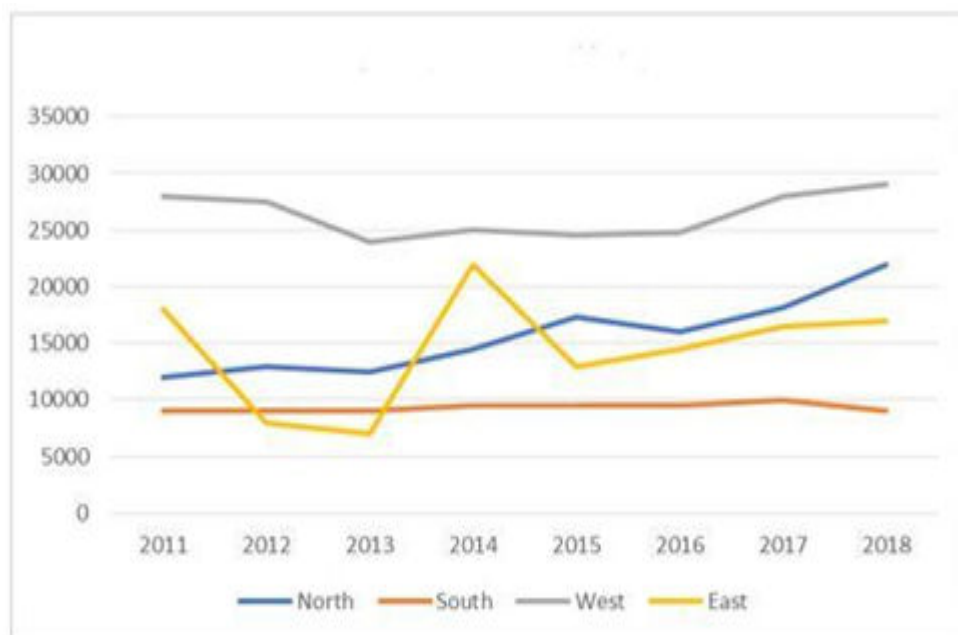
- Straight-line graphs can be used to explain potential future contract markets and business prospects.
- To determine the precise strength of medications, a straight-line graph is employed in both medicine and pharmacy.
- The government uses straight-line graphs for both research and budgetary planning.
- Chemistry and biology both use linear graphs.
- To determine whether our body weight is acceptable for our height, straight-line graphs are employed.

Best Practices

- Only connecting adjacent values along an interval scale should be done with lines.
- In order to provide correct insights, intervals should be of comparable size.
- Select a baseline that makes sense for your set of data; a zero baseline might not adequately capture changes in the data.
- Line graphs are only helpful for comparing data sets if the axes have the same scales.

Example:

Sales	2011	2012	2013	2014	2015	2016	2017	2018
North	12000	13000	12500	14500	17300	16000	18200	22000
South	9000	9000	9000	9500	9500	9500	10000	9000
West	28000	27500	24000	25000	24500	24750	28000	29000
East	18000	8000	7000	22000	13000	14500	16500	17000



Check Your Progress 9:

1. What is a line graph?

.....

.....

.....

2. Where can we use a line graph?

.....

.....

.....

3. Draw a line chart from the following information:

A	B	C	D	E	F	G	H	I	J	K	L	M
Product	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Affordable Segment	173	153	195	147	120	144	148	109	174	130	172	131
Luxury Segment	189	189	105	112	173	109	151	197	174	145	177	161
Super Luxury Segment	185	185	126	134	196	153	112	133	200	145	167	110

8.12 PIE CHART

A pie chart, often referred to as a circle chart, is a style of graph that can be used to summarise a collection of nominal data or to show the many values of a single variable (e.g. percentage distribution). Such a chart resembles a circle that has been divided into a number of equal halves. Each segment corresponds to a specific category. The overall size of the circle is divided among the segments in the same proportion as the category's share of the whole data set.

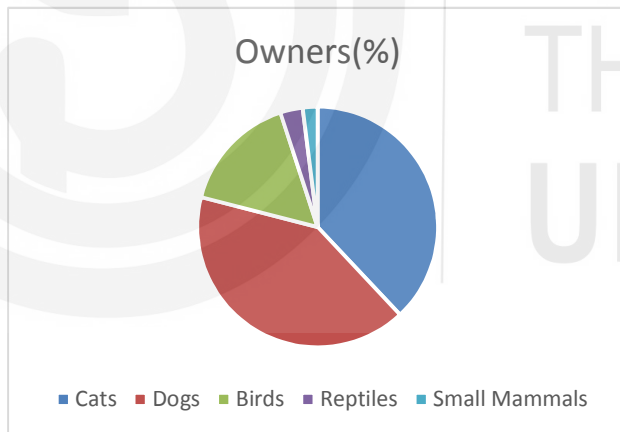
A pie chart often depicts the individual components that make up the whole. In order to bring attention to a particular piece of information that is significant, the illustration may, on occasion, show a portion of the pie chart that is cut away from the rest of the diagram. This type of chart is known as an exploded pie chart.

Types of a Pie Chart: There are mainly two types of pie charts one is a 2D pie chart, and another is a 3D pie chart. This can be further classified into following categories:

1. **Simple Pie Chart:** The most fundamental kind of pie chart is referred to simply as a pie chart and is known as a simple pie chart. It is an illustration that depicts a pie chart in its most basic form.

Example:

Pets	Owners (%)
Cats	38
Dogs	41
Birds	16
Reptiles	3
Small Mammals	2

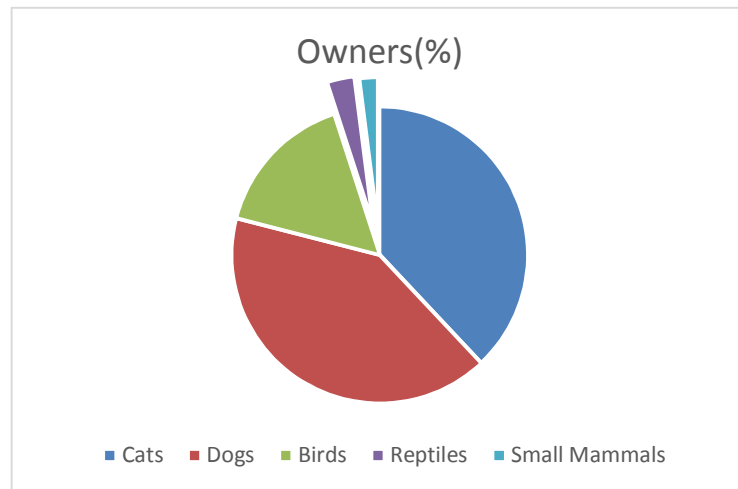


2. **Exploded Pie Chart:** To create an exploding pie chart, you must first separate the pie from the chart itself, as opposed to merging the two elements together. It is common practice to do this in order to draw attention to a certain section or slice of a pie chart.

Example:

Pets	Owners (%)
Cats	38
Dogs	41
Birds	16

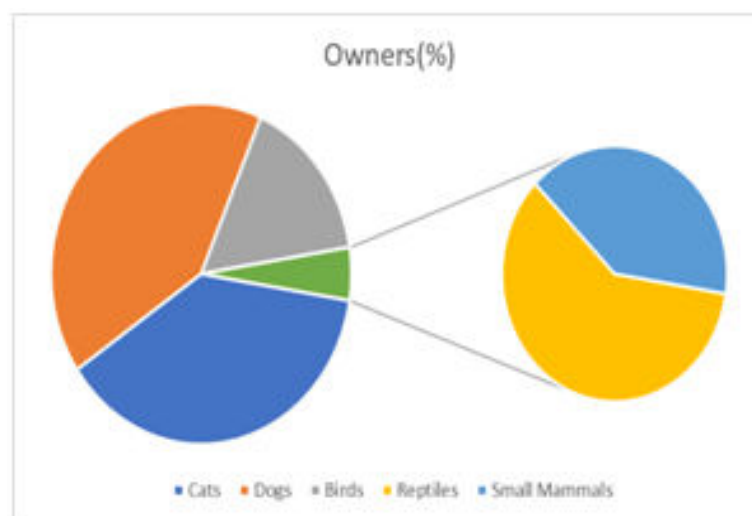
Reptiles	3
Small Mammals	2



3. **Pie of Pie:** A pie-of-pie method is a straightforward approach that enables more categories to be represented on a pie chart without producing an overcrowded and difficult-to-read graph. A pie chart that is generated from an already existing pie chart is referred to as a "pie of pie".

Example:

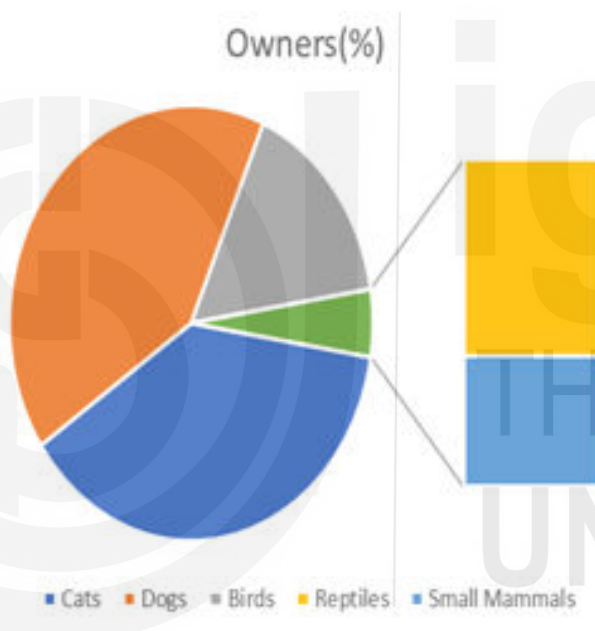
Pets	Owners (%)
Cats	38
Dogs	41
Birds	16
Reptiles	3
Small Mammals	2



3. **Bar of Pie:** A bar of pie is an additional straightforward method for showing additional categories on a pie chart while minimising space consumption on the pie chart itself. The expansion that was developed from the already existing pie chart was a bar graph rather than a pie of pie, despite the fact that both serve comparable objectives.

Example:

Pets	Owners (%)
Cats	38
Dogs	41
Birds	16
Reptiles	3
Small Mammals	2



Constructing a Pie chart: "The total value of the pie is always 100%."

To compute the percentage and angle for making a pie chart, follow the steps given below:

- Categorise the data
- Calculate the total
- Divide the categories
- Convert into percentages
- Finally, calculate the degrees

You may use the pie chart formula – $\text{angle} = \left(\frac{\text{Given Data}}{\text{Total value of Data}} \right) \times 360^\circ$

Use cases: If you want your audience to get a general idea of the part-to-whole relationship in your data, and comparing the exact sizes of the slices is

not as critical to you, then you should use pie charts. And indicate that a certain portion of the whole is disproportionately small or large.

- Voting preference by age group
- Market share of cloud providers

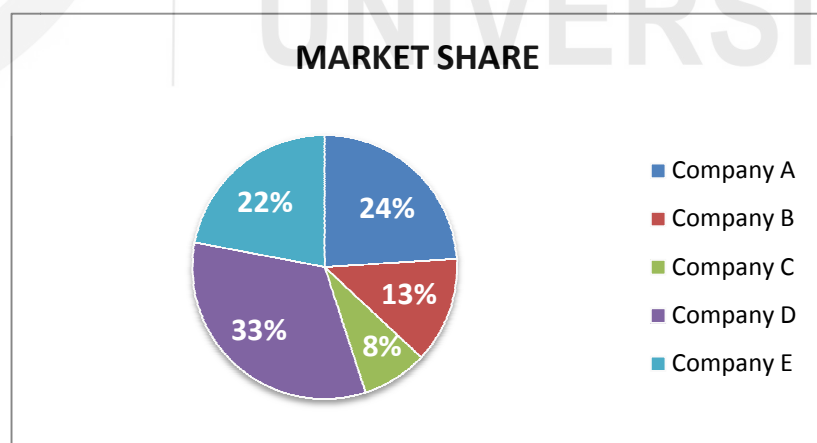
Best Practices

- **Fewer pie wedges are preferred:** The observer may struggle to interpret the chart's significance if there are too many proportions to compare. Similar to this, keep the overall number of pie charts on dashboards to a minimum.

Overlay pies on maps: Pie charts can be used to further deconstruct geographic tendencies in your data and produce an engaging display.

Example

COMPANY	MARKET SHARE
Company A	24%
Company B	13%
Company C	8%
Company D	33%
Company E	22%



Check Your Progress 10:

1. What is the pie chart?

.....

.....

.....

2. What are the different types of pie charts?

.....

.....

.....

3. Draw a pie chart from the following information:

<i>Table: Favorite Type of Movie</i>				
Comedy	Action	Romance	Drama	SciFi
4	5	6	1	4

8.13 DOUGHNUT CHART

Pie charts have been superseded by a more user-friendly alternative called a doughnut chart, which makes reading pie charts much simpler. It is recognised that these charts express the relationship of 'part-to-whole,' which is when all of the parts represent one hundred percent when collected together. It presents survey questions or data with a limited number of categories for making comparisons.

In comparison to pie charts, they provide far more condensed and straightforward representations. In addition, the centre hole can be used to assist in the display of relevant information. You might use them in segments, where each arc would indicate a proportional value associated with a different piece of data.

Constructing a Doughnut chart: A doughnut chart, like a pie chart, illustrates the relationship of individual components to the whole, but unlike a pie chart, it can display more than one data series at the same time. A ring is added to a doughnut chart for each data series that is plotted within the chart itself. The beginning of the first data series can be seen near the middle of the chart. A specific kind of pie chart called a doughnut chart is used to show the percentages of categorical data. The amount of data that falls into each category is indicated by the size of that segment of the doughnut. The creation of a doughnut chart involves the use of a string field and a number, count of features, or rate/ratio field.

There are two types of doughnut charts: one is a normal doughnut chart, and another is an exploded doughnut chart. Exploding doughnut charts, much like exploded pie charts, highlight the contribution of each value to a total while emphasising individual values. However, unlike exploded pie charts, exploded doughnut charts can include more than one data series.

Use cases: Doughnut charts are good to use when comparing sets of data. By using the size of each component to reflect the percentage of each category, they are used to display the proportions of categorical data. A string field and a count of features, number, rate/ratio, or fields are used to make a doughnut chart. Use cases of the Doughnut chart are:

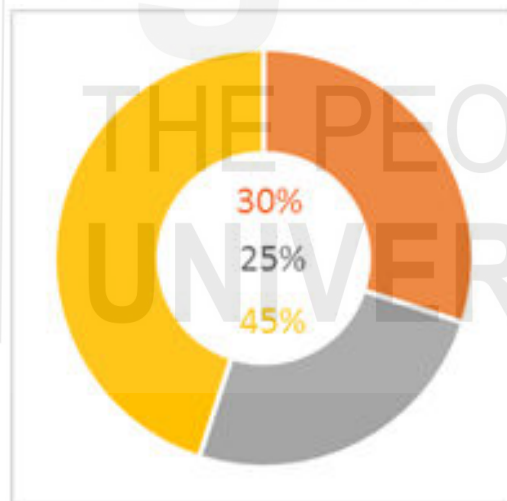
- Android OS market share
- Monthly sales by channel

Best Practices

- Stick to five slices or less because thinner and long-tail slices become unreadable and uncomparable.
- Use this chart to display one point in time with the help of the filter legend.
- Well-formatted and informative labels are essential because the information conveyed by circular shapes alone is not enough and is imprecise.
- It is a good practice to sort the slices to make it more clear for comparison.

Example:

Project Status	
Completed	30%
Work in progress	25%
Incomplete	45%



Check Your Progress 11:

1. What is the doughnut chart?

.....

.....

.....

2. What distinguishes a doughnut chart from a pie chart?

.....

.....

.....

3. Draw a doughnut chart from the following information:

Product	2020	2021
x	40	50
y	30	60
z	60	70

8.14 AREA CHART

An area chart, a hybrid of a line and bar chart, shows the relationship between the numerical values of one or more groups and the development of a second variable, most often the passage of time. The inclusion of shade between the lines and a baseline, similar to a bar chart's baseline, distinguishes a line chart from an area chart. An area chart has this as its defining feature.

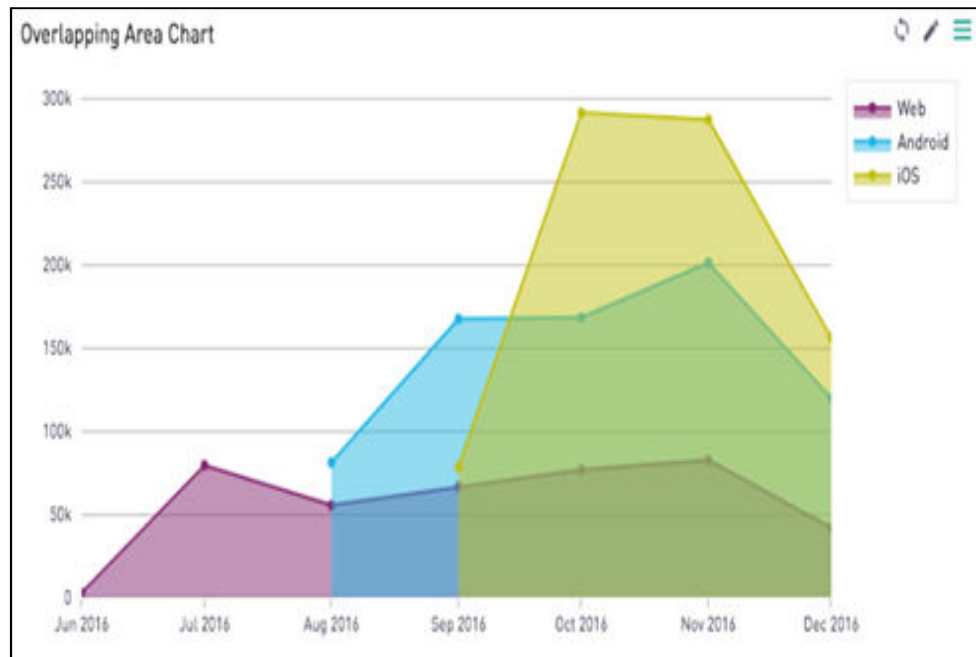
Types of Area Chart:

Overlapping area chart: An overlapping area chart results if we wish to look at how the values of the various groups compare to one another. The conventional line chart serves as the foundation for an overlapping area chart. One point is plotted for each group at each of the horizontal values, and the height of the point indicates the group's value on the vertical axis variable.

All of the points for a group are connected from left to right by a line. A zero baseline is supplemented by shading that is added by the area chart between each line. Because the shading for different groups will typically overlap to some degree, the shading itself incorporates a degree of transparency to ensure that the lines delineating each group may be seen clearly at all times.

The shading brings attention to a group that has the highest value by highlighting the group's pure hue. Take care that one series is not always higher than the other, as this could cause the plot to become confused with the stacked area chart, which is the other form of an area chart. In circumstances like these, the most prudent course of action will consist of sticking to the traditional line chart.

Months (2016)	Web	Android	IOS
June	0	-	
July	70k	-	
Aug	55k	80k	
Sep	60k	165k	80k
Oct	70k	165k	295k
Nov	80k	200k	290k
Dec	40k	125k	155k

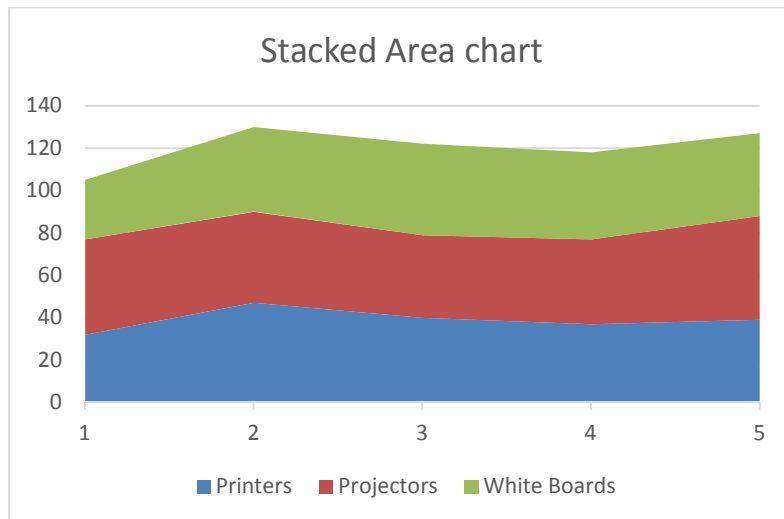


Stacked area chart: The stacked area chart is what is often meant to be conveyed when the phrase "area chart" is used in general conversation. When creating the chart of overlapping areas, each line was tinted based on its vertical value all the way down to a shared baseline. Plotting lines one at a time creates the stacked area chart, which uses the height of the most recent group of lines as a moving baseline. Therefore, the total that is obtained by adding up all of the groups' values will correspond to the height of the line that is entirely piled on top.

When you need to keep track of both the total value and the breakdown of that total by groups, you should make use of a stacked area chart. This type of chart will allow you to do both at the same time. By contrasting the heights of the individual curve segments, we are able to obtain a sense of how the contributions made by the various subgroups stack up against one another and the overall sum.

Example:

A	B	C	D
	Printers	Projectors	White Boards
2017	32	45	28
2018	47	43	40
2019	40	39	43
2020	37	40	41
2021	39	49	39



Use Cases: In most cases, many lines are drawn on an area chart in order to create a comparison between different groups (also known as a series) or to illustrate how a whole is broken down into its component pieces. This results in two distinct forms of area charts, one for each possible application of the chart.

- **The trend of the magnitude of a single quantitative variable** - An increase in a public company's revenue reserves, programme enrollment from a qualified subgroup by year, and trends in mortality rates over time by primary causes of death are just a few examples.
- **Comparison of the contributions made by different category members (or groups)-** the variation in staff sizes among departments or support tickets opened for various problems.
- Birth and death rates over time for a region, the magnitudes of cost versus revenue for a business, the magnitudes of export versus import over time for a country

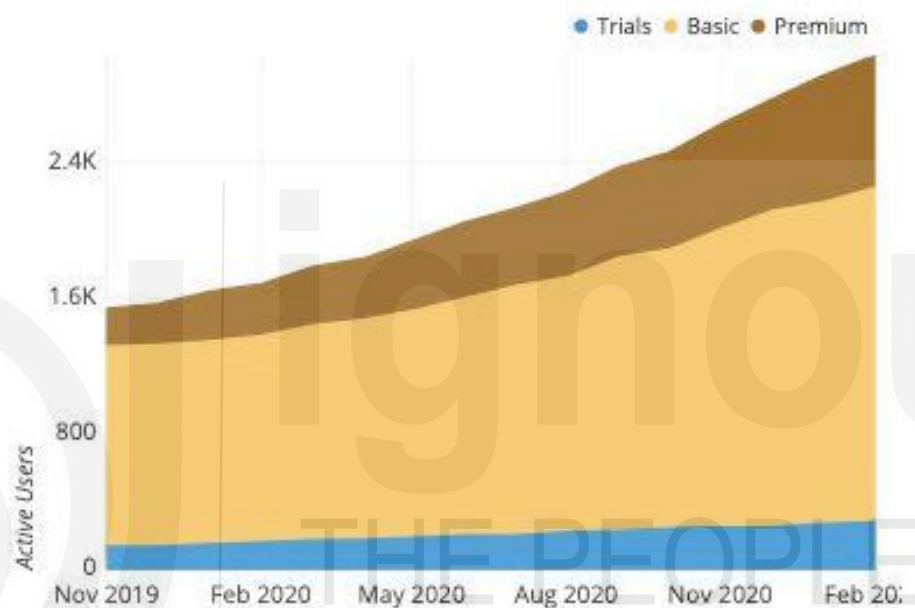
Best Practices:

- To appropriately portray the proportionate difference in the data, start the y-axis at 0.
- To boost readability, choose translucent, contrasting colours.
- Keep highly variable data at the top of the chart and low variable data at the bottom during stacking.
- If you need to show how each value over time contributes to a total, use a stacked area chart.
- However, it is recommended to utilise 100% stacked area charts if you need to demonstrate a part-to-whole relationship in a situation where the cumulative total is unimportant.

Example:

Example of data structure

MONTH	TRIALS	BASIC	PREMIUM
2019-11	154	1180	201
2019-12	157	1186	219
2020-01	170	1195	270
2020-02	180	1213	285



The above Stacked area chart belongs to tele-service offered by various television-based applications. In this data, there are different types of subscribers who are using the services provided by tele-applications in different months.

Check Your Progress 12:

1. What is an area chart?

.....

.....

.....

.....

2. What are the types of area charts?

.....

.....

.....

.....

3. Draw an area chart from the following information:

	Product A	Product B	Product C
2017	2000	600	75
2018	2200	450	85
2019	2100	500	125
2020	3000	750	123

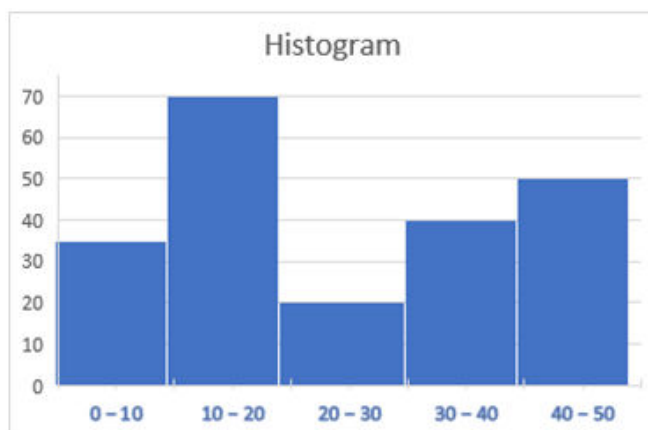
8.15 SUMMARY

This Unit introduces you to some of the basic charts that are used in data science. The Unit defines the characteristics of Histograms, which are very popular in univariate frequency analysis of quantitative variables. It then discusses the importance and various terms used in the box plots, which are very useful while comparing quantitative variables over some qualitative characteristics. Scatter plots are used to visualise the relationships between two quantitative variables. The Unit also discusses the heat map, which are excellent visual tools for comparing values. In case three variables are to be compared, then you may use bubble charts. The Unit also highlights the importance of bar charts, distribution plots, pair plots and line graphs. In addition, it highlights the importance of Pie charts, doughnut charts and area charts for visualising different kinds of data. In addition, there are many different kinds of charts that are used in different analytical tools. You may read about them from references.

8.16 ANSWERS

Check Your Progress 1

1. A bar graph is a pictorial representation using vertical and horizontal bars in a graph. The length of the bars is proportional to the measure of data. It is also called a bar chart. A histogram is also a pictorial representation of data using rectangular bars that are adjacent to each other. It is used to represent grouped frequency distribution with continuous classes.
- 2.



3. It is used to summarise continuous or discrete data that is measured on an interval scale. It is frequently used to depict the key characteristics of the data distribution conveniently.
4. A histogram is a graphic depiction of data points arranged into user-specified ranges. The histogram, which resembles a bar graph in appearance, reduces a data series into an intuitive visual by collecting numerous data points and organising them into logical ranges or bins.

Check Your Progress 2

1. Follow these instructions to interpret a box plot:

Step 1: Evaluate the major characteristics. Look at the distribution centre and spread. Examine the potential impact of the sample size on the box plot's visual appeal.

Step 2: Search for signs of anomalous or out-of-the-ordinary data. Skewed data suggest that data may not be normal. Other situations in your data may be indicated by outliers.

Step 3: Evaluate and compare groups. Evaluate and compare the centre and spread of groups if your box plot contains them.

2. A box plot is a common method of showing data distribution based on a five-number summary ("minimum," first quartile ("Q1"), median ("Q3"), and "maximum"). You can learn more about your outliers' values from it.
3. Box plots are generally used for three purposes -
 - Finding outliers in the data
 - Finding the dispersion of data from a median
 - Finding the range of data
4. The box plot distribution will reveal the degree to which the data are clustered, how skewed they are, and also how symmetrical they are.
 - Positively Skewed: The box plot is positively skewed if the distance from the median to the maximum is greater than the distance from the median to the minimum.
 - Negatively skewed: Box plots are said to be negatively skewed if the distance from the median to the minimum is higher than the distance from the median to the maximum.
 - Symmetric: When the median of a box plot is equally spaced from both the maximum and minimum values, the box plot is said to be symmetric.

Check Your Progress 3

1.
 - The most practical method for displaying bivariate (2-variable) data is a scatter plot.
 - A scatter plot can show the direction of a relationship between two variables when there is an association or interaction between them (positive or negative).
 - The linearity or nonlinearity of an association or relationship can be

ascertained using a scatter plot.

- A scatter plot reveals anomalies, questionably measured data, or incorrectly plotted data visually.

2.

- The Title- A brief description of what is in your graph is provided in the title.
- The Legend- The meaning of each point is explained in the legend.
- The Source- The source explains how you obtained the data for your graph.
- Y-Axis.
- The Data.
- X-Axis.

3. A scatter plot is composed of a horizontal axis containing the measured values of one variable (independent variable) and a vertical axis representing the measurements of the other variable (dependent variable). The purpose of the scatter plot is to display what happens to one variable when another variable is changed.

4.

- Positive Correlation.
- Negative Correlation.
- No Correlation (None)

Check Your Progress 4

1. Three main types of input exist to plot a heatmap: wide format, correlation matrix, and long format.

Wide format: The wide format (or the untidy format) is a matrix where each row is an individual, and each column is an observation. In this case, the heatmap makes a visual representation of the matrix: each square of the heatmap represents a cell. The colour of the cell changes according to its value.

Correlation matrix: Suppose you measured several variables for n individuals. A common task is to check if some variables are correlated. You can easily calculate the correlation between each pair of variables, and plot this as a heatmap. This lets you discover which variable is related to the other.

Long format: In the tidy or long format, each line represents an observation. You have 3 columns: individual, variable name, and value (x , y and z). You can plot a heatmap from this kind of data.

2. A heat map is a two-dimensional visualisation of data in which colours stand in for values. A straightforward heat map offers a quick visual representation of the data. The user can comprehend complex data sets with the help of more intricate heat maps.
3. Using one variable on each axis, heatmaps are used to display relationships between two variables. You can determine if there are any

trends in the values for one or both variables by monitoring how cell colours vary across each axis.

Check Your Progress 5

1. A bubble chart is a variant of a scatter chart in which the data points are swapped out for bubbles, with the size of the bubbles serving as a representation of an additional dimension of the data. A bubble chart horizontal and vertical axes are both value axes.
2. To identify whether at least three numerical variables are connected or exhibit a pattern, bubble charts are utilised. They could be applied in specific situations to compare categorical data or demonstrate trends across time.
3. In scatter charts, one numeric field is displayed on the x-axis and another on the y-axis, making it simple to see the correlation between the two values for each item in the chart. A third numerical field in a bubble chart regulates the size of the data points.
4. Any bubbles between 0 and 5 pts on this scale will appear at 5 pt, and all the bubbles on your chart will be between 5 and 20 pts. To construct a chart that displays many dimensions, combine bubble size with colour by value.

Check Your Progress 6

Answer 1:

In the process of statistics development, bar charts are typically employed to display the data. The following is a list of some of the applications of the bar chart:

To clearly illustrate the relationships between various variables, bar charts are typically utilised. When presented in a pictorial format, the parameters can be more quickly and easily envisioned by the user.

Bar charts are the quickest and easiest way to display extensive amounts of data while also saving time.

The method of data representation that is most commonly utilised. As a result, it is utilised in a variety of different sectors.

When studying trends over extended amounts of time, it is helpful to have this information.

Answer 2:

Charts are primarily divided into two categories:

Horizontal Bar Charts:

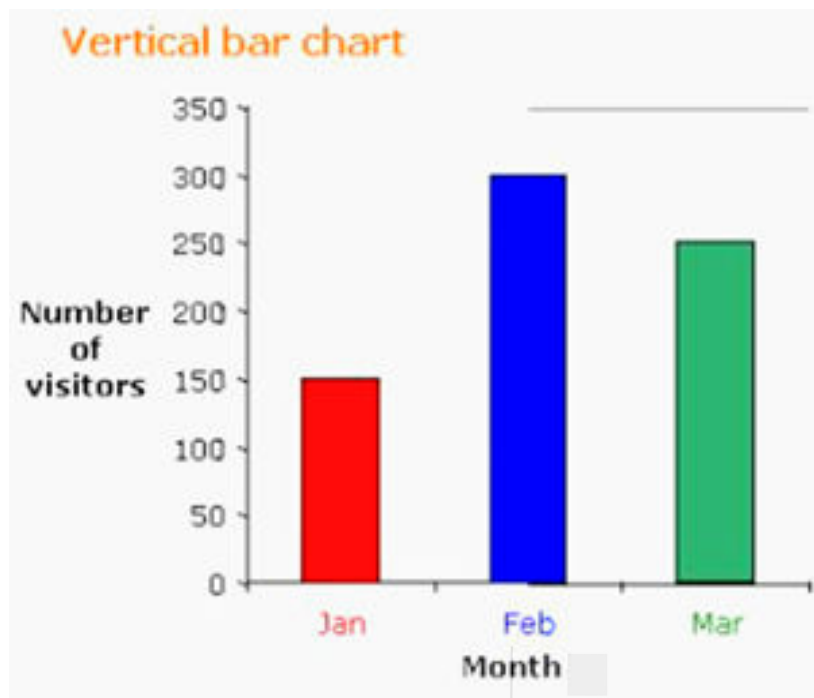
Vertical Bar Charts

We can further divide into two types:

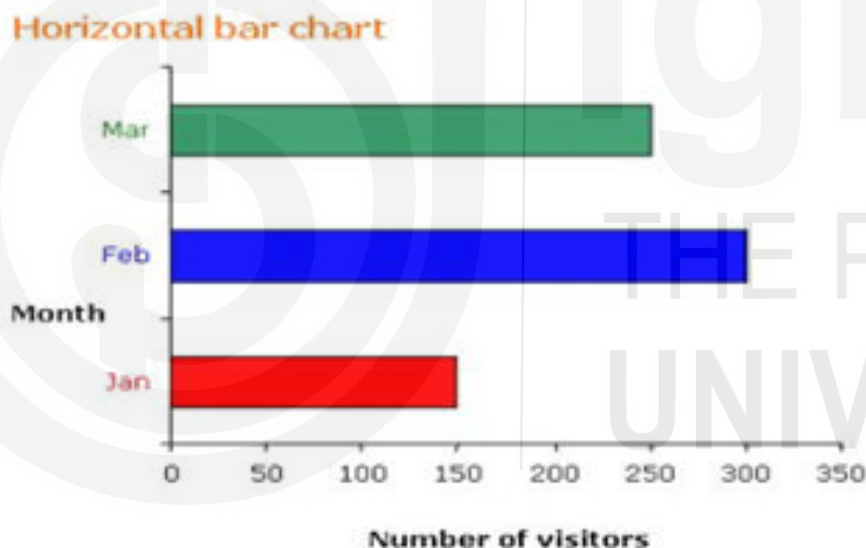
Grouped Bar Charts

Stacked Bar Charts

Answer 3:



Answer4:



Check Your Progress 7:

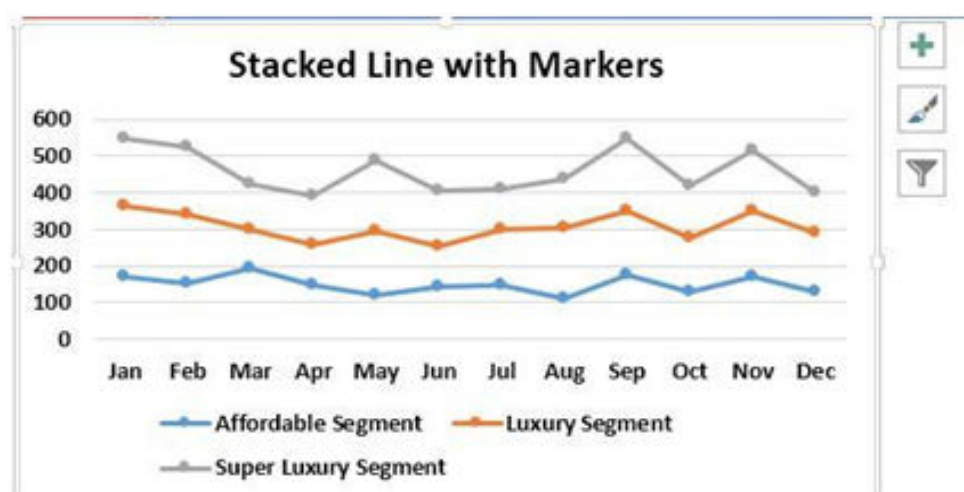
1. For visually assessing the distribution of sample data, you can draw distribution charts. Using these charts, you can contrast the actual distribution of the data with the theoretical values expected from a certain distribution.
2. The distribution plot is useful for analysing the relationship between the range of a set of numerical data and its distribution. You are only allowed to use one or two dimensions and one measure when creating a distribution graphic.
3. These graphs show - how the data is distributed; how the data is composed; how values relate to one another.

Check Your Progress 8:

1. We can visualise pairwise relationships between variables in a dataset using pair plots. By condensing a lot of data into a single figure, this gives the data a pleasant visual representation and aids in our understanding of the data.
2. A scatter plot of a and b, one of a and c, and finally one of a and d are shown in the first line. b and a (symmetric to the first row) are in the second row, followed by b and c, b and d, and so on. In pairs, no sums, mean squares, or other calculations are performed. That is in your data frame if you discover it in your pairings plot.
3. Pair plots are used to determine the most distinct clusters or the best combination of features to describe a connection between two variables. By creating some straightforward linear separations or basic lines in our data set, it also helps to create some straightforward classification models.

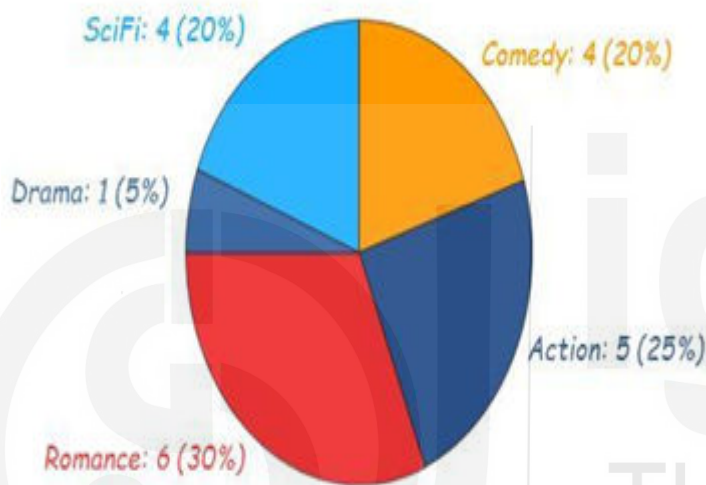
Check Your Progress 9:

1. A graph that depicts change over time by means of points and lines is known as a line graph, line chart, or line plot. It is a chart that depicts a line uniting numerous points or a line that illustrates the relation between the points. The line or curve used to depict quantitative data between two changing variables in the graph combines a sequence of succeeding data points to create a representation of the graph.
2. Tracking changes over a short as well as a long period of time is one of the most important applications of line graphs. Additionally, it is used to compare the modifications that have occurred for various groups throughout the course of the same period of time. When dealing with data that has only minor variations, using a line graph rather than a bar graph is strongly recommended. For instance, the finance team at a corporation may wish to chart the evolution of the cash balance that the company now possesses throughout the course of time.
- 3.



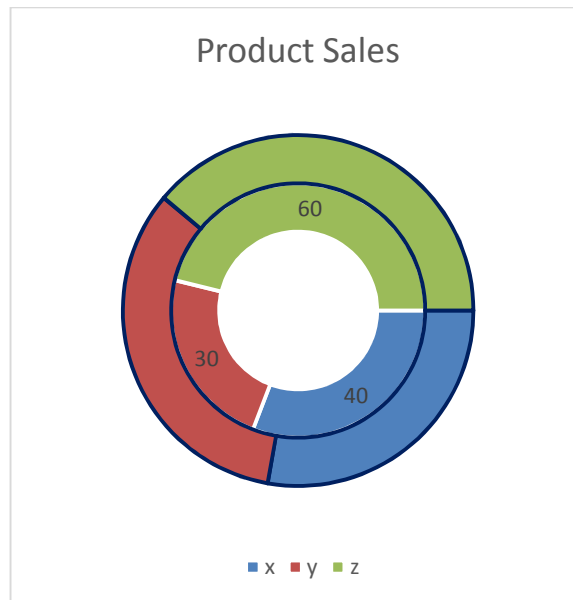
Check Your Progress 10:

1. A pie chart, often referred to as a circle chart, is a style of graph that can be used to summarise a collection of nominal data or to show the many values of a single variable. (e.g. percentage distribution).
2. There are mainly two types of pie charts: one is 2D pie chart and another is 3D pie chart. This can be further classified into following categories:
 1. Simple Pie Chart
 2. Exploded Pie Chart
 3. Pie of Pie
 4. Bar of Pie
- 3.



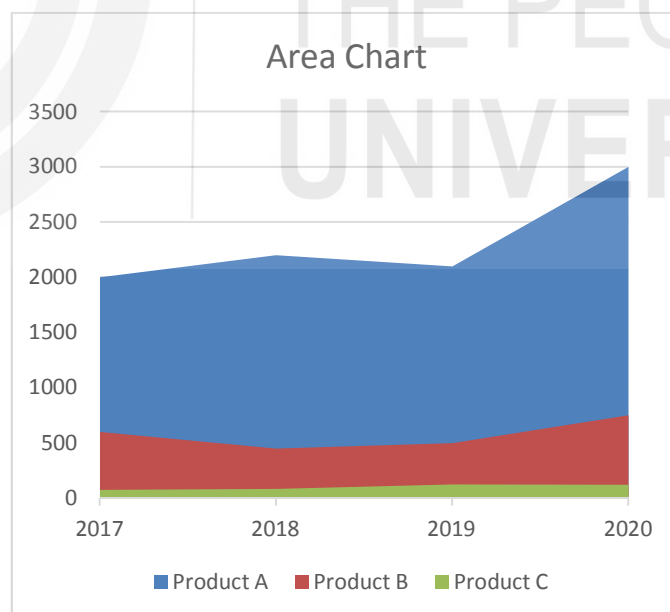
Check Your Progress 11:

1. Pie charts have been superseded by a more user-friendly alternative called a doughnut chart, which makes reading pie charts much simpler. It is recognised that these charts express the relationship of 'part-to-whole,' which is when all of the parts represent one hundred percent when collected together. In comparison to pie charts, they provide for more condensed and straightforward representations.
2. A donut chart is similar to a pie chart, with the exception that the centre is cut off. When you want to display particular dimensions, you use arc segments rather than slices. Just like a pie chart, this form of chart can assist you in comparing certain categories or dimensions to the greater overall; nevertheless, it has a few advantages over its pie chart counterpart.
- 3.



Check Your Progress 12:

1. An area chart shows how the numerical values of one or more groups change in proportion to the development of a second variable, most frequently the passage of time. It combines the features of a line chart and a bar chart. A line chart can be differentiated from an area chart by the addition of shading between the lines and a baseline, just like in a bar chart. This is the defining characteristic of an area chart.
2. Overlapping area chart and Stacked area chart
- 3.



8.17 REFERENCES

- Useful Ways to Visualise Your Data (With Examples). Pdf
- Data Visualisation Cheat Sheet. Pdf
- Which chart or graph is right for you? Pdf
- <https://www.excel-easy.com/examples/frequency-distribution.html>
- <https://analyticswithlohr.com/2020/09/15/556/>
- <https://www.fusioncharts.com/line-charts>
- <https://evolytics.com/blog/tableau-201-make-stacked-area-chart/>
- <https://chartio.com/learn/charts/area-chart-complete-guide/>
- <https://www.lifewire.com/exploding-pie-charts-in-excel-3123549>
- <https://www.formpl.us/resources/graph-chart/line/>
- <https://www150.statcan.gc.ca/n1/edu/power-pouvoir/ch9/bargraph-diagrammeabarres/5214818-eng.htm>
- <https://sixsigmamania.com/?p=475>
- <https://study.com/academy/lesson/measures-of-dispersion-variability-and-skewness.html>

UNIT 9 EXCEL FOR DESCRIPTIVE DATA ANALYSIS

Structure

- 9.0 Introduction
- 9.1 Objectives
- 9.2 Importing Data from Various File Formats
- 9.3 Data Handling and Pre-processing
- 9.4 Frequency Distribution
- 9.5 Central Tendency and Dispersion
- 9.6 Descriptive Data Analysis and its Interpretation
- 9.7 Summary
- 9.8 Solutions/Answers

9.0 INTRODUCTION

In the world of data analysis, the ability to summarize and interpret data effectively is crucial. Descriptive data analysis serves as the foundation for understanding datasets, allowing analysts to uncover patterns, trends, and insights that may not be immediately obvious. Microsoft Excel, one of the most widely used tools for data analysis, provides a versatile platform for performing descriptive statistics.

This unit will guide you through the essentials of using Excel for descriptive data analysis, covering various tools and functions that enable you to explore and summarize data efficiently. From calculating basic summary statistics such as mean, median, and standard deviation to creating visualizations like histograms and box plots, Excel offers an intuitive interface to analyze and present data.

Whether you're a beginner looking to get acquainted with Excel's analytical capabilities or an experienced user seeking to refine your skills, this chapter will equip you with the knowledge needed to perform a comprehensive descriptive analysis of any dataset. You will learn how to leverage Excel's built-in functions, data tools, and visualization options to transform raw data into meaningful insights that can guide informed decision-making.

By the end of this chapter, you will be able to confidently navigate Excel's data analysis features and use them to perform essential descriptive statistics that are critical in data-driven environments.

9.1 OBJECTIVES

After finishing this unit, you will be able to:

- Describe the meaning

9.2 IMPORTING DATA FROM VARIOUS FILE FORMATS

In the field of data analysis, the ability to seamlessly import data from a variety of file formats is an essential skill. Data often comes in different forms, such as spreadsheets, text files, databases, or even web-based sources. Each of these file types presents unique challenges and opportunities when it comes to importing data into Excel for analysis.

This section will focus on how to import data from various commonly used file formats into Excel. Whether you are working with CSV, TXT, or database files like Access or SQL, Excel provides a range of options to facilitate smooth data importation. Understanding how to properly bring external data into Excel is crucial for ensuring the integrity and accuracy of your analysis.

We will explore the step-by-step processes for importing data from different formats, discussing best practices, potential pitfalls, and tips for maintaining data consistency and quality throughout the import process. By mastering this skill, you'll be able to work with diverse datasets from a variety of sources, making Excel a powerful tool for your data analysis tasks.

Here's how you can import data from different file formats into Excel:

1. Importing from a Text or CSV File

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **From Text/CSV** in the **Get & Transform Data** group.
- c) **Browse** and select your file, then click **Import**.
- d) **Preview the data** and make any necessary changes in the Import Wizard.
- e) Click **Load** to import the data into Excel.

2. Importing from an XML File

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **Get Data**, then **From File**, and select **From XML**.
- c) **Browse** and select your XML file, then click **Import**.
- d) **Select the elements** you want to import and click **OK**.
- e) Click **Load** to import the data into Excel.

3. Importing from an Access Database

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **Get Data**, then **From Database**, and select **From Microsoft Access Database**.
- c) **Browse** and select your Access database file, then click **Import**.
- d) **Select the tables** you want to import and click **Load**.

4. Importing from a Web Page

- a) **Open Excel** and go to the **Data** tab.

- b) Click on **Get Data**, then **From Other Sources**, and select **From Web**.
- c) **Enter the URL** of the web page you want to import data from and click **OK**.
- d) **Select the data** you want to import from the Navigator window.
- e) Click **Load** to import the data into Excel.

5. Importing from JSON

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **Get Data**, then **From File**, and select **From JSON**.
- c) **Browse** and select your JSON file, then click **Import**.
- d) **Select the data** you want to import from the Navigator window.
- e) Click **Load** to import the data into Excel.

These steps should cover most common file formats you might want to import into Excel.

Check Your Progress 1

- 1. What file types can Excel import data from?
.....
.....
.....
- 2. What is the importance of maintaining data consistency during import?
.....
.....
.....
- 3. What steps are followed to import data from a CSV file?
.....
.....
.....
- 4. How can data be imported from a web page into Excel?
.....
.....
.....

9.3 DATA HANDLING AND PRE-PROCESSING

Data handling and pre-processing are essential steps in the data analysis pipeline, though they refer to different aspects of preparing data for analysis. While both involve transforming raw data into a clean, structured form, each process has a distinct focus.

Data Handling refers to the management and organization of data within a dataset. It involves tasks such as sorting, filtering, and structuring data to make it easier to analyze. For instance, data handling can include organizing columns, merging multiple datasets, or converting data into a more useful format. The goal is to ensure that the dataset is structured logically and efficiently, often by removing or consolidating irrelevant or redundant information.

On the other hand, **Data Pre-Processing** focuses more on cleaning the dataset to ensure its quality and reliability. This includes handling missing values, correcting errors, dealing with outliers, and ensuring consistency across the data. Pre-processing may also involve transforming variables or normalizing data to make it comparable across different units of measurement. The objective is to remove any issues that could distort or skew analysis results, ultimately ensuring the data is accurate and suitable for the intended analysis.

While both data handling and pre-processing are critical to the success of any data analysis project, the main difference lies in their focus. Data handling is primarily concerned with the organization and structuring of data, whereas data pre-processing focuses on cleaning and preparing the data to ensure its quality. In practice, the two processes often overlap, with one leading naturally into the other. For example, handling missing values (a pre-processing task) often involves reorganizing or filtering the dataset (a data handling task).

Before any meaningful analysis can be performed, the raw data must be properly prepared and cleaned. Data handling and pre-processing are critical steps in ensuring that the information you are working with is accurate, consistent, and ready for analysis. In Excel, this process involves organizing, transforming, and cleaning data to eliminate errors, deal with missing values, and ensure that the dataset is structured correctly.

This section will guide you through the essential techniques for handling and pre-processing data within Excel. We will cover methods for cleaning datasets, such as removing duplicates, handling missing values, and correcting inconsistencies. Additionally, you will learn how to reshape data, filter out irrelevant information, and apply transformations that make the data more useful for analysis.

Effective data handling and pre-processing not only improve the quality of your analysis but also save time and effort in the long run. By the end of this section, you will be equipped with the tools and strategies needed to prepare your data for further analysis, ensuring that your findings are built on a solid foundation of clean, reliable information.

In short, both data handling and pre-processing are integral to preparing data for analysis. Properly handling and cleaning data in these early stages helps analysts avoid pitfalls that could lead to inaccurate conclusions, ultimately ensuring more reliable and insightful results from the analysis.

Data Handling: Data handling refers to the broader process of managing data throughout its lifecycle. It includes tasks like:

- **Data Collection:** Gathering data from various sources.
- **Data Storage:** Organizing and storing data in databases, spreadsheets, or other storage systems.
- **Data Access:** Retrieving and making data available for analysis or other purposes.
- **Data Management:** Ensuring data quality, consistency, and security.
- **Data Sharing:** Distributing data to stakeholders as needed.

Data Preprocessing: Data preprocessing is a subset of data handling focused specifically on preparing raw data for analysis. It involves tasks like:

- **Data Cleaning:** Removing duplicates, correcting errors, handling missing values, and normalizing data.
- **Data Transformation:** Converting data into a suitable format, such as aggregating, encoding, or scaling data.
- **Data Integration:** Combining data from different sources to create a unified dataset.
- **Data Reduction:** Reducing the volume of data by selecting relevant features or using dimensionality reduction techniques.

Key Differences between Data Handling and Data Preprocessing are:

- **Scope:** Data handling is broader and encompasses the entire lifecycle of data, while data preprocessing is specifically about preparing data for analysis.
- **Focus:** Data handling includes management, storage, and sharing aspects, whereas preprocessing focuses on cleaning and transforming data for analytical purposes.
- **Stage:** Data handling happens at all stages of the data lifecycle, while preprocessing is typically an initial step before data analysis.

Some frequently used Excel functions, commands and formulas for Data Handling: there is a fundamental difference between the functions, commands and formulas for Data Handling, to begin with, firstly understand the difference among them.

1. **Excel Functions for Data Handling : Functions** in Excel are predefined formulas that perform specific operations on data. They are built-in operations that can be directly used in cells to carry out calculations, manipulate text, or analyze data. Functions are used within formulas to perform calculations, making them a fundamental part of data handling.

For example, **TEXT functions** such as =TRIM(A1) and =LEFT(A1, 3) can clean or extract parts of text data. Statistical functions like =SUM(A1:A10) and =AVERAGE(B1:B10) help you quickly summarize data. Lookup functions, such as =VLOOKUP(A1, B1:C10, 2, FALSE), allow you to search for data within a range. Logical functions, such as =IF(A1 > 10, "Yes", "No"), help you evaluate conditions in your data.

Excel functions are essential because they allow you to manipulate and retrieve data directly inside cells without needing to manually process it. They are predefined operations that return a specific value based on the arguments provided.

2. **Excel Commands for Data Handling: Commands** are actions or tools within the Excel interface that enable users to perform data-related tasks through the ribbon, context menu, or keyboard shortcuts. Unlike functions, commands don't require entering formulas in cells but are instead applied to the dataset via Excel's built-in features.

For example, **Sort** is a command that organizes data in ascending or descending order, which can be accessed through the "Data" tab. The **Filter** command helps display only rows that meet certain criteria, also found in the "Data" tab. The **Remove Duplicates** command eliminates duplicate entries in a dataset, ensuring that only unique values remain. **Text to Columns** is another command that splits data from one column into multiple columns based on a delimiter, such as commas or spaces.

Commands are used to alter the structure of data or modify how it is displayed, often without needing to write any formulas. They provide quick ways to reorganize, clean, or modify data directly through the Excel interface.

3. **Excel Formulas for Data Handling: Formulas** in Excel combine one or more functions, cell references, and operators to perform more complex data manipulations or calculations. A formula is an expression entered into a cell, which can be as simple as adding two values or as complex as combining multiple functions for advanced data analysis.

For example, a basic formula such as `=A1 + B1` adds the values in two cells. More complex formulas might combine functions, such as `=IF(A1 > 10, "Pass", "Fail")`, which evaluates a condition and returns one result if true and another if false. Nested formulas, like `=SUM (IF (A1:A10 > 5, B1:B10))`, allow for multiple levels of conditions. **Array formulas**, such as `{=SUM (A1:A10 * B1:B10)}`, handle multiple values simultaneously and perform operations on entire arrays.

Formulas provide the flexibility to customize data handling, allowing you to create specific calculations or manipulations by combining various functions and operators. They are dynamic and can be adapted to a wide range of tasks, depending on the needs of the data analysis.

Key Differences between Function, Command and Formula:

- **Functions** are predefined tools in Excel that are used within formulas to perform specific operations on data. They are written inside cells and return a result based on their input.
- **Commands** are actions performed through the Excel interface (via ribbon, menus, or shortcuts). They modify data directly but don't require typing formulas in cells.
-

- **Formulas** are combinations of functions, cell references, and operators. They allow users to perform complex calculations and data manipulations by combining multiple elements into a single expression.

Each of these tools—**functions**, **commands**, and **formulas**—play a crucial role in data handling within Excel, making it easier to organize, analyze, and manipulate datasets according to specific needs.

Frequently Used Excel Functions for Data Handling & Pre-Processing

These are built-in formulas that help with organizing and manipulating data directly in the cells:

1. **TEXT Functions** (for manipulating and cleaning text data):
 - **LEFT**: Extracts a specified number of characters from the beginning of a text string.
 - Example: =LEFT(A1, 5)
 - **RIGHT**: Extracts characters from the end of a text string.
 - Example: =RIGHT(A1, 3)
 - **MID**: Extracts a substring from the middle of a text string.
 - Example: =MID(A1, 3, 4)
 - **TRIM**: Removes extra spaces from a text string.
 - Example: =TRIM(A1)
 - **UPPER**: Converts text to uppercase.
 - Example: =UPPER(A1)
 - **LOWER**: Converts text to lowercase.
 - Example: =LOWER(A1)
 - **PROPER**: Capitalizes the first letter of each word.
 - Example: =PROPER(A1)
 - **TEXTJOIN**: Joins text from multiple cells using a specified delimiter.
 - Example: =TEXTJOIN(" ", TRUE, A1:A5)
2. **Lookup and Reference Functions** (for retrieving and referencing data):
 - **VLOOKUP**: Searches for a value in the first column and returns a value in the same row from another column.
 - Example: =VLOOKUP(A2, B2:D10, 3, FALSE)
 - **HLOOKUP**: Searches for a value in the first row and returns a value in the same column from another row.
 - Example: =HLOOKUP(A1, B1:F3, 2, FALSE)
 - **INDEX**: Returns the value of a cell in a specified row and column from a given range.
 - Example: =INDEX(A1:C10, 2, 3)

- **MATCH**: Searches for a value in a range and returns the relative position.
 - Example: =MATCH("John", A2:A10, 0)
- **OFFSET**: Returns a cell or range of cells that is a specified number of rows and columns from a given reference.
 - Example: =OFFSET(A1, 2, 3)

3. Logical Functions (for conditional handling):

- **IF**: Performs a logical test and returns one value if true and another if false.
 - Example: =IF(A1 > 50, "Pass", "Fail")
- **COUNTIF**: Counts the number of cells that meet a specific condition.
 - Example: =COUNTIF(A1:A10, ">5")
- **SUMIF**: Sums the values in a range based on a specific condition.
 - Example: =SUMIF(A1:A10, ">5", B1:B10)
- **AND/OR**: Returns TRUE if all/any of the conditions are true.
 - Example: =AND(A1 > 50, B1 < 100)

4. Aggregation Functions (for summarizing data):

- **SUM**: Adds up all the values in a range.
 - Example: =SUM(A1:A10)
- **AVERAGE**: Calculates the average of a range.
 - Example: =AVERAGE(A1:A10)
- **MAX/MIN**: Returns the highest or lowest value in a range.
 - Example: =MAX(A1:A10) and =MIN(A1:A10)
- **MEDIAN**: Returns the median value of a range.
 - Example: =MEDIAN(A1:A10)
- **SUBTOTAL**: Returns a subtotal for a range, allowing different aggregation functions (SUM, AVERAGE, etc.) depending on the function number.
 - Example: =SUBTOTAL(9, A1:A10) (9 represents the SUM function)

5. Unique and Distinct Data Functions:

- **UNIQUE** (Excel 365 and Excel 2021): Returns a list of unique values from a range.
 - Example: =UNIQUE(A1:A10)
- **REMOVE DUPLICATES**: Removes duplicate values from a selected range (Command, not a function).

Frequently Used Excel Commands for Data Handling & Pre-Processing

These are features you can access via Excel's ribbon, context menus, or keyboard shortcuts for organizing and manipulating data:

1. **Sort:** Organizes data in ascending or descending order based on one or more columns.
 - How to use: Go to the "Data" tab and select "Sort."
2. **Filter:** Displays only the rows that meet specific criteria.
 - How to use: Go to the "Data" tab and click "Filter."
3. **Remove Duplicates:** Eliminates duplicate entries in a dataset, ensuring each record is unique.
 - How to use: Go to the "Data" tab and click "Remove Duplicates."
4. **Text to Columns:** Splits data from one column into multiple columns based on delimiters (such as commas or spaces).
 - How to use: Go to the "Data" tab and select "Text to Columns."
5. **Flash Fill:** Automatically fills in data based on patterns detected from your initial entries.
 - How to use: Start typing a pattern and press Ctrl + E to trigger Flash Fill.
6. **Group and Outline:** Organizes data into groups that can be expanded or collapsed.
 - How to use: Select rows/columns, right-click, and choose "Group."
7. **Remove Blank Cells:** Select and remove blank cells in a range.
 - How to use: Use the "Go To Special" command (F5 or Ctrl + G), select "Blanks," and delete or replace them.
8. **Consolidate:** Combines data from multiple ranges into one summary report.
 - How to use: Go to the "Data" tab and select "Consolidate."
9. **Data Validation:** Restricts the type of data that can be entered into a cell.
 - How to use: Go to the "Data" tab and click on "Data Validation."
10. **PivotTables:** Summarizes and analyzes large datasets by dynamically grouping and aggregating data.
 - How to use: Go to the "Insert" tab and click on "PivotTable."
11. **Hyperlink:** Adds a clickable link to a specific URL or location within a workbook.
 - How to use: Go to the "Insert" tab and select "Hyperlink."

Frequently Used Excel Formulas for Data Handling & Pre-Processing

These formulas are used to combine functions and logic to handle complex data tasks.

1. **Concatenate (or CONCAT):** Combines the contents of two or more cells into one.
 - Formula: =CONCAT(A1, " ", B1) or =CONCATENATE(A1, " ", B1)

2. **IFERROR:** Returns a specified value if a formula results in an error.
 - Formula: =IFERROR(VLOOKUP(A2, B2:B10, 1, FALSE), "Not Found")
3. **INDEX and MATCH Combination:** An alternative to VLOOKUP for more flexible data referencing.
 - Formula: =INDEX(B2:B10, MATCH(A2, A1:A10, 0))
4. **SUMPRODUCT:** Multiplies corresponding values in ranges or arrays and returns the sum of the products.
 - Formula: =SUMPRODUCT(A1:A10, B1:B10)
5. **ARRAY Formulas:** Allows for complex calculations on multiple values at once.
 - Formula: {=SUM(A1:A10 * B1:B10)} (entered using Ctrl + Shift + Enter)

The data pre-processing also involves various tasks including data cleaning, transformation, filtering, formatting, integration, aggregation, and visualization also. The excel possesses function, formulas and commands for each, bellow is a brief description for the key pre-processing tasks you can perform in excel:

1. Data Cleaning:

- **Removing Duplicates:** Use the **Remove Duplicates** feature in the Data tab.
- **Handling Missing Values:** Replace, fill, or remove missing values using functions like **IFERROR**, **ISBLANK**, and **IFNA**.
- **Data Validation:** Create rules to ensure data integrity by using the **Data Validation** feature.

2. Data Transformation

- **Text Functions:** Use functions like **CONCATENATE**, **LEFT**, **RIGHT**, **MID**, and **TEXT** for text manipulation.
- **Date and Time Functions:** Use functions like **DATEDIF**, **DATE**, **YEAR**, **MONTH**, and **DAY** for handling dates and times.
- **Mathematical Functions:** Apply functions such as **SUM**, **AVERAGE**, **ROUND**, **FLOOR**, and **CEILING** for numerical data processing.

3. Data Filtering and Sorting

- **Filters:** Use the **Filter** feature to narrow down data based on specific criteria.
- **Sorting:** Organize data in ascending or descending order based on one or more columns.

4. Data Formatting

- **Conditional Formatting:** Highlight cells that meet specific criteria using color scales, data bars, and icon sets.
- **Cell Formatting:** Adjust cell formats to enhance data readability,

such as changing number formats, font styles, and alignment.

5. Data Integration

- **Merging Data:** Use **VLOOKUP**, **HLOOKUP**, **INDEX**, and **MATCH** to combine data from different sources.
- **Appending Data:** Copy and paste data from multiple worksheets into a single worksheet.

6. Data Aggregation

- **Pivot Tables:** Summarize large datasets and generate insights using **Pivot Tables**.
- **Grouping:** Group data by categories or time periods to make analysis easier.

7. Data Visualization

- **Charts and Graphs:** Create various types of charts (like bar, line, pie) to visualize data trends and patterns.
- **Slicers and Timelines:** Use slicers and timelines to filter data dynamically in Pivot Tables and Pivot Charts.

8. Power Query: For more advanced preprocessing, use **Power Query**:

- **Data Import and Transformation:** Import data from various sources and perform complex transformations.
- **Advanced Cleaning:** Automate data cleaning tasks like removing duplicates, unpivoting data, and splitting columns.
- **Data Shaping:** Reshape and reorganize data to fit your analysis needs.

Excel's robust set of features makes it a powerful tool for data preprocessing. Whether you're dealing with small datasets or large ones, it has the tools you need to clean, transform, and prepare your data for analysis.

☞ Check Your Progress 2

1. How do data handling and pre-processing differ?

.....

.....

.....

2. What are common Excel functions for handling text data?

.....

.....

.....

3. What is the role of commands like 'Remove Duplicates' and 'Sort'?

.....

.....

.....

4. What Excel features help with missing or incorrect data?

.....

.....

.....

9.4 FREQUENCY DISTRIBUTION

A **frequency distribution** is a summary of how often different values or ranges of values occur in a dataset. It helps in understanding the distribution and patterns in data, such as how frequently each value or range appears.

To understand the process of generation of frequency distribution in excel consider a dataset given in the Example-1 below:

Example-1: Below, Table 1 - captures the lives of 100 electric bulbs (in Hrs.), and say, you are required to construct a frequency distribution and create histogram from this data using Spreadsheet package.

Table 1: Lives of 100 Electric Bulbs

511.6	977.7	600.2	1099.7	803.7
923.4	1108.3	906.7	759.6	1111.9
918.3	1051.1	992.5	817.2	665.3
1143.6	848.4	939.8	1163.0	715.2
936.1	750.5	991.2	1199.5	950.2
1161.7	1027.7	995.1	966.5	1146.5
848.0	956.8	1100.0	955.2	1023.0
900.5	982.3	699.2	1069.8	1245.3
1059.5	1091.0	850.7	1219.3	1012.6
1053.2	939.5	777.8	749.6	980.8
1091.3	1016.3	930.4	1242.2	1131.4
1314.7	1137.2	763.1	1294.4	917.3
1204.1	980.1	922.3	1057.7	907.2
808.0	857.7	1127.1	934.3	1262.3
965.4	873.4	955.1	806.5	1033.0
1068.3	950.3	930.6	1000.1	898.5
1293.1	940.9	1293.8	1035.2	706.0
880.9	912.2	803.5	922.6	846.1
1092.3	1182.0	985.2	945.3	835.0
1001.5	1048.8	895.1	1067.2	1062.8

Solution: Now to summarize the pattern in dataset we need to develop a **Frequency Distribution, which** is the most common method for summarizing and presentation of data, which enables us to quickly assess how frequently any value occurs in the given data set.

Figure 1 shows the spreadsheet (for the example, MS-Excel has been used, but you are free to use any spreadsheet package). Enter the given data in the cells $A2....E21$, that needs to be used to construct the frequency distribution. In the next step, you are required to specify class intervals to construct a frequency distribution. In this context, observe that the minimum and maximum values for the raw data are 511.6 and 1314.7 respectively (use $\min(A2:E21)$, $\max(A2:E21)$), as these are useful information in deciding about the classes to be formed. Here, the data under bins shows the class categories and they can be read as 0 to 510.5 (included), 510.6 to 590.5 (included) and so on. The last category is 1390.6 and above. Please note that for the purpose of calculation of frequencies, the upper limits of the classes ($H2....H13$) alone are required.

Once the basic data is entered, you need to calculate the Frequency distribution for these class intervals. For this, you need to enter an array formula in the spreadsheet. An array formula performs calculations on an array of data and may produce an array of results. To enter array formula, you may need to press CTRL+SHIFT+ENTER into a worksheet. The formula for calculating frequency is also an array formula. To enter this formula, perform the following steps:

- Select the cells in which result is expected in our case these are $J2....J14$.
- Enter the formula: $=\text{frequency}(A2:E21,H2:H13)$ and press CTRL+SHIFT+ENTER keys.
- You can also find the Cumulative frequencies, use simple addition formula for this purpose in cells $K2....K14$. Write your own formula for this purpose.

On completion of these operations, the screen will look like as that of Figure 1.

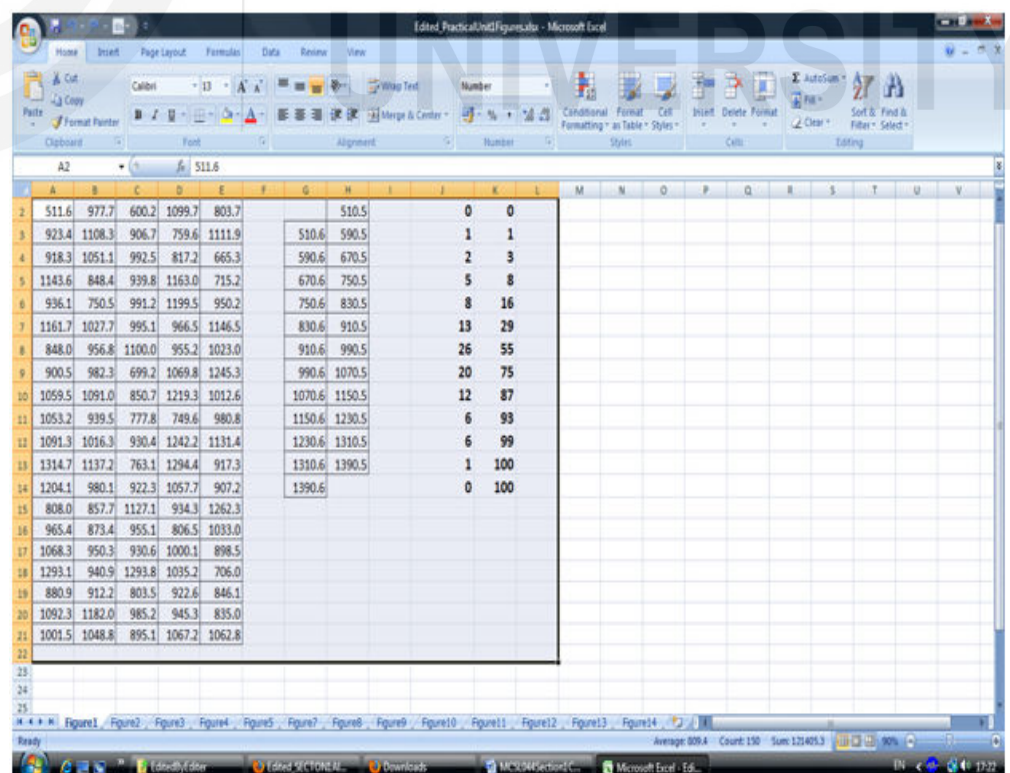


Figure 1: Calculating Frequencies in Spreadsheet

Now, to create a histogram as well as frequency distribution, you may also use *Data → Data Analysis* Menu. In some of your worksheets this option may not be displayed. In such case follow the following steps to *Load the Analysis ToolPak*.

Steps to Activate Data Analysis Toolpak

For Windows (Excel 2016, 2019, 2021, Microsoft 365):

1. Open Excel.
2. Go to **File > Options**.
3. In the **Excel Options** dialog box, select **Add-ins**.
4. At the bottom, next to **Manage**, select **Excel Add-ins** from the dropdown, then click **Go**.
5. In the **Add-Ins** box, check **Analysis ToolPak**.
6. Click **OK**.

Once activated, you'll see "**Data Analysis**" on the **Data** tab (usually on the far right). On loading this, you will be able to see the *Data Analysis* option in the *Data* Menu option.

In order to make the histogram, you just need to use the original raw data and bins, so delete all other data in Figure 1 except keep the columns *A* to *E* and *H*. Moreover, transfer the column *H* data to *G* by incrementing previous *G* data by 0.1. Now to make the histogram perform the following steps:

1. Select *Data → Data Analysis* and then select *Histogram* in the resulting dialog box and press **Ok** button.
2. In the resulting dialog box, set the *Input Range* to *A2..E21*, *Bin Range* to *G2..G13* and *Output Range* to *I1..J14*. Please do not forget to check the *Chart Output* check box.
3. Click **Ok**

The Histogram will be displayed as shown in the Figure 2. You need to modify the format of the histogram to make it look as per your need. For example, to remove gaps in between bars:

1. Click on Bar → *Right Click → Format Data Series...*
2. *Series Options → Gap Width 0%*
3. *Fill → Pattern Fill*
4. *Border Colour → Solid Line etc..*

Notice that horizontal axis labels now depict the upper class boundaries having class width of 80 hours in Figure 2. Can you now plot the frequency polygon and the less than type to give for the same frequency distribution You can also plot Bar diagram for appropriate data as shown in the Unit.

You can also explore the fact that all spreadsheet packages allow you to modify the Title of the chart, Axis titles, colour of bars, size of bars, size of chart and many other formatting features. For this you should refer to documentation of the spreadsheet package that you are using.

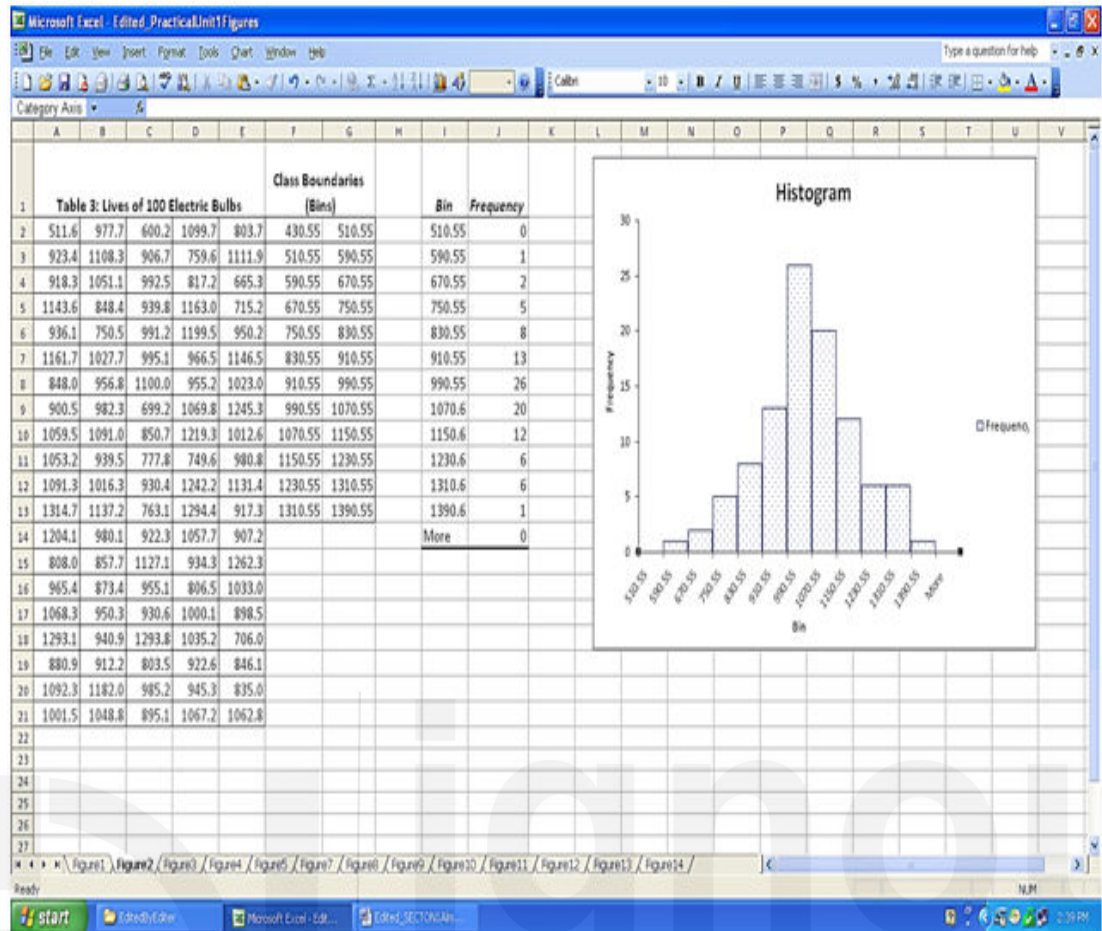


Figure 9.2: Histogram using the Data → Data Analysis options

☞ Check Your Progress 3

1. What is the purpose of a frequency distribution?

.....

.....

.....

2. How is the FREQUENCY function used in Excel?

.....

.....

.....

3. What tool is used in Excel to create histograms?

.....

.....

.....

4. How do you determine class intervals for a frequency distribution?

.....

.....

.....

9.5 CENTRAL TENDENCY AND DISPERSION

Both Central tendency and dispersion are the fundamental concepts in statistics used to summarize and describe datasets.

Central tendency refers to the measure that identifies the center or typical value within a dataset. It provides a single value that attempts to describe a set of data by indicating the position around which most values cluster. The most common measures of central tendency are the mean, which is the arithmetic average of all values; the median, which is the middle value when the data is arranged in order; and the mode, which is the value that appears most frequently. These measures give a general idea of where the data points tend to concentrate.

To find the mean, which is the average of a dataset, you can use the formula `=AVERAGE(range)`, where "range" refers to the set of cells containing your data. For example, `=AVERAGE(A2:A10)` will return the average value of the numbers in cells A2 through A10.

To determine the median, or the middle value when your data is sorted, use `=MEDIAN(range)`. This function is especially useful when your data has outliers that may skew the mean.

To find the mode, which is the most frequently occurring number in a dataset, use `=MODE.SNGL(range)`. If your dataset has more than one mode, you can use `=MODE.MULT(range)` to return multiple results.

On the other hand, dispersion refers to the degree of spread or variability in the data. While central tendency indicates the center of the data, dispersion shows how much the values differ from each other and from the central value. Key measures of dispersion include the range, which is the difference between the highest and lowest values; the variance, which measures the average of the squared differences from the mean; and the standard deviation, which is the square root of the variance and represents the average amount by which values deviate from the mean. Another important measure is the interquartile range (IQR), which describes the spread of the middle 50% of the data. Together, central tendency and dispersion provide a more complete understanding of the data's distribution and variability.

As Dispersion measures the spread or variability of your data. A simple measure is the range, which can be calculated using `=MAX(range)-MIN(range)`, which subtracts the smallest value from the largest.

For a more statistical measure of spread, you can calculate variance. Use `=VAR.S(range)` for a sample variance or `=VAR.P(range)` for the population variance. These functions compute the average of the squared deviations from the mean.

The standard deviation is the square root of the variance and provides a more interpretable measure of spread in the same units as the data. Use `=STDEV.S(range)` for sample standard deviation or `=STDEV.P(range)` for population standard deviation.

If you need to calculate the interquartile range (IQR), which measures the middle 50% of your data and is useful for identifying outliers, you can use $=\text{QUARTILE.EXC}(\text{range},3)-\text{QUARTILE.EXC}(\text{range},1)$, which subtracts the first quartile (Q1) from the third quartile (Q3).

The central tendency and dispersion of data can be determined by Using the Data Analysis Toolpak also, many of these statistics can be computed automatically using the Data Analysis Toolpak. After activating Data Analysis Toolpak via File > Options > Add-ins, go to the Data tab and select Data Analysis. Choose Descriptive Statistics, select your data range, check the Summary Statistics box, and Excel will generate a table that includes the mean, median, standard deviation, minimum, maximum, range, and more.

Microsoft Excel - Edited_PracticalUnit1Figures

File Edit View Insert Format Tools Data Window Help

Type a question for help

D33

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Table 3: Lives of 100 Electric Bulbs						Class Boundaries (Bins)		Frequency (f)	Average Life In Hours (x)	$y = (x - 950.5) / 80$	$f \cdot y$	$f \cdot y^2$							
2	511.6	977.7	600.2	1099.7	803.7		430.55	510.55	0	470.55	-6	0	0							
3	923.4	1108.3	906.7	759.6	1111.9		510.55	590.55	1	550.55	-5	-5	25							
4	918.3	1051.1	992.5	817.2	665.3		590.55	670.55	2	630.55	-4	-8	32							
5	1143.6	848.4	939.8	1163.0	715.2		670.55	750.55	5	710.55	-3	-15	45							
6	936.1	750.5	991.2	1199.5	950.2		750.55	830.55	8	790.55	-2	-16	32							
7	1161.7	1027.7	995.1	966.5	1146.5		830.55	910.55	13	870.55	-1	-13	13							
8	848.0	956.8	1100.0	955.2	1023.0		910.55	990.55	26	950.55	0	0	0							
9	900.5	982.3	699.2	1069.8	1245.3		990.55	1070.55	20	1030.55	1	20	20							
10	1059.5	1091.0	850.7	1219.3	1012.6		1070.55	1150.55	12	1110.55	2	24	48							
11	1053.2	939.5	777.8	749.6	980.8		1150.55	1230.55	6	1190.55	3	18	54							
12	1091.3	1016.3	930.4	1242.2	1131.4		1230.55	1310.55	6	1270.55	4	24	96							
13	1314.7	1137.2	763.1	1294.4	917.3		1310.55	1390.55	1	1350.55	5	5	25							
14	1204.1	980.1	922.3	1057.7	907.2		More		0	1430.55	6	0	0							
15	808.0	857.7	1127.1	934.3	1262.3				100			34	390							
16	965.4	873.4	955.1	806.5	1033.0				$\mu' =$		0.34									
17	1068.3	950.3	930.6	1000.1	898.5				Mean $\mu =$		977.75									
18	1293.1	940.9	1293.8	1035.2	706.0				Variance $\sigma^2 =$		24220.16									
19	880.9	912.2	803.5	922.6	846.1				Standard Deviation $\sigma =$		155.63									
20	1092.3	1182.0	985.2	945.3	835.0															
21	1001.5	1048.8	895.1	1067.2	1062.8															
22																				
23																				
24																				
25																				
26																				
27																				

Figure1 / Figure2 / Figure3 / Figure4 / Figure5 / Figure7 / Figure8 / Figure9 / Figure10 / Figure11 / Figure12 / Figure13 / Figure14

Ready

start EditedByEditor Microsoft Excel - Edited_SECTION1Aim...

NUM

2:45 PM

Figure 9.3: Applying the measures of Central tendency and dispersion

Let us discuss them with the help of the Example-1 of section 9.4 above. The following figure shows an implementation.

Please note that we have used the following procedure in the worksheet to arrive at the result shown in Figure 3:

- In the cell I2 to I14, you can use the array formula as explained in the last section to calculate the frequencies of each class.
- For calculating the Average life in hours (x_j) or the class mid-points, you may enter in the cell J2 the formula $=(G2+H2)/2$.
- Copy this formula from cell J3 to J13. Notice that you need to find the value for cell J14 using different formula.
- Calculate the value of y_j in the cell K2 as per the formula $=(J2-\$J\$8)/80$. Copy this formula to cells K3 to K14.

- Now enter the formula for calculating $y_j f_j$ and $y_j^2 f_j$ (you can do it yourself).
- Enter the formula for the μ' (see page 30, Unit 1) - the mean for x -observation, variance for x -observations and Standard deviation for x -observation.

You can compare the results so obtained with the results that you can obtain by directly applying the formula of mean (AVERAGE), median (MEDIAN) and standard deviation (STDEV) on the cell range A2:E21 containing the original data.

☞ Check Your Progress 4

1. Construct a frequency distribution for the data given below, using spreadsheet package.

Table 9.2: Bike Running Data (Km) per Month

511.6	977.7	600.2	1099.7	803.7
923.4	1108.3	906.7	759.6	1111.9
918.3	1051.1	992.5	817.2	665.3
1143.6	848.4	939.8	1163.0	715.2
936.1	750.5	991.2	1199.5	950.2
1161.7	1027.7	995.1	966.5	1146.5
848.0	956.8	1100.0	955.2	1023.0
900.5	982.3	699.2	1069.8	1245.3
1059.5	1091.0	850.7	1219.3	1012.6
1053.2	939.5	777.8	749.6	980.8
1091.3	1016.3	930.4	1242.2	1131.4
1314.7	1137.2	763.1	1294.4	917.3
1204.1	980.1	922.3	1057.7	907.2
808.0	857.7	1127.1	934.3	1262.3
965.4	873.4	955.1	806.5	1033.0
1068.3	950.3	930.6	1000.1	898.5
1293.1	940.9	1293.8	1035.2	706.0
880.9	912.2	803.5	922.6	846.1
1092.3	1182.0	985.2	945.3	835.0
1001.5	1048.8	895.1	1067.2	1062.8

.....

.....

.....

.....

2. Create the histogram from this data using Spreadsheet package.
.....
.....
.....
3. Create the measures of central tendencies and measure of dispersion for this data.
.....
.....
.....
4. What are the three main measures of central tendency?
.....
.....
.....
5. How can you calculate standard deviation in Excel?
.....
.....
.....
6. What does the interquartile range (IQR) measure?
.....
.....
.....
7. What Excel tool provides multiple descriptive statistics in one output?
.....
.....
.....

9.6 DESCRIPTIVE DATA ANALYSIS AND ITS INTERPRETATION

Descriptive data analysis is the foundational step in understanding the characteristics and structure of a dataset. It involves summarizing and organizing data using statistical measures such as mean, median, mode, standard deviation, and frequency distributions. Through tables, charts, and graphical representations, this analysis provides an overview of the data's central tendencies, dispersion, and patterns. The interpretation of these descriptive statistics is crucial, as it offers insights into trends, anomalies, and relationships within the data, guiding further analysis and decision-making processes.

explain with an example to use excel for descriptive data analysis also interpret the results obtained after descriptive data analysis is done in excel.

Since Computer is a discipline, that facilitates the working of our day-to-day life, we choose an example/Case from our daily life in this context and use it to interpret the respective results. Through the case study. This case will help us to understand, how to use Descriptive statistics as a tool for data analysis.

CASE DESCRIPTION: As a pilot project, a private company involved in hardware and maintenance related activity planned to launch its own brand of assembled computers. The company identified a small colony having 12 families in all. Each of the family is having members of different age group. For the sake of their brand performance related study, they decided to issue one computer system to each family as a single unit. They got the system installed with several applications suitable for each age group viz., Computer Games, Movies, Application Softwares, E-Books, Internet, and Security related software. The company wants to study the application usage pattern and the satisfaction level of the families, so that they can customize their product accordingly. In order to monitor the system usage time, they installed an automatic clock with the system, which records the usage time. This will help them to analyze the questionnaire feedback, which they wish to get filled, at the end of their study.

On the basis of an appraisal of the case cited above, you may identify the objectives of the study to be as under:

- What is the customer satisfaction level?
- Which software utility is performed exhaustively in most of the families?

In view of these objectives, you are required to draft a Questionnaire, where the responses to the questions are either direct or indirect, as desired. They can be categorized into one among the following scales, viz.,

- Nominal
- Ordinal
- Quantitative.

A sample questionnaire is given below, this will let you understand the Excel operations, to be performed at the later stages

QUESTIONNAIRE

Family No.:

What is the Average System Utilization time(minutes) ? :

15 30 45 60 75 90 105 120 135 150 165 180

What are the Activities Performed by your family while utilizing the system?

- a) Computer Games, b) Internet, c) E-Books,
d) Application Software's, e) Movies f) Security related software

What is the level of satisfaction, achieved by your family ?

-2 = Very dissatisfied, 1 = dissatisfied, 0 = indifference,

1 = satisfied, 2 = Very satisfied.

Would your family like to purchase the product?

Yes

No

We got the questionnaire filled from respective families, finally we have the following data from the computer savvy families. The collected data is through following variables

- *Family* (Nominal scale) is the observation/serial number of the questionnaire. In the present case, one questionnaire per household is filled by each family.
- *Time* (Quantitative scale) is a quantitative data type which is measured in minutes. It is the software utilization time for a family.
- *System Utilization Activity* (Nominal scale) is a nominal data type consisting of 6 choices of activity in the park:
 1. Computer Games,
 2. Internet,
 3. E-Books,
 4. Application Software's,
 5. Movies
 6. Security related software

To this question, there may be multiple choices that for each family, one corresponding to each of the several activities they perform.

- *Product Satisfaction* (Ordinal scale) It measures family satisfaction toward the Product. It is measured in the ordinal scale with 5 values as options listed below:
 - 2 = Very dissatisfied,
 - 1 = dissatisfied,
 - 0 = indifference,
 - 1 = satisfied,
 - 2 = Very satisfied.
- *Product Purchase* (Nominal scale) is measured in the nominal scale (Yes or No) regarding the final decision about purchase of the product.

Based on the responses to the questions framed in the questionnaire, collected data is presented in the following table:

Raw Data

Family	System Utilization Time	System Utilization Activity	Product Satisfaction	Product Purchase
1	30	1, 2, 3	0	N
2	30	4,6	1	Y
3	60	1, 2	2	Y
4	45	5	-1	N
5	30	6	1	N
6	60	2	2	Y
7	30	4	1	N
8	45	3, 4	-1	N
9	15	6	1	Y
10	60	2	2	Y
11	180	1, 2, 3, 4	2	Y
12	120	1,2,4	2	Y

Raw data presented in the table above is coded into the following form, which is amenable for further statistical analysis using spreadsheet software.

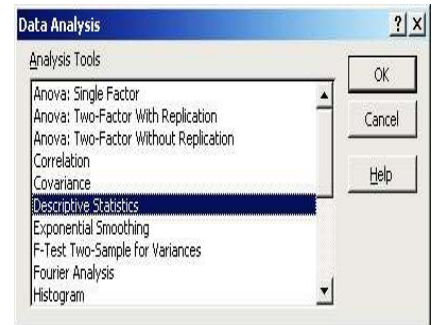
Raw Data coded for analysis in Formatted Excel Sheet

Family	System Utilization Time	System Utilization Activity						Product Satisfaction	Product Purchase
		1	2	3	4	5	6		
1	30	1	1	1	0	0	0	0	0
2	30	0	0	0	1	0	1	1	1
3	60	1	1	0	0	0	0	2	1
4	45	0	0	0	0	1	0	-1	0
5	30	0	0	0	0	0	1	1	0
6	60	0	1	0	0	0	0	2	1
7	30	0	0	0	1	0	0	1	0
8	45	0	0	1	1	0	0	-1	0
9	15	0	0	0	0	0	1	1	1
10	60	0	1	0	0	0	0	2	1
11	180	1	1	1	1	0	0	2	1
12	120	1	1	0	1	0	0	2	1

After the excel file with data in the format shown above has been created from the Raw Data table, notice that the responses under the head “*Product Purchase*” are marked 0 and 1, which represents the codes for responses “Y”

and “N”, respectively. The same also applies to the responses under “*System Utilization Activity*” head of Raw Data table.

Please note that Family, System Utilization Activity, Product Purchase, are variables measured in NOMINAL SCALE. So, we should NOT calculate sum, average, variance etc. Instead the best is to calculate frequency, percentage, plot graphs only. But here we are going to demonstrate that how you can use the tool of data analysis tool pack to get the desired answers, related to the marked objectives of your study. Now to get the answer for the stated objectives, we use the Descriptive Statistics Tool of Data Analysis ToolPak.



Enable the Data Analysis ToolPak:

- Go to File > Options > Add-ins.
- In the Manage box, select Excel Add-ins and click Go.
- Check the Analysis ToolPak box and click OK.

Data analysis through Excel:

Perform Following Steps:

1. Tabulate the collected data as given above in Excel Spreadsheet.
2. Click DATA TAB → DATA ANALYSIS → DESCRIPTIVE STATISTICS → OK

We present below some of the relevant screenshots in the present case.

Results of DESCRIPTIVE STATISTICS Procedure

<

Statistics	System Utilization Time	System Utilization Activity						Product Satisfaction	Product Purchase
		1	2	3	4	5	6		
Mean	58.75	0.33	0.50	0.25	0.42	0.08	0.25	1	0.58
Standard Error	13.50	0.14	0.15	0.13	0.15	0.08	0.13	0.33	0.15
Median	45	0	0.5	0	0	0	0	1	1
Mode	30	0	1	0	0	0	0	2	1
Standard Deviation	46.76	0.49	0.52	0.45	0.51	0.29	0.45	1.13	0.51
Sample Variance	2186.93	0.24	0.27	0.20	0.27	0.08	0.20	1.27	0.27
Kurtosis	3.8	-1.65	-2.44	-0.33	-2.26	12	-0.33	-0.34	-2.26
Skewness	2.0	0.81	0	1.33	0.39	3.46	1.33	-0.91	-0.39
Range	165	1	1	1	1	1	1	3	1
Minimum	15	0	0	0	0	0	0	-1	0
Maximum	180	1	1	1	1	1	1	2	1
Sum	705	4	6	3	5	1	3	12	7
Count	12	12	12	12	12	12	12	12	12
Sample size = (Z*Std Dev./Std Error) ²	46.0992	46.099	46.099	46.099	46.099	46.099	46.099	46.099	46.0992
Z=1.96 @ 95%CL & 1.65 @ 90% CL						46.099	46.099	46.0992	46.0992

DATA INTERPRETATION

Now, let us interpret the results of descriptive statistics and analyse, what they say about our objectives viz.

- What is the customer satisfaction level?
- Which software utility is performed exhaustively in most of the families?

So far as *customer satisfaction level* is concerned, the tabulated data under head of *Mean* and *Mode* gives the result as 1 and 2 respectively i.e., on an average the customers are satisfied and in fact they are “most frequently” - very satisfied. To interpret, we are to refer our data coding standards. As in this case they are :

- *Product Satisfaction* is an ordinal scale with 5 values:
-2 = Very dissatisfied,

- 1 = dissatisfied,
- 0 = indifference,
- 1 = satisfied,
- 2 = Very satisfied.

Similarly, in order to study the objective “*Which software utility is performed exhaustively in most of the families? Refer to the results under the head SUM for the **System Utilization Activity**. Here, the activity 2 is identified to be performed the most i.e. Families are mostly using the Computer system for internet related activities.*

To interpret, we are to refer our data coding standards. As in this case, they are:

- *System Utilization Activity* is a nominal data type consist of 6 choices of activity in the park:
 1. Computer Games,
 2. Internet,
 3. E-Books,
 4. Application Software’s,
 5. Movies
 6. Security related software

Now, you might be wondering that the purpose of this session is to talk about sampling and we are discussing about the descriptive statistics. This is intentional, we want you to realize that sampling is not simply a formula but it is related to various factors, as in this case the number of families are limited in the target area i.e. target population is limited. By using the formula for Sample size calculation i.e. $Sample\ size = (Z * Std\ Dev. / Std\ Error)^2$; we can calculate the respective sample size as

$$Sample\ size = (Z * Std\ Dev. / Std\ Error)^2$$

You might have located this expression $Sample\ size = (Z * StdDev. / Std\ Error)^2$, its just another formula for calculating the sample size, when standard deviation is known. However when standard deviation is not known, we use another formula, we are not going to discuss that, here we are going to concentrate on $Sample\ size = (Z * StdDev. / Std\ Error)^2$

We calculated the sample size to be 46 families but our study is considering only 12. So for the reliability of the study the collected data is insufficient and thus we cannot conclude that the interpretation of the results related to our objectives is fine enough.

You might be in the position to realize the importance of sampling now, because from the above case, you identified that by limited data you might find that the objectives of the study are fulfilled but which might not be the fact as the sample size is inappropriate.

9.7 SUMMARY

This unit titled "Excel for Descriptive Data Analysis" provides a comprehensive guide on how to utilize Microsoft Excel for performing descriptive statistical analysis. It begins by explaining the importance of descriptive statistics in data analysis, emphasizing how Excel can be used to import, clean, and handle datasets. It covers the processes of importing data from various formats like CSV, XML, Access, and web sources. The document elaborates on data handling and pre-processing techniques such as sorting, filtering, dealing with missing values, and transforming data for analysis. It highlights frequently used Excel functions, formulas, and commands that assist in organizing and manipulating data efficiently.

Further, the unit delves into key statistical concepts such as frequency distribution, central tendency, and dispersion, explaining how to compute and interpret these using Excel. A practical case study is provided to demonstrate descriptive data analysis through a real-world example involving software usage and customer satisfaction. The study uses Excel's built-in functions and the Data Analysis Toolpak to calculate statistical measures like mean, standard deviation, and frequency, along with visualizations like histograms. The unit concludes by stressing the importance of sample size in making reliable statistical inferences, thereby linking descriptive analysis with broader statistical practices like sampling and inference.

9.8 ANSWERS TO CHECK YOUR PROGRESS

☞ Check Your Progress 1

1. What file types can Excel import data from?

A1: CSV, TXT, XML, JSON, Access, and web pages.

2. What is the importance of maintaining data consistency during import?

A2: To prevent data distortion and ensure accurate analysis.

3. What steps are followed to import data from a CSV file?

A3: Go to Data tab → From Text/CSV → Select file → Import → Load.

4. How can data be imported from a web page into Excel?

A4: Use Data → Get Data → From Web → Enter URL → Select data → Load.

☞ Check Your Progress 2

1. How do data handling and pre-processing differ?

A1: Data handling focuses on structuring data; pre-processing ensures data quality.

2. What are common Excel functions for handling text data?

A2: TRIM, LEFT, RIGHT, MID, and PROPER.

3. What is the role of commands like 'Remove Duplicates' and 'Sort'?

A3: They clean and organize data directly through Excel's interface without formulas.

4. What Excel features help with missing or incorrect data?

A4: Functions like IFERROR, ISBLANK, and Data Validation help clean such issues.

☞ **Check Your Progress 3**

1. What is the purpose of a frequency distribution?

A1: To show how often different values or ranges occur in a dataset.

2. How is the FREQUENCY function used in Excel?

A2: As an array formula: =FREQUENCY(data_range, bins_range), using CTRL+SHIFT+ENTER.

3. What tool is used in Excel to create histograms?

A3: The Data Analysis Toolpak's Histogram option.

4. How do you determine class intervals for a frequency distribution?

A4: Use the dataset's minimum and maximum values and decide on equal-width bins.

☞ **Check Your Progress 4**

1. Construct a frequency distribution for the data given below, using spreadsheet package.

Table 2: Bike Running Data (Km) per Month

511.6	977.7	600.2	1099.7	803.7
923.4	1108.3	906.7	759.6	1111.9
918.3	1051.1	992.5	817.2	665.3
1143.6	848.4	939.8	1163.0	715.2
936.1	750.5	991.2	1199.5	950.2
1161.7	1027.7	995.1	966.5	1146.5
848.0	956.8	1100.0	955.2	1023.0
900.5	982.3	699.2	1069.8	1245.3
1059.5	1091.0	850.7	1219.3	1012.6
1053.2	939.5	777.8	749.6	980.8
1091.3	1016.3	930.4	1242.2	1131.4
1314.7	1137.2	763.1	1294.4	917.3
1204.1	980.1	922.3	1057.7	907.2
808.0	857.7	1127.1	934.3	1262.3
965.4	873.4	955.1	806.5	1033.0
1068.3	950.3	930.6	1000.1	898.5
1293.1	940.9	1293.8	1035.2	706.0
880.9	912.2	803.5	922.6	846.1
1092.3	1182.0	985.2	945.3	835.0
1001.5	1048.8	895.1	1067.2	1062.8

A1. Refer section 9.5

2. Create the histogram from this data using Spreadsheet package.

A2. Refer section 9.5

3. Create the measures of central tendencies and measure of dispersion for this data.

A3. Refer section 9.5

4. What are the three main measures of central tendency?

A4: Mean, median, and mode.

5. How can you calculate standard deviation in Excel?

A5: Use =STDEV.S(range) for sample data or =STDEV.P(range) for population data.

6. What does the interquartile range (IQR) measure?

A6: The spread of the middle 50% of data ($Q3 - Q1$).

7. What Excel tool provides multiple descriptive statistics in one output?

A7: The Descriptive Statistics option in the Data Analysis Toolpak.



ignou
THE PEOPLE'S
UNIVERSITY

UNIT 10 EXCEL FOR INFERENTIAL DATA ANALYSIS

Structure

- 10.1 Introduction
- 10.2 Objectives
- 10.3 Sampling
- 10.4 Hypothesis Testing
 - 10.4.1 t-test
 - 10.4.2 chi-square Test
 - 10.4.3 F-test
 - 10.4.4 Z-test
 - 10.4.5 ANOVA
- 10.5 Correlation and Regression Analysis
 - 10.5.1 Univariate
 - 10.5.2 Multivariate
- 10.6 Time Series and Forecasting Using Excel
- 10.7 Summary
- 10.8 Solutions/Answers
- 10.9 References

10.1 INTRODUCTION

Inferential data analysis is a powerful statistical method that allows researchers and analysts to make predictions and draw conclusions about a population based on a sample of data. Unlike descriptive statistics, which merely summarize data, inferential statistics enable us to infer trends and test hypotheses, providing deeper insights into the data.

Microsoft Excel, a widely-used spreadsheet software, offers a range of tools and functions that make it an excellent choice for performing inferential data analysis. With its user-friendly interface and robust capabilities, Excel allows users to conduct various statistical tests, including t-tests, z-tests, ANOVA, and regression analysis, without the need for complex programming.

In this Unit, we will explore how to leverage Excel's features to perform inferential data analysis. We will cover essential topics such as:

- **Sampling Data:** Understanding how to collect and prepare sample data for analysis.
- **Hypothesis Testing:** Using Excel to test hypotheses and draw conclusions about populations.
- **Regression Analysis:** Implementing regression techniques to identify relationships between variables.

- **ANOVA (Analysis of Variance):** Comparing means across multiple groups to determine if there are significant differences.
- **Correlation Analysis:** Measuring the strength and direction of relationships between variables.

By the end of this Unit, you will have a solid understanding of how to use Excel for inferential data analysis, enabling you to make informed decisions based on your data.

10.2 OBJECTIVES

The objective of this Unit is to equip users with the knowledge and skills necessary to effectively utilize Microsoft Excel for performing inferential data analysis. By the end of this Unit, users will be able to:

- Grasp the basic concepts of inferential statistics and their importance in data analysis.
- Perform various hypothesis tests using Excel to make informed decisions based on sample data.
- Use Excel's tools to conduct regression analysis, identifying and interpreting relationships between variables.
- Apply Analysis of Variance (ANOVA) techniques to compare means across multiple groups and determine statistical significance.
- Measure and interpret the strength and direction of relationships between different variables using Excel.
- Accurately interpret the results of inferential statistical tests and communicate findings effectively.

By achieving these objectives, users will be able to leverage Excel's capabilities to perform comprehensive inferential data analysis, enhancing their ability to draw meaningful conclusions from data and support data-driven decision-making.

10.3 SAMPLING

Sampling is a crucial step in inferential data analysis, as it allows us to draw conclusions about a larger population based on a subset of data. In Excel, you can perform sampling using the Data Analysis ToolPak, which provides various tools for statistical analysis.

Steps to Perform Sampling in Excel:

1. **Enable the Data Analysis ToolPak:**
 - Go to File > Options > Add-ins.
 - In the Manage box, select Excel Add-ins and click Go.
 - Check the Analysis ToolPak box and click OK.
2. **Access the Sampling Tool:**
 - Go to the Data tab on the Ribbon.

- Click on Data Analysis in the Analysis group.
 - Select Sampling from the list and click OK.
3. **Configure Sampling Options:**
- **Input Range:** Specify the range of data you want to sample from.
 - **Sampling Method:** Choose between Random or Periodic sampling.
 - **Random Sampling:** Selects a specified number of samples randomly from the data.
 - **Periodic Sampling:** Selects samples at regular intervals (e.g., every 5th data point).
 - **Number of Samples:** Enter the number of samples you want to draw.
 - **Output Range:** Specify where you want the sampled data to appear (new worksheet, new workbook, or a specific range).
4. **Generate the Sample:**
- Click OK to generate the sample based on your specified criteria.

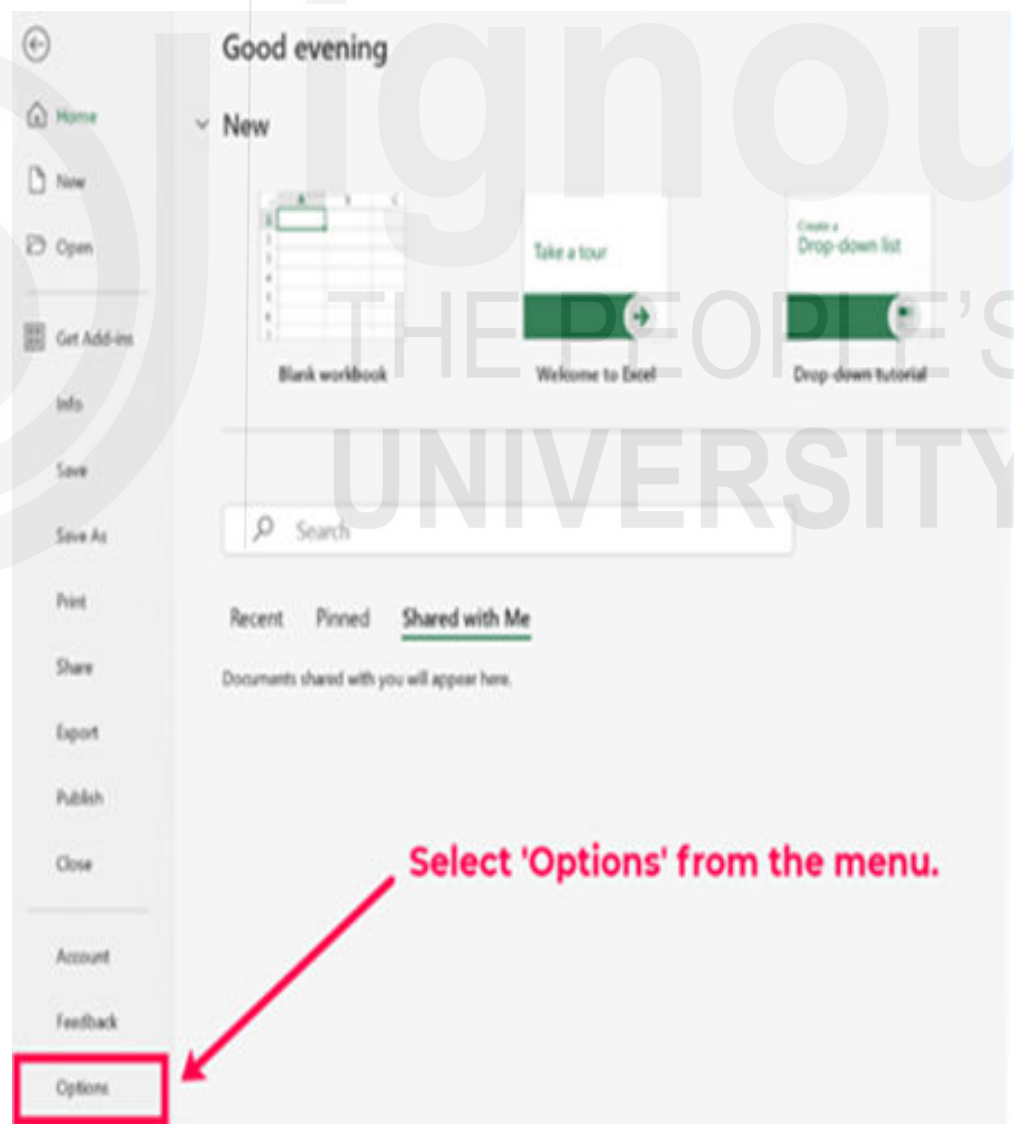


Fig. 10.1: Option Menu

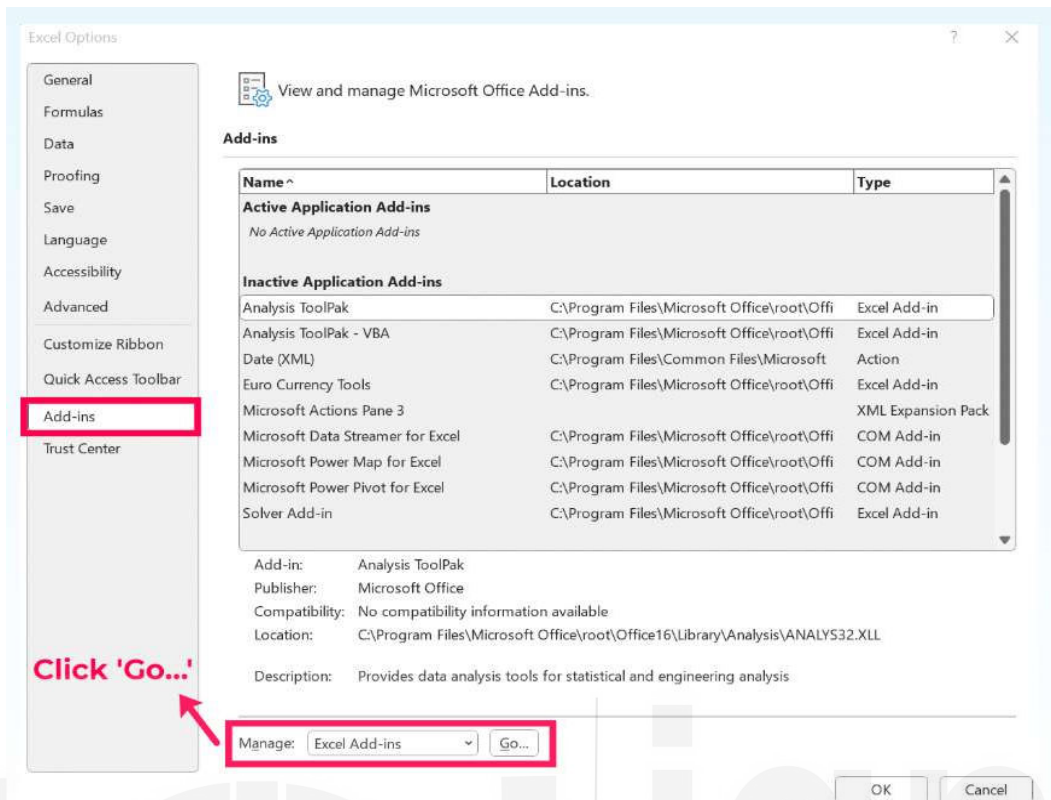


Fig. 10.2: Add_In

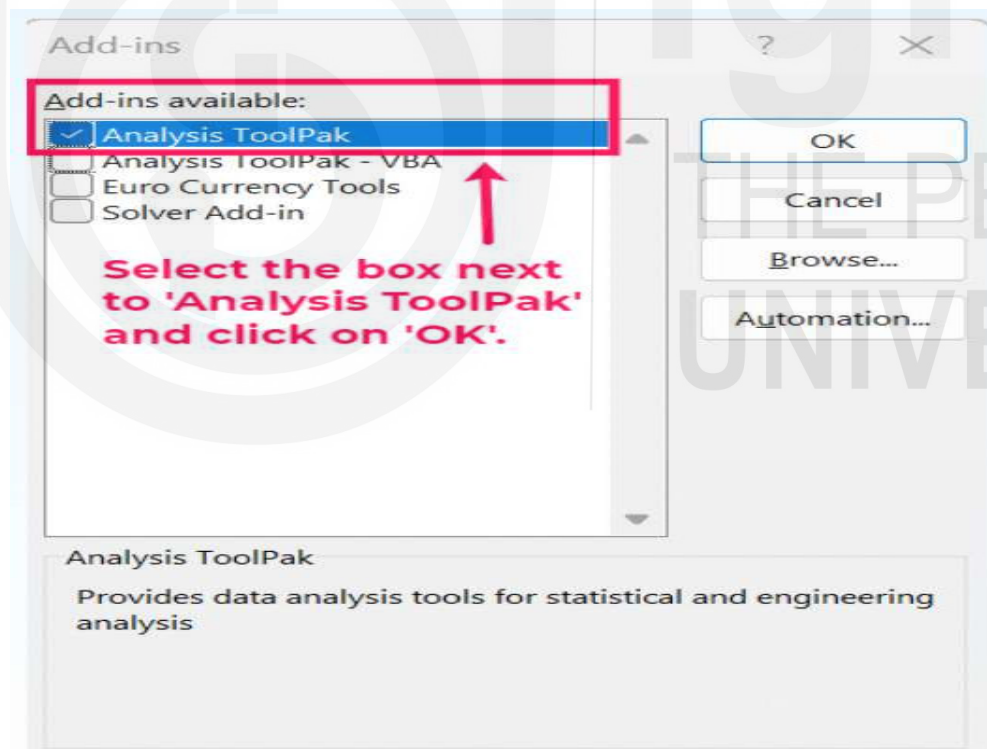


Fig. 10.3: Analysis ToolPak

Features of the Analysis ToolPak in Excel

The Analysis ToolPak in Excel offers various powerful tools for performing statistical analysis. Below is a summary of its key features and their applications:

Features of the Analysis ToolPak in Excel

The Analysis ToolPak in Excel offers various powerful tools for performing statistical analysis. Below is a summary of its key features and their applications:

Table 10.1: Summary of Analysis ToolPak Functions

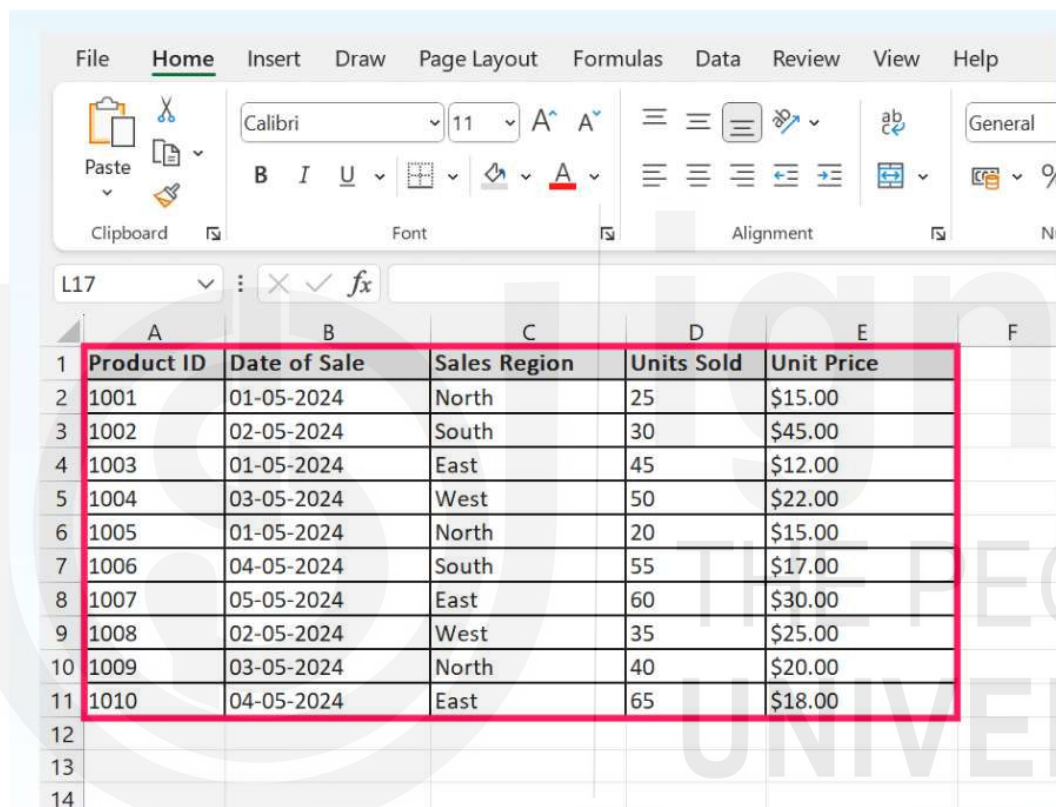
Tool	Description	Application
Histograms	Creates a graphical representation of data frequency distribution.	Ideal for visualizing data trends, identifying central tendencies, and detecting outliers.
Descriptive Statistics	Provides summary statistics like mean, median, mode, etc.	Essential for quickly understanding the key characteristics of a data set.
t-Tests	Compares the means of paired or independent samples.	Used to assess differences between two groups or conditions in experiments.
Analysis of Variance (ANOVA)	Tests for significant differences between the means of three or more groups.	Determines whether group means differ significantly from one another.
Regression Analysis	Performs linear regression and forecasting.	Analyzes relationships between variables and aids in trend prediction and forecasting.
Correlation	Measures the strength and direction of the relationship between two variables.	Assesses the degree of association between two variables.
Covariance	Evaluates the directional relationship between two variables.	Helps in understanding how two variables vary together.
Fourier Analysis	Transforms time series data into the frequency domain.	Applied in signal processing and system analysis to detect periodic patterns.
Random Number Generation	Produces random numbers from various statistical distributions.	Useful in simulations, Monte Carlo methods, and generating test data.
Exponential Smoothing	A weighted moving average method used for smoothing data series.	Commonly used for forecasting by reducing data volatility and highlighting trends.
F-Test	Compares statistical models to determine the best fit for the data.	Assesses the variance between samples and evaluates the significance of predictors in regression analysis.
Moving Average	Computes the moving average for a given data set.	Helps identify trends in time series data by smoothing out short-term fluctuations.

Example:

Suppose you have a dataset of 1000 customer transactions and you want to randomly sample 100 transactions for analysis:

1. **Input Range:** A1:A1000
2. **Sampling Method:** Random
3. **Number of Samples:** 100
4. **Output Range:** B1:B100

After configuring these options and clicking OK, Excel will generate a random sample of 100 transactions in the specified output range.



The screenshot shows the Excel interface with the 'Home' tab selected. A table of sample data is highlighted with a red border. The table has 5 columns: Product ID, Date of Sale, Sales Region, Units Sold, and Unit Price. The data is as follows:

	A	B	C	D	E
1	Product ID	Date of Sale	Sales Region	Units Sold	Unit Price
2	1001	01-05-2024	North	25	\$15.00
3	1002	02-05-2024	South	30	\$45.00
4	1003	01-05-2024	East	45	\$12.00
5	1004	03-05-2024	West	50	\$22.00
6	1005	01-05-2024	North	20	\$15.00
7	1006	04-05-2024	South	55	\$17.00
8	1007	05-05-2024	East	60	\$30.00
9	1008	02-05-2024	West	35	\$25.00
10	1009	03-05-2024	North	40	\$20.00
11	1010	04-05-2024	East	65	\$18.00

Fig. 10.4: Sample Data

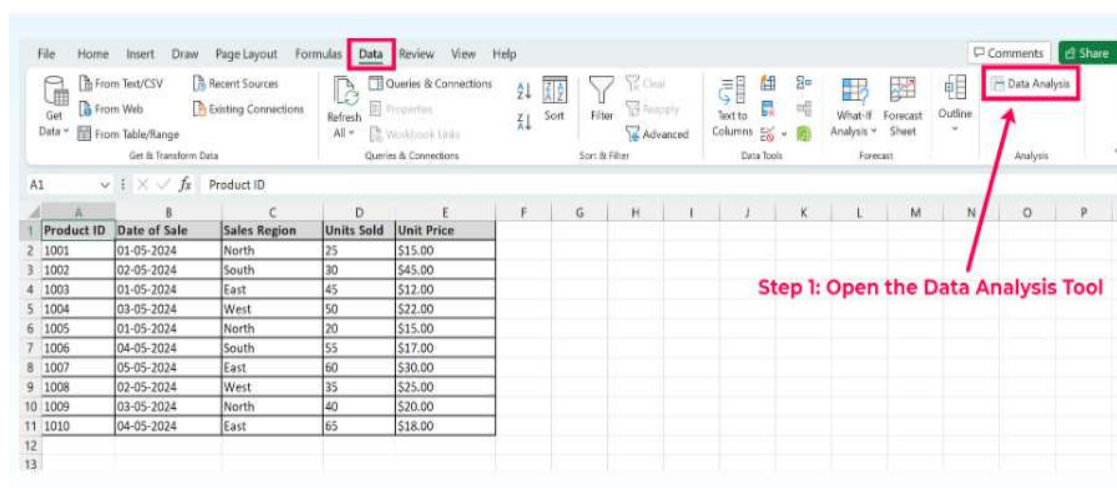


Fig. 10.5: Data Tab

Data Visualization and Interpretation

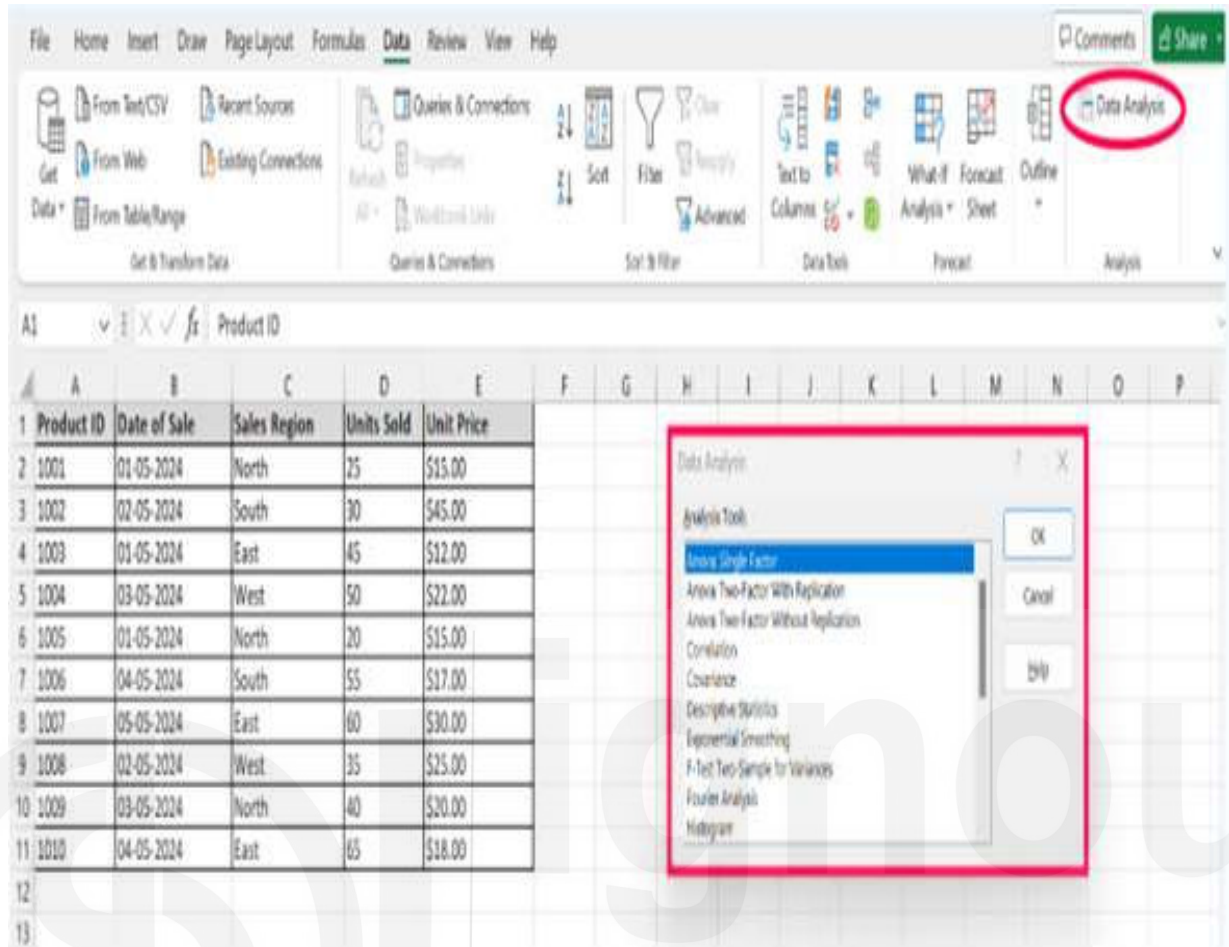


Fig. 10.6: Data Analysis Tab



Fig.10.7: Descriptive Statistics Dialog Box

Next, choose the input range for "Units Sold," check the "Labels in First Row" option since the data includes headers, and select the desired output options. Additionally, choose the "Summary Statistics" option.

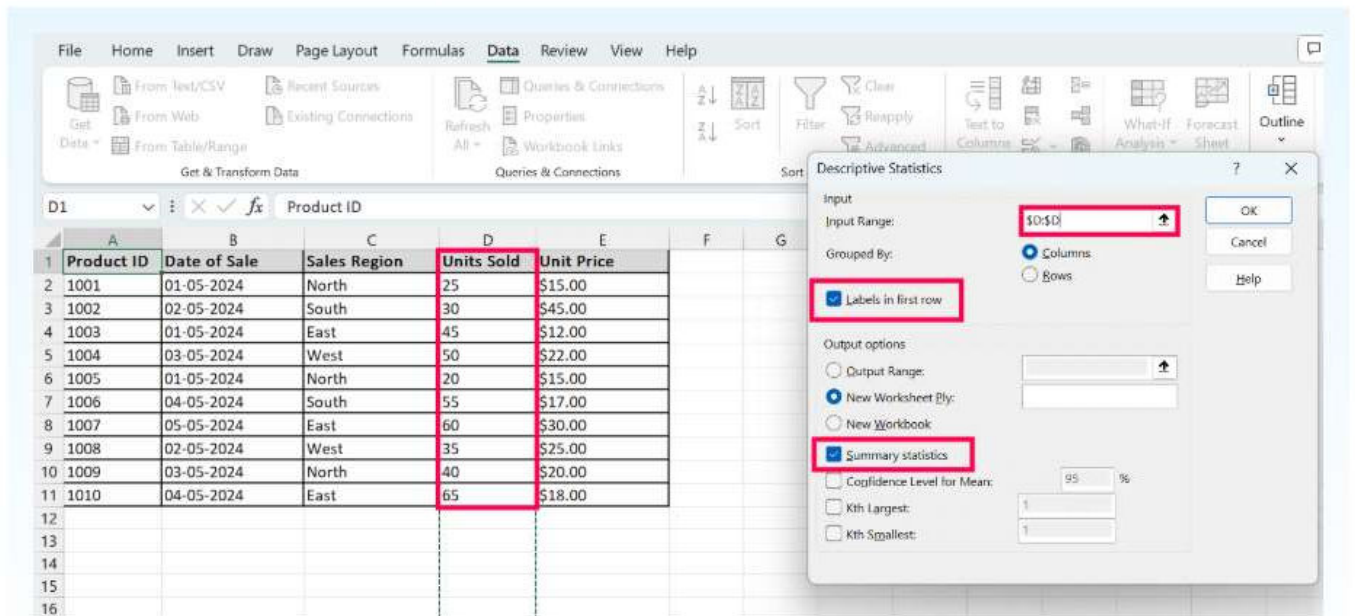


Fig. 10.8: Descriptive Statistics Tab

The output provides a summary of the data, highlighting measures of central tendency and variability, which aid in analyzing the distribution and scale of the sales data.

Units Sold	
Mean	42.5
Standard Error	4.787135539
Median	42.5
Mode	#N/A
Standard Deviation	15.13825177
Sample Variance	229.1666667
Kurtosis	-1.2
Skewness	0
Range	45
Minimum	20
Maximum	65
Sum	425
Count	10

Fig. 10.9: Result of Descriptive Statistics

Importance of Sampling:

- **Efficiency:** Sampling reduces the amount of data to be analyzed, making the process more manageable and less time-consuming.
- **Cost-Effective:** Analysing a sample is often more cost-effective than analysing the entire population.
- **Accuracy:** Proper sampling techniques ensure that the sample is representative of the population, leading to accurate and reliable inferences.

By mastering sampling in Excel, you can effectively perform inferential data analysis, making informed decisions based on your sample data.

Example 2: Stratified Sampling

While Excel doesn't have a built-in tool for stratified sampling, you can achieve it by manually dividing your data into strata and then performing random sampling within each stratum. For instance, if you have sales data categorized by region (North, South, East, West), you can:

1. **Filter** the data by each region.
2. **Perform Random Sampling** within each filtered region using the steps outlined previously.
3. **Combine** the samples from each region to form your stratified sample.

Example 3: Bootstrapping

Bootstrapping is a resampling technique used to estimate the distribution of a statistic by sampling with replacement. Here's how you can perform bootstrapping in Excel:

1. **Input Range:** A1:A100 (original sample of 100 data points)
2. **Sampling Method:** Random
3. **Number of Samples:** 100 (same as the original sample size)
4. **Output Range:** B1:B100

Repeat the random sampling process multiple times (e.g., 1000 iterations) to generate different bootstrap samples. You can then calculate the desired statistic (e.g., mean, median) for each bootstrap sample and analyze the distribution of these statistics.

Example 4: Cluster Sampling

Cluster sampling involves dividing the population into clusters and then randomly selecting clusters for analysis. Suppose you have a dataset of 1000 students divided into 10 classes (clusters):

1. **Randomly Select Clusters:** Use Excel's RAND function to assign a random number to each class and sort the classes by these random numbers.
2. **Select Clusters:** Choose a specified number of clusters (e.g., 3 out of 10).

3. **Combine Data:** Combine the data from the selected clusters for analysis.

Example 5: Systematic Sampling

Systematic sampling involves selecting every k-th element from the population. Suppose you have a dataset of 2000 survey responses and you want to select every 50th response:

1. **Input Range:** A1:A2000
2. **Sampling Method:** Periodic
3. **Period:** 50
4. **Output Range:** B1:B40

This will generate a sample of every 50th survey response from the dataset.

These examples illustrate different sampling techniques you can use in Excel to perform inferential data analysis. Each method has its own advantages and is suitable for different types of data and analysis objectives.

10.4 HYPOTHESIS TESTING

Hypothesis testing is a statistical method used to make decisions or inferences about a population parameter based on a sample. It involves setting up two competing hypotheses: the null hypothesis (H_0), which states that there is no effect or difference, and the alternative hypothesis (H_1), which suggests there is an effect or difference.

The process involves calculating a test statistic and comparing it to a critical value from a known distribution to either accept or reject the null hypothesis.

10.4.1 t-test

The **t-test** is used to compare the means of two groups to see if they are statistically different from each other. There are different types of t-tests: the independent sample t-test (for comparing means from two different groups), paired sample t-test (for comparing means from the same group at different times), and one-sample t-test (for comparing a sample mean to a known value).

Example:

Let's say we want to test whether a new teaching method improves students' performance. We take two groups of students: one taught with the traditional method and the other with the new method. We measure their scores after the training.

Data:

- Traditional method: Mean = 70, SD = 10, n = 30
- New method: Mean = 75, SD = 12, n = 30

The null hypothesis (H_0) is that there is no difference in means, and the alternative hypothesis (H_1) is that the means are different.

Using a **t-test**, we calculate the t-statistic and compare it to a critical value from the t-distribution table to determine if we reject or accept H_0 .

t-statistic formula for independent samples:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

If the calculated t-value exceeds the critical value for the given degree of freedom, we reject H_0 .

Problem Statement:

Suppose a company claims that the average weight of its product is 500 grams. However, a customer suspects that the actual average weight is less than 500 grams. To test this, the customer collects a random sample of 10 products, which has the following weights (in grams):

Sample Weights: 490, 495, 485, 500, 492, 488, 491, 487, 489, 486.

The customer wants to test the claim using a **one-sample t-test** with a significance level (α) of 0.05.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The average weight is 500 grams.
 - $H_0: \mu = 500$
- **Alternative Hypothesis (H_1):** The average weight is less than 500 grams.
 - $H_1: \mu < 500$ (one-tailed test)

Step 2: Enter Data into Excel

1. Open Excel and enter the sample data into a column. For example, enter the weights in cells A2 to A11.

Step 3: Calculate the Sample Mean and Standard Deviation

To calculate the sample mean:

- In a new cell (e.g., B2), enter the formula:
`=AVERAGE (A2:A11)`

To calculate the sample standard deviation:

- In another cell (e.g., B3), enter the formula:
`=STDEV.S(A2:A11)`

Step 4: Perform the t-Test Using Excel's T.TEST Function

Now, we will use the T.TEST function to perform a one-sample t-test.

1. In another cell (e.g., C2), enter the formula:
`=T.TEST (A2:A11, 500, 1, 1)`
 - A2:A11 is the range of sample data.

- 500 is the population mean under the null hypothesis.
- 1 specifies a one-tailed test.
- 1 specifies that this is a paired t-test (as we are comparing one sample to a known value).

This function will return the **p-value** for the test.

Step 5: Interpret the Results

The t-test function will return the p-value, which you compare against the significance level ($\alpha = 0.05$).

- **If p-value < 0.05:** Reject the null hypothesis (the average weight is likely less than 500 grams).
- **If p-value $\geq$ 0.05:** Fail to reject the null hypothesis (there is no sufficient evidence to claim the average weight is less than 500 grams).

10.4.1.1 By using Descriptive Statistics routine

To perform this task using the Descriptive Statistics routine, follow these steps:

1. Navigate to Data > Data Analysis > Descriptive Statistics, and click OK to access the Descriptive Statistics dialog box.
2. In the dialog box, specify the range for the Satisfaction variable in the Input Range field. If the column header is included, check the "Labels in first row" option. Select the desired output location (the default and recommended choice is "New Worksheet Ply").
3. Next, enable the "Summary Statistics" and "Confidence Level for Mean" options (refer to Figure 10.10). The default confidence level is set at 95%, which is standard in management research. However, you may adjust this setting if needed.

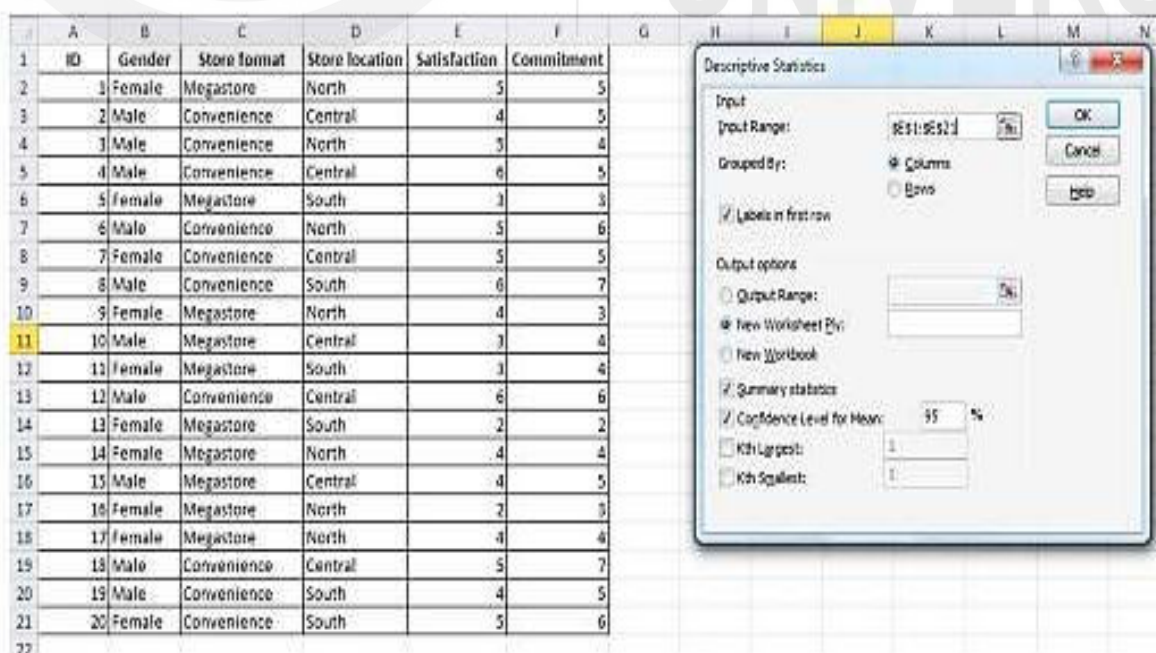


Fig. 10.10: Descriptive statistics dialogue box

4. Click "OK" to produce the output (Figure 10.11). Excel's Descriptive Statistics output provides the mean (4.25) along with a value labeled as the "Confidence Level (95%)." This label can be misleading, as it actually represents the margin of error for the confidence interval of the mean (0.565606).

	A	B	C
1	<i>Satisfaction</i>		
2			
3	Mean	4.25	
4	Standard Error	0.270234	
5	Median	4	
6	Mode	5	
7	Standard Deviation	1.208522	
8	Sample Variance	1.460526	
9	Kurtosis	-0.53737	
10	Skewness	-0.33546	
11	Range	4	
12	Minimum	2	
13	Maximum	6	
14	Sum	85	
15	Count	20	
16	Confidence Level(95.0%)	0.565606	
17			
18			

Fig. 10.11: Descriptive statistics output

5. This data can now be used to determine the confidence interval for the mean (rounded to two decimal places) as follows:
 - a. Lower Confidence Limit = mean – margin of error = $4.25 - 0.565606 = 3.68$.
 - b. Upper Confidence Limit = mean + margin of error = $4.25 + 0.565606 = 4.82$.
6. The results can be presented in a table alongside other relevant statistics for the variable, or they can be reported in text format with the mean, using the following notation: 95% CI [Lower Limit, Upper Limit]. For this example, it would be written as 95% CI [3.68, 4.82].

10.4.1.2 Independent two-sample t-test with equal variance assumption

Excel offers two types of t-tests for independent samples: one assumes equal variances, while the other does not. Using the customer satisfaction dataset (available on line), we will demonstrate the test to evaluate the following hypotheses:

H1: Male and female customers have different satisfaction levels.
H0: There is no difference in satisfaction levels between male and female customers.

A) Do the variances match?

Before performing the t-test, you can check the assumption of equal variances using the F-Test Two-Sample for Variances in the Data Analysis

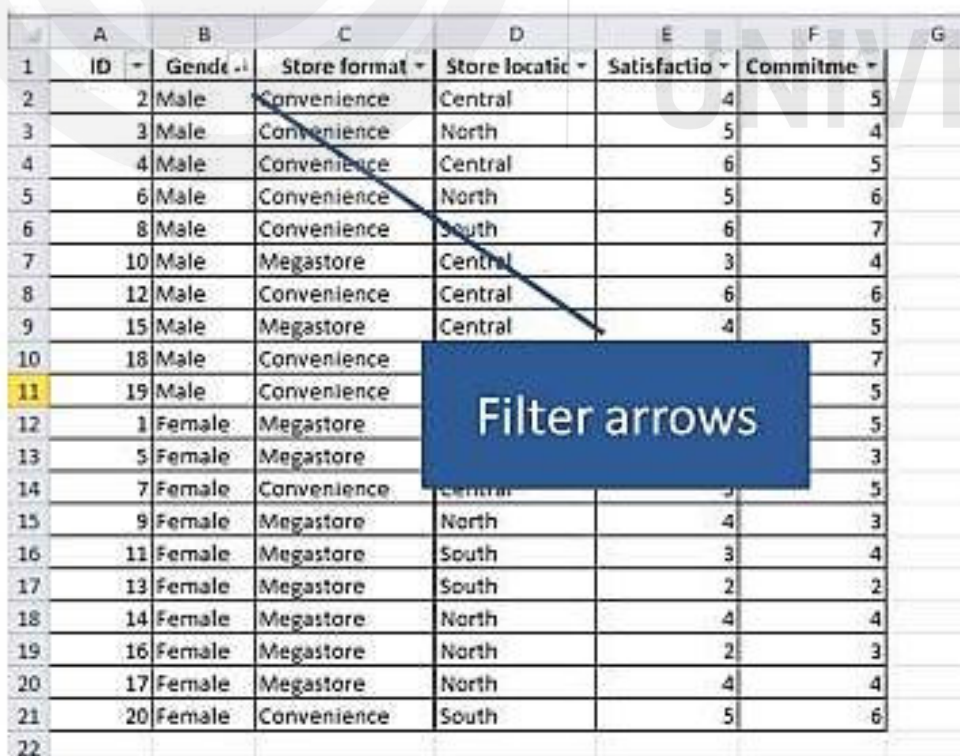
ToolPak. This test is conducted similarly to the t-test. Note that it provides a one-tailed p-value, which must be doubled to obtain the required two-tailed p-value. If the two-tailed p-value is below 0.05 (or if the assumption cannot be tested), you should assume unequal variances and use the corresponding version of the t-test. In this case, we will use the equal variances version because the F-Test in Excel returned a one-tailed p-value of 0.37, resulting in a two-tailed p-value of 0.74. Since 0.74 is greater than 0.05, the assumption of equal variances is valid.

B) Performing the t-test

To run the t-test, you need to input the range for the two variables (i.e., the responses of male and female customers) into the test dialogue box. Your dataset must be appropriately formatted for this. If it isn't, you have two options:

Option 1: Sort the dataset so that rows for each group (e.g., males and females) are adjacent. To do this:

- Click on a column header in the dataset.
- Go to *Home > Sort & Filter > Filter* to apply filters to the column headers.
- Click the filter arrow in the relevant column header (in this case, Gender).
- Select *Sort A to Z* or *Sort Z to A*. In this example, *Sort Z to A* was chosen to arrange the dataset in the order of Males followed by Females. This organizes the dataset such that all Male and Female cases are grouped together (see Figure 10.12).



	A	B	C	D	E	F	G
1	ID	Gender	Store format	Store location	Satisfaction	Commitment	
2	2	Male	Convenience	Central	4	5	
3	3	Male	Convenience	North	5	4	
4	4	Male	Convenience	Central	6	5	
5	6	Male	Convenience	North	5	6	
6	8	Male	Convenience	South	6	7	
7	10	Male	Megastore	Central	3	4	
8	12	Male	Convenience	Central	6	6	
9	15	Male	Megastore	Central	4	5	
10	18	Male	Convenience			7	
11	19	Male	Convenience			5	
12	1	Female	Megastore			5	
13	5	Female	Megastore			3	
14	7	Female	Convenience	Central	5	5	
15	9	Female	Megastore	North	4	3	
16	11	Female	Megastore	South	3	4	
17	13	Female	Megastore	South	2	2	
18	14	Female	Megastore	North	4	4	
19	16	Female	Megastore	North	2	3	
20	17	Female	Megastore	North	4	4	
21	20	Female	Convenience	South	5	6	
22							

Fig. 10.12 Sorted dataset by categories in variable Gender

- e. You can now specify the range for each group by entering the corresponding data range in the Input Range dialogue box.
- f. (Tip: After completing the test, you can restore the dataset to its original layout by sorting it using the ID variable. Select 'Sort A to Z' to return the dataset to its initial order based on the ID variable.)

Option 2: involves creating a duplicate of the data where the values for each group being compared are placed in separate columns. To do this, first sort the data as described in Option 1, then copy the relevant data from the main dataset and paste it into a new worksheet, creating two new columns (e.g., one for males and one for females). Be sure to assign appropriate column headers (see Figure 10.13). This approach is particularly useful for large datasets and helps keep the data and test results organized. (Tip: Assign a meaningful name to the new worksheet to make managing the workbook easier.)

	A	B	C
1	Male	Female	
2	4	5	
3	5	3	
4	6	5	
5	5	4	
6	6	3	
7	3	2	
8	6	4	
9	4	2	
10	5	4	
11	4	5	
12			
13			

Fig. 10.13: Worksheet displaying satisfaction levels of male and female respondents

For clarity of presentation, we will use option 2 in our example.

To run the test, follow these steps:

1. Select *Data > Data Analysis > t-Test: Two Sample Assuming Equal Variances > OK* to open the test dialogue box.
2. In the dialogue box:
 - a. Enter the data ranges for the two variables in the *Input* fields. If both columns include headers (as in this case), you can include them in the range if preferred.
 - b. Enter 0 in the *Hypothesized Mean Difference* box. This indicates to Excel that you are testing the null hypothesis (H_0), which assumes no difference between the group means.
 - c. If you included column headers, check the *Labels* box.
 - d. Set the desired *Alpha* (significance level) for the test. The default value is 0.05.
 - e. Choose the *Output* option. If you used option 1 to prepare your data, select the default *New Worksheet Ply* to display the output in a new worksheet. If you used option 2 (as shown here), select an appropriate *Output Range* in the active worksheet. (Tip: Ensure the

output range does not overlap with the dataset, or it will overwrite the data.) The completed dialogue box is shown in Figure 10.14.

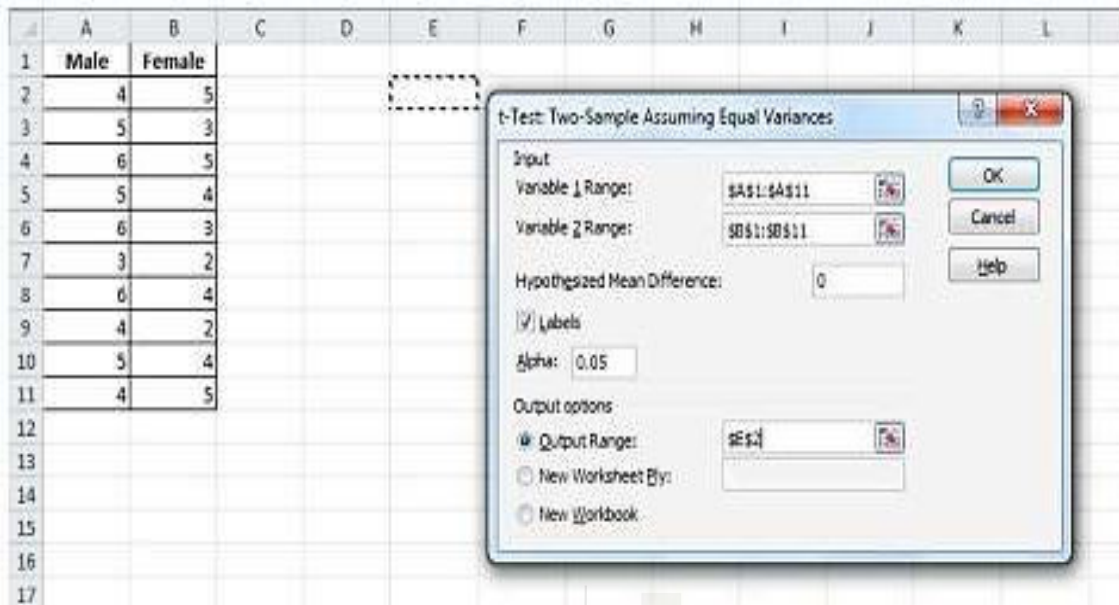


Fig. 10.14: Dialogue box for t-test

- f. Click OK. The output that appears is shown in Figure 10.15, with adjusted column spacing for improved readability.

	Male	Female
Mean	4.8	3.7
Variance	1.066667	1.344444
Observations	10	10
Pooled Variance	1.205556	
Hypothesized Mean	0	
df	18	
t Stat	2.240186	
P(T<=t) one-tail	0.018968	
t Critical one-tail	1.734064	
P(T<=t) two-tail	0.037935	
t Critical two-tail	2.100922	

Fig. 10.15 –Output for t-test

3. The next step is to interpret the output. Focus on the following key points:
 - a. The means of each group, which show the size of the difference between the groups.
 - b. The group sizes (both are 10), which can be checked to ensure no cases were inadvertently excluded from the sample.
 - c. The test statistic (t Stat = 2.240186).

- d. The degrees of freedom ($df = 18$).
 - e. The two-tailed p-value (0.037935).
4. Compare the p-value to your chosen significance level (Alpha), which is 0.05 in this case. Since 0.037935 is less than 0.05, we reject H_0 , accept H_1 , and conclude that there is a statistically significant difference between the two groups.
 5. The results can be presented in a simplified table or, if mentioned in text, using the format $t(df) = t \text{ Stat}$ rounded to two decimal places, $p = p\text{-value}$ rounded to two decimal places, as shown: $t(18) = 2.24, p = 0.04$.

(Note: Our example uses a very small sample size for illustration. In practice, such small sample sizes may not be sufficient to detect differences in the population. Additionally, it's important to ensure that the samples meet the assumption of normal distribution.)"

10.4.2 Chi-Square Test

The **chi-square test** is used for categorical data to assess how likely it is that any observed difference between sets of categories is due to chance. It compares observed frequencies to expected frequencies under the null hypothesis.

Example:

Suppose a company wants to know if there is a significant difference in customer preferences for three product colors: red, blue, and green. They survey 150 customers, and the results are:

- Red: 60
- Blue: 40
- Green: 50

The null hypothesis (H_0) is that customers have no color preference, and all colors are equally preferred.

We calculate the expected frequencies assuming the null hypothesis is true (i.e., each color should have an expected frequency of 50, given 150 total responses). We then compute the **chi-square statistic**:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where O_i are the observed frequencies and E_i are the expected frequencies? If the chi-square value is larger than the critical value from the chi-square distribution, we reject H_0 .

Problem Statement:

A local grocery store wants to know if customer preference for certain fruit types (apple, banana, and orange) is evenly distributed. The store randomly surveys 60 customers, and the observed preferences are as follows:

- Apple: 25
- Banana: 20
- Orange: 15

The expected distribution is equal for each fruit, meaning that under the null hypothesis, the store expects the preferences to be equally distributed among the three fruits.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** Customer preferences are equally distributed among apples, bananas, and oranges.
 - H_0 : The distribution of preferences is equal.
- **Alternative Hypothesis (H_1):** Customer preferences are not equally distributed.
 - H_1 : The distribution of preferences is not equal.

Step 2: Organize Data

Create a table to show the observed and expected frequencies.

Fruit	Observed (O)	Expected (E)
Apple	25	20
Banana	20	20
Orange	15	20

Step 3: Calculate Chi-Square Statistic

The formula for the **Chi-Square statistic** is:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where:

- O = Observed frequency
- E = Expected frequency

Step 4: Apply the Formula

For **Apple**:

$$\frac{(25-20)^2}{20} = \frac{(5)^2}{20} = \frac{25}{20} = 1.25$$

For **Banana**:

$$\frac{(20-20)^2}{20} = \frac{(0)^2}{20} = 0$$

For **Orange**:

$$\frac{(15-20)^2}{20} = \frac{(-5)^2}{20} = \frac{25}{20} = 1.25$$

Now, sum these values:

$$\chi^2 = 1.25 + 0 + 1.25 = 2.5$$

Step 5: Determine Degrees of Freedom

Degrees of freedom (df) for a Chi-Square test is calculated as:

$$df = \text{number of categories} - 1 = 3 - 1 = 2$$

Step 6: Find the Critical Value

Assume a significance level (α) of 0.05. Using a Chi-Square distribution table, the critical value for $\alpha=0.05$ and $df = 2$ is **5.991**.

Step 7: Compare the Chi-Square Statistic with the Critical Value

- **Calculated Chi-Square value** = 2.5
- **Critical value** = 5.991

Since 2.5 is less than 5.991, we **fail to reject the null hypothesis**.

Step 8: Conclusion

There is **no sufficient evidence** to suggest that customer preferences for apples, bananas, and oranges are not equally distributed.

- **Chi-Square Statistic (χ^2):** 2.5
- **Degrees of Freedom:** 2
- **Critical Value (from Chi-Square table):** 5.991
- **Conclusion:** Fail to reject the null hypothesis; the preferences are likely equally distributed.

Performing the Chi-Square Test in Excel:

- 1) Enter the **Observed Frequencies** in one column (e.g., cells A2 to A4: 25, 20, 15).
- 2) Enter the **Expected Frequencies** in the next column (e.g., cells B2 to B4: 20, 20, 20).
- 3) Use the formula to calculate the **Chi-Square statistic**:
 - In another column, calculate the individual chi-square values using the formula:
$$=(A2-B2)^2/B2$$
- 4) Sum the individual chi-square values to get the total χ^2 statistic.

You can also use the **CHISQ.TEST** function in Excel to automatically calculate the p-value:

- Formula:
$$=CHISQ.TEST(A2:A4, B2:B4)$$

This will return the p-value directly, which can be compared with the significance level (0.05). If the p-value is less than 0.05, you reject the null hypothesis; otherwise, you fail to reject it.

10.4.2.1 Chi-square test for association

A chi-square test of association can be performed using either the CHISQ.DIST. RT function or the CHISQ.TEST function. Both methods require data manipulation, and we will demonstrate both, starting with the former, as it better illustrates the test's principles and provides additional valuable information.

A) Chi-square test of association with the CHISQ.DIST. RT function

We will demonstrate this test using the same dataset which reports the frequency of car driving based on household income in the form of a contingency table (Figure 10.16). The following hypotheses will be tested at the 5% significance level ($\alpha = 0.05$):

H1: There is an association between household income and frequency of car driving.

H0: There is no association between household income and frequency of car driving.

For this test, your data must be in the form of a contingency table containing counts (not percentages). If your data is not already formatted this way, refer to *Data Exploration with Microsoft Excel: Analysing More Than One Variable* for guidance on creating contingency tables.

	A	B	C	D	E	F	G
1	Observed frequencies (O)						
2			Driving frequency				
3			Every day	At least 3 times per week	1-2 times per day	Less than once per week	Total
4	Annual household income	Up to £15000	205	110	55	23	393
5		£15000 to £25000	228	72	36	13	349
6		Over £25000	350	82	36	10	478
7		Total	783	264	127	46	1220
8							

Fig. 10.16 – Frequency of car driving by household income

The chi-squared test operates by comparing the observed frequencies (from the contingency table) with the expected frequencies to compute a test statistic. This statistic is then compared to the relevant chi-squared distribution to determine the p-value. Excel's CHISQ.DIST. RT function will calculate the p-value, but you must first calculate the test statistic and degrees of freedom. We will walk through each step of the process.

B) Calculating the expected frequencies

The formula can be used to calculate the expected frequencies (E) for each cell in the table.

$$E = \frac{\text{Row total} * \text{Column total}}{\text{Total Sample Size}}$$

To perform this in Excel, start by copying and pasting the original contingency table into the worksheet. Then, apply the formula above to calculate the expected value for each cell, as shown in Figure 10.17. (Tip: use Paste Special > Values to paste only the numerical values, not the formulas, and delete the observed frequency cells, leaving just the row and column totals for the calculation.)

	A	B	C	D	E	F	G	H
1	Observed frequencies (O)							
2			Driving frequency					
3				At least 3 times per week	1-2 times per day	Less than once per week	Total	
4			Every day					
5	Annual	Up to £15000	205	110	55	23	393	
6	household	£15000 to £25000	228	72	36	13	349	
7	income	Over £25000	350	82	36	10	478	
8		Total	783	264	127	46	1220	
9	Expected frequencies (E)							
10			Driving frequency					
11				At least 3 times per week	1-2 times per day	Less than once per week	Total	
12			Every day					
13	Annual	Up to £15000	252	85	41	15	393	
14	household	£15000 to £25000	224	76	36	13	349	
15	income	Over £25000	307	103	50	18	478	
16		Total	783	264	127	46	1220	
17	$E = \frac{\text{row total} \times \text{column total}}{\text{total sample size}}$							

Fig. 10.17: Observed and expected frequencies

C) Calculating the chi-squared test statistic

The chi-squared test statistic is represented by the following equation:

$$\text{chi-squared test statistic} = \sum \frac{(O - E)^2}{E}$$

Here, O represents the observed frequencies for each cell, and E denotes the corresponding expected frequencies. The symbol Σ indicates that you should sum the calculations for all cells. A straightforward approach is to create a table that includes the value of $(O - E)^2 / E$ for each cell, and then sum all the values to get the final result, as shown in Figure 10.18. In this example, the test statistic is 43.685. (Tip: Use the ^2 operator in the Excel formula bar to calculate the square of a value.)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Observed frequencies (O)					Calculating the test statistic $(O-E)^2/E$									
2			Driving frequency												
3				At least 3 times per week	1-2 times per day	Less than once per week									
4			Every day												
5	Annual	Up to £15000	205	110	55	23	393		Up to £15000	4.3413597	7.324218	4.852271814	4.517778		
6	household	£15000 to £25000	228	77	35	13	349		£15000 to £25000	0.0718131	0.164167	0.001003425	0.001922		
7	Income	Over £25000	150	82	35	10	478		Over £25000	8.0881577	4.842405	1.806547311	1.571432		
8		Total	783	264	127	46	1220								
9	Expected frequencies (E)														
10			Driving frequency												
11				At least 3 times per week	1-2 times per day	Less than once per week									
12			Every day												
13	Annual	Up to £15000	252	85	41	15	393								
14	household	£15000 to £25000	224	78	35	13	349								
15	Income	Over £25000	307	103	50	18	478								
16		Total	783	264	127	46	1220								
17	$E = \frac{\text{row total} \times \text{column total}}{\text{total sample size}}$														
18															

Fig. 10.18 – Calculating the test statistic

Next, you need to calculate the degrees of freedom (df) for the test, which is determined using the following equation.

$$df = (r - 1) * (c - 1)$$

Here, r represents the number of rows in the contingency table, and c represents the number of columns. In our example, there are 3 rows and 4 columns (excluding the row and column totals), so:

$$df = (3 - 1) * (4 - 1) = 2 * 3 = 6$$

Therefore, the degrees of freedom for the test is 6.

D) Determining the p-value using the CHISQ.DIST. RT function

We now have the necessary information to use the CHISQ.DIST. RT function to calculate the p-value for the test.

1. Choose a cell where the function output will appear.
2. Go to Formulas > More Functions > Statistical > CHISQ.DIST. RT to open the Function Arguments dialog box.
3. In the dialog box, enter the chi-squared test statistic (or the reference to the cell with the statistic) in the X field (likely meant to be the letter χ), and the number of degrees of freedom (or the cell reference) in the Deg_freedom field (Figure 10.19).



Fig. 10.19: CHISQ.DIST. RT dialogue box

4. Click OK to obtain the output (Figure 10.20). Keep in mind that, similar to the ANOVA test, the p-value is displayed in scientific notation (e.g., 8.53401E-8).

113 • =CHISQ.DIST.RT(L8,L9)

Observed frequencies (O)

		Driving frequency				
		Every day	At least 3 times per week	1-2 times per day	Less than once per week	Total
Annual household Income	Up to £15000	265	110	55	23	453
	£15000 to £25000	128	72	36	13	249
	Over £25000	350	82	36	10	478
	Total	743	264	127	46	1180

Expected frequencies (E)

		Driving frequency				
		Every day	At least 3 times per week	1-2 times per day	Less than once per week	Total
Annual household Income	Up to £15000	252	85	41	15	393
	£15000 to £25000	224	76	36	13	349
	Over £25000	307	103	50	18	478
	Total	783	264	127	46	1120

Calculating the test statistic $(O-E)^2/E$

	Every day	At least 3 times per week	1-2 times per day	Less than once per week
Up to £15000	8.8433597	7.324218	4.852271814	4.517778
£15000 to £25000	0.0718131	0.164187	0.003003455	0.001322
Over £25000	6.0883577	4.842405	3.804547513	3.571432

Test statistic $(= \sum (O-E)^2/E) =$

43.685

Degrees of freedom $(= (r-1)(c-1)) =$

6 (r = number of rows;
 c = number of columns)

Results

p-value using CHISQ.DIST.RT

8.54832E-09

$$E = \frac{\text{row total} \times \text{column total}}{\text{total sample size}}$$

Fig. 10.20: CHISQ.DIST.RT output

5. The next step is to interpret the output. Compare the p-value with your chosen significance level (Alpha), which is 0.05 in this case. Since the calculated p-value is less than 0.05, we reject H_0 , accept H_1 , and conclude that there is a statistically significant association between household income and frequency of car driving.

6. The results can be presented in a simple table showing the test statistic, degrees of freedom, sample size, and p-value, or in text using the format $\chi^2(\text{degrees of freedom, sample size}) = \text{chi-squared statistic (to two decimal places), } p = \text{p-value (to two decimal places)}$. For example, in this case: $\chi^2(6, 60) = 43.69, p < 0.001$.

E) Chi-Squared test of association using the CHISQ.TEST function

You can perform the same test using the CHISQ.TEST function as an alternative. In this case, you only need to create a table of expected cell frequencies, similar to when using the CHISQ.DIST.RT function. However, you won't need to calculate the chi-squared statistic or degrees of freedom manually.

To use the CHISQ.TEST function:

1. Organize your data in a contingency table with observed frequencies.
2. Create a table of expected frequencies.
3. Choose a cell for the function output.
4. Go to Formulas > More Functions > Statistical > CHISQ.TEST to open the Function Arguments dialog box.
5. In the dialog box, enter the range of observed frequencies from the contingency table in the Actual_range field, and the expected frequencies in the Expected_range field (remember not to include row and column totals in the range). See Figure 10.21.

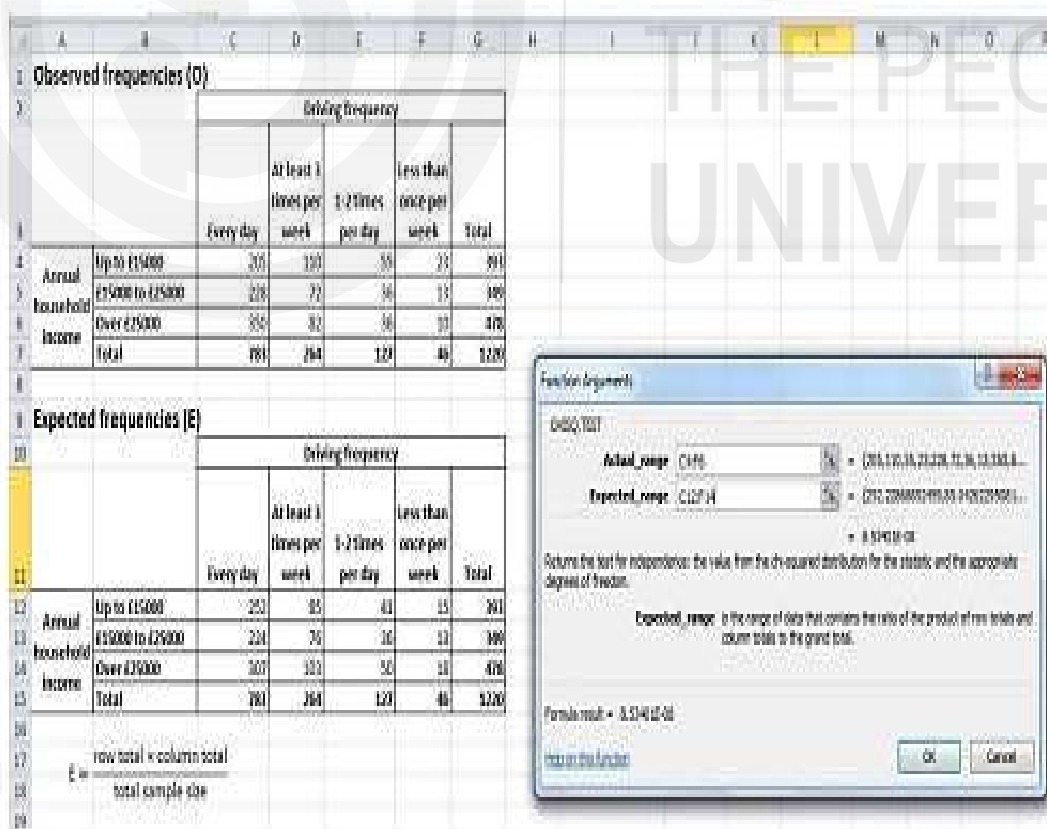


Fig. 10.21: CHISQ.TEST Function Argument dialogue box

- Click OK. The output, shown in Figure 10.22, is identical to that from the CHISQ.DIST.RT function and should be interpreted in the same manner.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Observed frequencies (O)											
2			Driving frequency									
3				At least 3 times per week	1-2 times per day	Less than once per week	Total					
4			Every day									
5	Annual household income	Up to £15000	205	110	55	23	393					
6		£15000 to £25000	228	72	36	13	349					
7		Over £25000	350	82	36	10	478					
8		Total	783	264	127	46	1220					
9	Expected frequencies (E)											
10			Driving frequency									
11				At least 3 times per week	1-2 times per day	Less than once per week	Total					
12			Every day									
13	Annual household income	Up to £15000	252	85	41	15	393					
14		£15000 to £25000	224	76	36	13	349					
15		Over £25000	307	103	50	18	478					
16		Total	783	264	127	46	1220					
17	Results											
18	p-value using CHISQ.TEST											8.53401E-08
19	E = $\frac{\text{row total} \times \text{column total}}{\text{total sample size}}$											

Fig. 10.22: CHISQ.TEST output

The CHISQ.TEST function is a bit quicker to use, but when reporting test results, it is standard practice to include the test statistic and degrees of freedom. Therefore, we recommend using the CHISQ.DIST.RT function or at least calculating the necessary details as demonstrated in the example for that function. You can also run both functions to verify the accuracy of the chi-squared statistic calculation.

10.4.3 F-test

The **F-test** is used to compare two variances to see if they are significantly different. It is commonly used in analysis of variance (ANOVA) and regression analysis.

Example:

Suppose we have two groups with the following variances:

- Group 1: Variance (S_1^2) = 15, n_1 = 20
- Group 2: Variance (S_2^2) = 10, n_2 = 20

The null hypothesis (H_0) is that the variances are equal, and the alternative hypothesis (H_1) is that they are not equal.

The **F-statistic** is calculated as:

$$F = \frac{S_1^2}{S_2^2}$$

In this case:

$$F = \frac{15}{10} = 1.5$$

We compare the F-value to the critical value from the F-distribution table to determine if we reject or accept H_0 .

Problem Statement:

A company wants to compare the variability in salaries between two departments: **Department A** and **Department B**. The company randomly samples 10 employees from each department and records their salaries (in \$1000s):

- **Department A Salaries** (Sample 1): 45, 48, 50, 52, 49, 46, 51, 47, 50, 48
- **Department B Salaries** (Sample 2): 55, 58, 54, 60, 57, 53, 59, 56, 55, 58

The company wants to know if there is a significant difference in the variances of salaries between the two departments. We'll perform an **F-test** to compare the variances.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The variances of the two departments' salaries are equal.
 - $H_0: \sigma_1^2 = \sigma_2^2$
- **Alternative Hypothesis (H_1):** The variances of the two departments' salaries are not equal.
 - $H_1: \sigma_1^2 \neq \sigma_2^2$

Step 2: Organize the Data

Department A (Sample 1):

- 45, 48, 50, 52, 49, 46, 51, 47, 50, 48

Department B (Sample 2):

- 55, 58, 54, 60, 57, 53, 59, 56, 55, 58

Step 3: Calculate the Sample Variances

1. **Variance of Department A (Sample 1):**

$$s_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1}$$

Where:

- x_i = data points for Department A
- $\bar{x}_1$ = sample mean of Department A
- n_1 = number of data points for Department A (10)

Use Excel or a calculator to calculate the variance of Department A:

$$s_1^2 = 4.22$$

2. Variance of Department B (Sample 2):

$$S_2^2 = \frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1}$$

Where:

- x_i = data points for Department B
- $\bar{x}_1$ = sample mean of Department B
- n_2 = number of data points for Department B (10)

Use Excel or a calculator to calculate the variance of Department B:

$$S_2^2 = 5.73$$

Step 4: Calculate the F-Test Statistic

The F-test statistic is calculated as the ratio of the two sample variances:

$$F = \frac{s_2^2}{s_1^2} = \frac{5.73}{4 \cdot 22} = 1.36$$

We assume that the larger variance is placed in the numerator to ensure that the F-ratio is greater than or equal to 1.

Step 5: Determine the Degrees of Freedom

Degrees of freedom for the F-test are based on the sample sizes:

- **Degrees of freedom for Department A (Sample 1):**

$$df_1 = n_1 - 1 = 10 - 1 = 9$$

- **Degrees of freedom for Department B (Sample 2):**

$$df_2 = n_2 - 1 = 10 - 1 = 9$$

Step 6: Find the Critical Value

Using an F-distribution table (or Excel), for a significance level (α) of 0.05 and degrees of freedom $df_1=9$ and $df_2=9$, the critical F-value is approximately **3.18**.

Step 7: Compare the F-Test Statistic with the Critical Value

- **F-test statistic** = 1.36
- **Critical F-value** = 3.18

Since 1.36 is less than 3.18, we **fail to reject the null hypothesis**.

Step 8: Conclusion

There is **no significant difference** in the variances of salaries between the two departments.

□ **F-Test Statistic (F):** 1.36

- **Degrees of Freedom:** $df_1=9$ and $df_2=9$
- **Critical Value:** 3.18 (for $\alpha=0.05$, $\alpha = 0.05$, $\alpha=0.05$)
- **Conclusion:** Fail to reject the null hypothesis, meaning that the variances of the salaries between the two departments are not significantly different.

10.4.3.1 The steps to conduct an F-test in Excel are outlined below:

Step 1: The data used for the F-test analysis is displayed in the following image.

	B	C
2	Girls	Boys
3	82	48
4	89	78
5	86	66
6	93	67
7	98	77
8	99	71
9	95	40
10	92	65
11	90	45
12	87	58
13	95	48
14	95	69
15		

Fig. 10.23: F_Test Sample Data

In the Analysis Toolpak workbook, you can practice performing the F-test analysis.

Step 2: On the Excel ribbon, go to the Data tab and click on "Data Analysis."

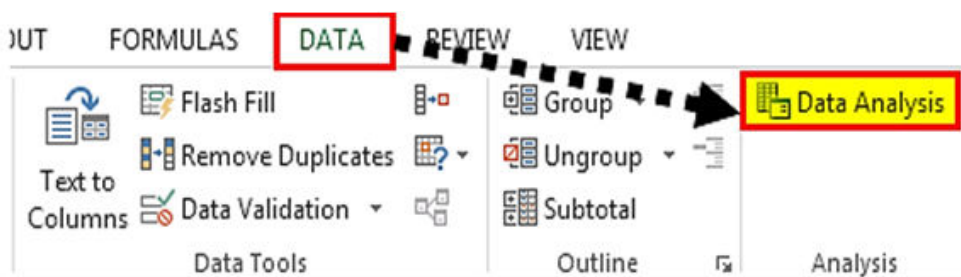


Fig. 10.24: Data Tab

Step 3: Once you click on "Data Analysis," a dialog box will appear. Select "F-test" and then click "OK" to activate the function.

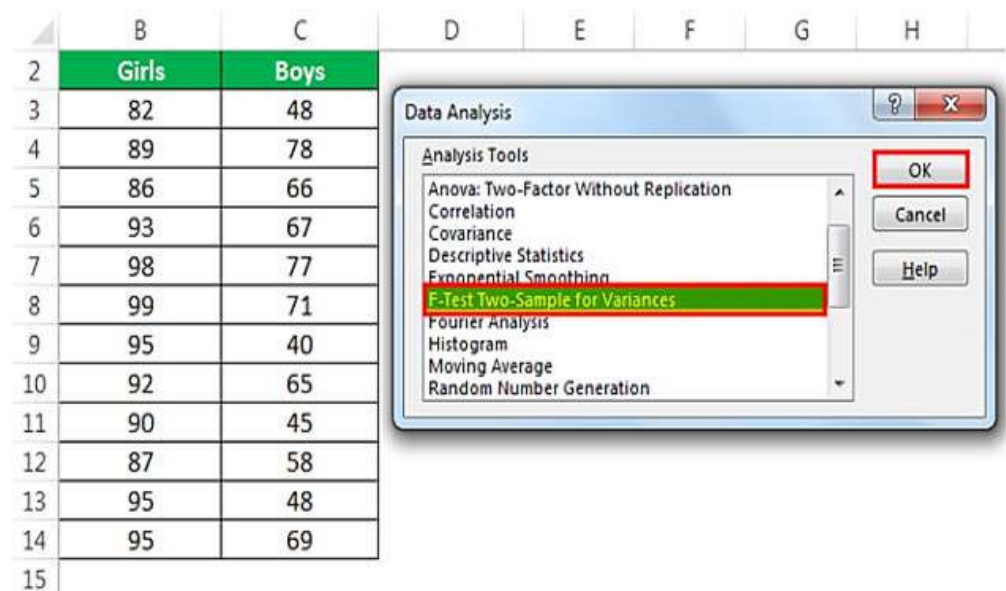


Fig. 10.25: F_Test Dialog Box

Step 4: Input the range for variable 1 and variable 2. To do this, select the cell range B3:B14 for variable 1 and C3:C14 for variable 2.

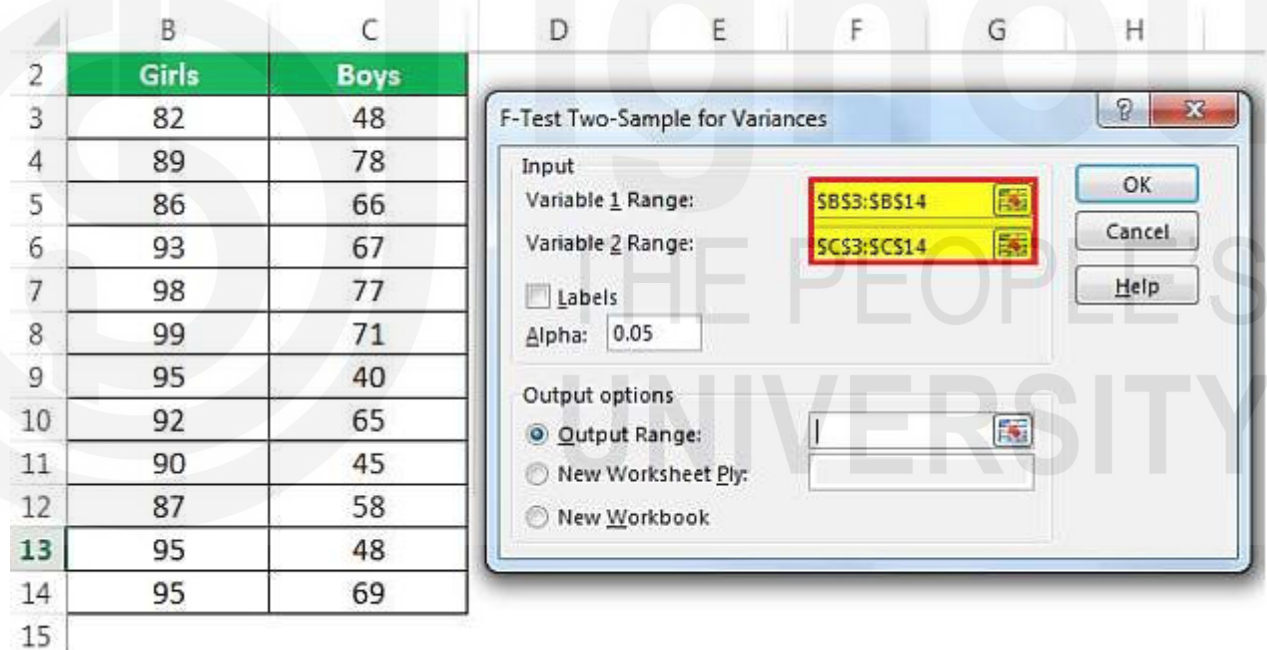


Fig. 10.26 – F_Test Two Sample for Variances

Step 5: Choose the "Output Range."

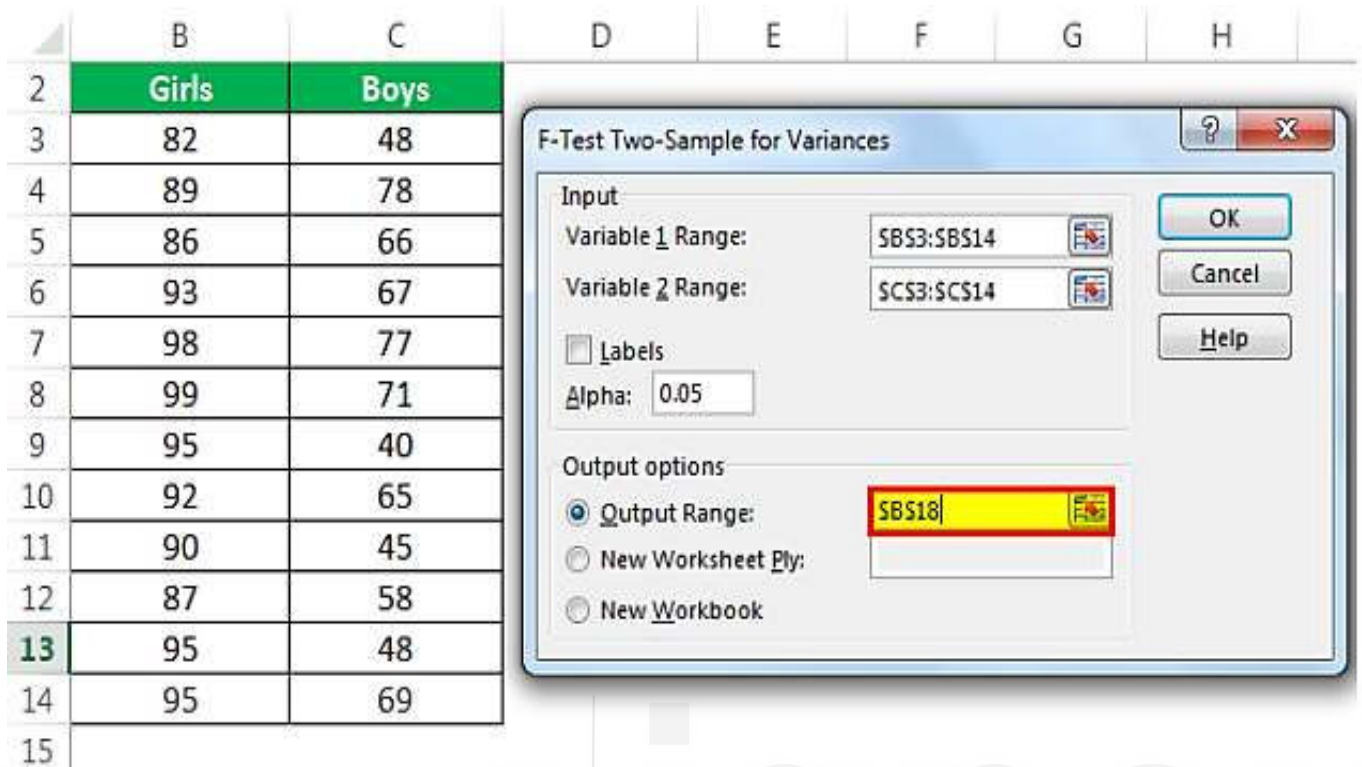


Fig. 10.27: F_Test Output Range

Step 6: Click "OK," and the data analysis will appear in the chosen cell.

	B	C	D	E
18	F-Test Two-Sample for Variances			
19				
20		<i>Variable 1</i>	<i>Variable 2</i>	
21	Mean	91.75	61	
22	Variance	26.0227273	166.363636	
23	Observations	12	12	
24	df	11	11	
25	F	0.15642077		
26	P(F<=f) one-tail	0.00232827		
27	F Critical one-t	0.35487036		
28				

Fig. 10.28: F_Test Output

A) How the F-Test Works

The following points provide a deeper understanding of the F-test function:

- The F-test is used to determine whether there is a significant difference between the variances of two datasets.
- The null hypothesis is rejected if the variances of the two datasets are unequal, and accepted if the variances are equal.
- The F-test calculates the probability or likelihood of variation.

- The F-test will return an error if:
 - The number of values in either array 1 or array 2 is less than two.
 - The variance of either array is zero.
- The F-test cannot be performed on a single sample; it requires two datasets.
- The F-test function ignores any text in the sample data and provides numerical results.

B) Examples of the F-Test

Here are some situations where the F-test can be applied:

- To compare the quality of lectures given by two professors teaching the same subject.
- To test two samples of bottle gourd under two different experimental conditions.
- To analyze the scores of two groups within the same field.

10.4.4 Z-test

The **Z-test** is used when comparing sample and population means to assess if they are significantly different, especially when the sample size is large ($n > 30$) and the population variance is known.

Example:

Suppose the average height of men in a country is 175 cm with a standard deviation of 5 cm. A researcher wants to test if a new sample of 50 men has a significantly different average height. The sample has a mean height of 177 cm.

The null hypothesis (H_0) is that the sample mean is equal to the population mean (175 cm), and the alternative hypothesis (H_1) is that the means are different.

We use the **Z-test** formula:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}=177$
- $\mu=175$
- $\sigma=5$
- $n=50$

$$z = \frac{177 - 175}{\frac{5}{\sqrt{50}}} = \frac{2}{0.707} = 2.83$$

If the Z-value exceeds the critical value (from the standard normal distribution), we reject H_0 .

	A
1	Scores
2	3
3	5
4	4
5	5
6	9
7	4
8	7
9	9
10	3
11	8
12	

Fig. 10.29: Z_Test Sample Data

We will use this data to determine the one-tailed probability value of the Z.TEST, assuming a population mean of 6 for the hypothesis.

The steps to apply the Z.TEST formula in Excel are as follows:

Begin by entering the Z.TEST formula in a cell within Excel.

		X	✓	<i>f_x</i>
				=Z.TEST(
	A	B	C	D
1	Scores			
2	3	=Z.TEST(		
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			

Fig. 10.30– Formula for Z_Test

Select the array as scores, A2 to A11.

	A	B	C	D
1	Scores			
2	3	=Z.TEST(A2:A11,		
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.31: Function Implementation of Z_Test

The next argument is "x." As we have already assumed the hypothesized population mean to be 6, use this value for this argument.

	A	B	C	D
1	Scores			
2	3	=Z.TEST(A2:A11,6)		
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.32: Function Range for Z_Test

The final argument is optional, so you can close the formula to obtain the Z.TEST value.

C2				
	A	B	C	D
1	Scores			
2	3		0.6561912	
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.33: Result for Z_Test Using Function

This is a one-tailed Z.TEST value. To calculate the two-tailed Z.TEST value, multiply this result by 2.

C3				
	A	B	C	D
1	Scores			
2	3		0.6561912	
3	5		1.3123825	
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.34: Result for Two-tailed Z_Test

Z TEST Using Data Analysis Option

We can perform a Z.TEST using Excel's "Data Analysis" option. Z.TEST is used to compare two means when the variance is known. In this process, we can define two hypotheses: the "Null Hypothesis" and the "Alternative Hypothesis." Their equations are as follows:

H0: $\mu_1 - \mu_2 = 0$ (Null Hypothesis)

H1: $\mu_1 - \mu_2 \neq 0$ (Alternative Hypothesis)

The alternative hypothesis (H1) suggests that the two population means are not equal.

For this example, we will use the scores of two students across multiple subjects.

	A	B
1	Student 1	Student 2
2	93	79
3	68	66
4	82	73
5	51	72
6	87	80
7	82	74
8	72	95
9	71	93
10	92	99
11	73	97
12		

Fig. 10.35: Sample Data

Step 1: Begin by calculating the variances of the two values using the VAR.P function.

E2					
	A	B	C	D	E
1	Student 1	Student 2			
2	93	79			
3	68	66			
4	82	73			
5	51	72			
6	87	80			
7	82	74			
8	72	95			
9	71	93			
10	92	99			
11	73	97			
12					

Varianc	146.49
Varianc	131.16

Fig. 10.36: Variances on Sample Data

Step 2: Navigate to the "Data" tab and select "Data Analysis."

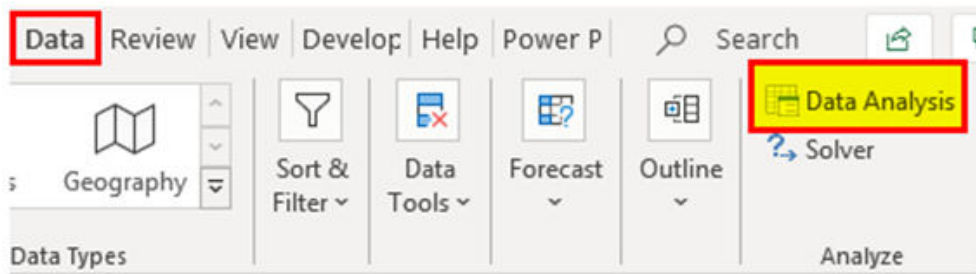


Fig. 10.37: Data to Data Analysis Tab

Scroll down, select "z-Test: Two Sample for Means," and click "OK."

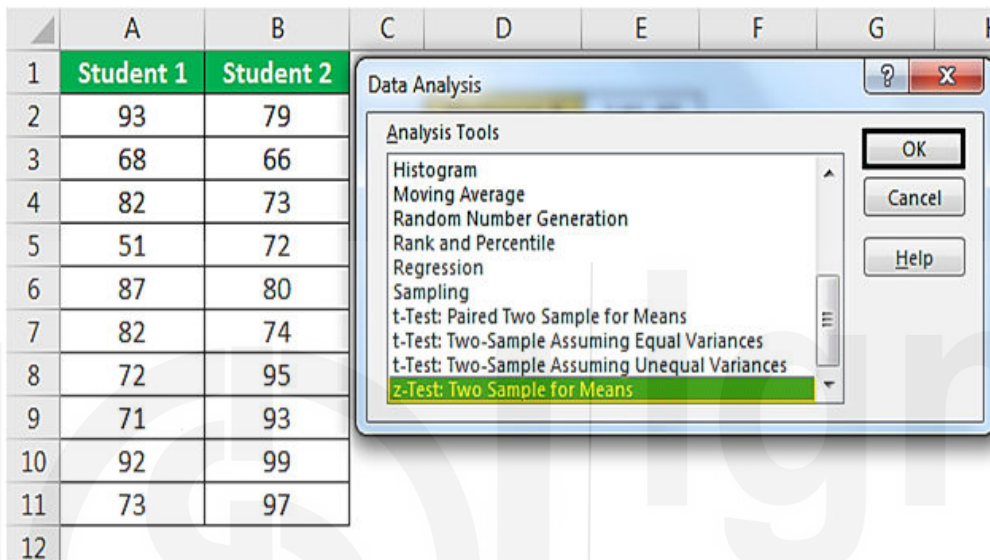


Fig. 10.38: Z_Test Dialog Box

Step 3: In the "Variable 1 Range," select the scores of "Student 1," and in the "Variable 2 Range," select the scores of "Student 2."

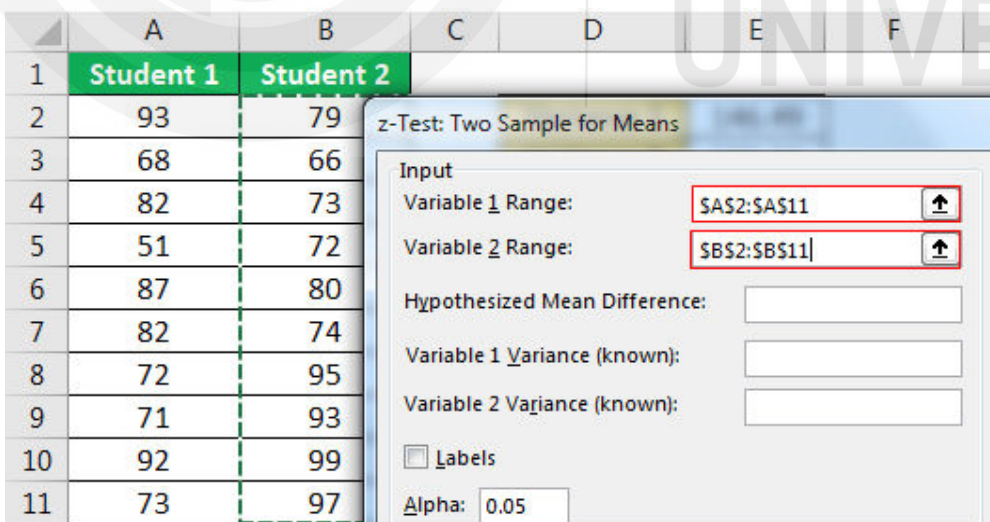


Fig. 10.39: Range values for Z_Test

Step 4: For "Variable 1 Variance (known)," input the variance score of "Student 1," and for "Variable 2 Variance (known)," input the variance score of "Student 2."

	A	B	C	D	E
1	Student 1	Student 2			
2	93	79		Variance 1	146.49
3	68	66		Variance 2	131.16
4	82				
5	51				
6	87				
7	82				
8	72				
9	71				
10	92				
11	73				
12					

z-Test: Two Sample for Means

Input

Variable 1 Range:
\$A\$2:\$A\$11
Variable 2 Range:
\$B\$2:\$B\$11
Hypothesized Mean Difference:
Variable 1 Variance (known):
146.49
Variable 2 Variance (known):
131.16
☐ Labels

Fig. 10.40: Z_Test Two Sample for Means

Step 5: Choose a cell for the "Output Range" and click "OK."

z-Test: Two Sample for Means

Input

Variable 1 Range:
\$A\$2:\$A\$11
Variable 2 Range:
\$B\$2:\$B\$11
Hypothesized Mean Difference:
Variable 1 Variance (known):
146.49
Variable 2 Variance (known):
131.16
☐ Labels
Alpha:
0.05

Output options
☒ Output Range:
\$G\$1
☐ New Worksheet Ply:
☐ New Workbook

OK
Cancel
Help

Fig. 10.41–Output Range for Z_Test Two Sample for Means

We obtained the result.

	G	H	I
1	z-Test: Two Sample for Means		
2			
3		<i>Variable 1</i>	<i>Variable 2</i>
4	Mean	77.1	82.8
5	Known Variance	146.49	131.16
6	Observations	10	10
7	Hypothesized Mean Difference	0	
8	z	-1.0817	
9	P(Z<=z) one-tail	0.13968	
10	z Critical one-tail	1.64485	
11	P(Z<=z) two-tail	0.27936	
12	z Critical two-tail	1.95996	
13			

Fig. 10.42–Result of Z_Test Two Sample for Means

We can reject the null hypothesis if $Z < -Z_{\text{Critical Two-Tailed}}$ or $Z > Z_{\text{Critical Two-Tailed}}$.

Based on the Z.TEST result, here are the findings:

$Z < -Z_{\text{Critical Two-Tailed}} = -1.080775083 > -1.959963985$

$Z > Z_{\text{Critical Two-Tailed}} = -1.080775083 < 1.959963985$

Since the results do not meet the criteria for rejection, we cannot reject the null hypothesis. Therefore, the means of the two students do not differ significantly.

Key Points to Remember:

- All arguments must be numerical values; otherwise, a #VALUE! error will occur.
- The array value must contain numbers; otherwise, a #N/A error will appear.
- The Z.TEST can be used with large data sets.

10.4.5 ANOVA

Analysis of Variance (ANOVA) is used to compare means across three or more groups. It helps determine if at least one group mean is significantly different from the others. There are two types of ANOVA: one-way (used when testing one independent variable) and two-way (used when testing two independent variables).

The owner of my company, which publishes computer books, wants to know whether the position of our books in the computer book section of bookstores influences sales. More specifically, does it really matter whether the books are placed in the front, back, or middle of the computer book section?

The publishing company wants to know whether its books sell better when a display is set up in the front, back, or middle of the computer book section. Weekly sales (in hundreds) were monitored at 12 different stores. At five stores, the books were placed in the front; at four stores, in the back; and at three stores, in the middle. Resulting sales are contained in the Signif worksheet in the file Onewayanova.xlsx, which is shown in Figure 10.43. Does the data indicate that the location of the books has a significant effect on sales?

	A	B	C	D
1	One-Way ANOVA			
2				
3		Front	Back	Middle
4		7		12
5		10		13
6		8		15
7		9		16
8		11		

Fig. 10.43: Book sales data.

You can assume that the 12 stores have similar sales patterns and are approximately the same size. This assumption allows you to use one-way ANOVA because you believe that at most one factor (the position of the display in the computer book section) is affecting sales. (If the stores were different sizes, you would need to analyze the data with two-way ANOVA.

To analyze the data, on the Data tab, click Data Analysis, and then select Anova: Single Factor. Fill in the dialog box as shown in Fig. 10.44.

The dialog box 'Anova: Single Factor' is shown with the following settings:

- Input Range:** \$B\$3:\$D\$8
- Grouped By:** Columns (selected), Rows
- Labels in first row:** ☒
- Alpha:** 0.05
- Output options:**
 - Output Range:** \$C\$12 (selected)
 - New Worksheet By:** (unselected)
 - New Workbook:** (unselected)

Fig. 10.44: Anova- Single Factor dialog box.

I used the following configurations:

- The data for our input range, including labels, is in cells B3:D8.
- Select the Labels In First Row option because the first row of the input range contains labels.
- I selected the Columns option because the data is organized in columns.
- I selected C12 as the upper-left cell of the output range.
- The selected alpha value is not important. You can use the default value. After clicking OK, you obtain the results shown in Fig. 10.45.

	C	D	E	F	G	H	I
13							
14	SUMMARY						
15	Groups	Count	Sum	Average	Variance		
16	Front	5	45	9	2.5		
17	Back	4	56	14	3.333333		
18	Middle	3	33	11	1		
19							
20							
21	ANOVA						
22	Source of Variation	SS	df	MS	F	P-value	F crit
23	Between Groups	55.6666667	2	27.83333	11.38636	0.00343	4.25649
24	Within Groups	22	9	2.444444			
25							
26	Total	77.6666667	11				
27							
28	est std error	1.56347192					

Fig. 10.45: One-way ANOVA results

In cells F16:F18, you see average sales depending on the location of the display. When the display is at the front of the computer book section, average sales are 900; when the display is at the back of the section, sales average 1,400; and when the display is in the middle, sales average 1,100. Because the p-value of 0.003 (in cell H23) is less than 0.15, you can conclude that these means are significantly different. If I am determining whether populations have significantly different means, why is the technique called analysis of variance? Suppose that the data in our book sales study is the data shown in the worksheet named Insig, shown in Fig.4 (and also in the file Onewayanova.xlsx). If you run a one-way ANOVA on this data, you obtain the results shown in Fig.10.46.

	B	C	D
3	Front	Back	Middle
4	7	2	3
5	20	16	19
6	8	25	11
7	8	13	
8	2		

Fig. 10.46: Book store data for which the null hypothesis is accepted

Note that the mean sales for each part of the store are exactly as before, yet the p-value of .66 indicates that you should accept the null hypothesis and conclude that the position of the display in the computer book section doesn't

affect sales. The reason for this strange result is that in the second data set, you have much more variation in sales when the display is at each position in the computer book section. In the first data set, for example, the variation in sales when the display is at the front is between 700 and 1,100, whereas in the second data set, the variation in sales is between 200 and 2,000. The variation of sales within each store position is measured by the sum of the squares of data within a group. This measure is shown in cell D24 in the first data set and in cell F24 in the second. In the first data set, the sum of squares of data within groups is only 22, whereas in the second data set, the sum of squares within groups is 574! This large variation within the data points at each store position masks the variation between the groups (store positions) themselves and makes it impossible to conclude for the second data set that the difference between sales in different store positions is significant.

	E	F	G	H	I	J	K
12	Anova: Single Factor		Overall mean				
13			11.1666667				
14	SUMMARY						
15	Groups	Count	Sum	Average	Variance		
16	Front	5	45	9	44		
17	Back	4	56	14	90		
18	Middle	3	33	11	64		
19							
20							
21	ANOVA						
22	Source of Variation	SS	df	MS	F	P-value	F crit
23	Between Groups	55.66667	2	27.83333	0.436411	0.659334	4.25649
24	Within Groups	574	9	63.77778			
25							
26	Total	629.6667	11				

Fig. 10.47: Anova results accepting the null hypothesis

How can I use the results of a one-way ANOVA for forecasting? If there is a significant difference between group means, the best forecast for each group is simply the group's mean. Therefore, in the first data set, you predict the following:

- Sales when the display is at the front of the computer book section will be 900 books per week.
- Sales when the display is at the back will be 1,400 books per week.
- Sales when the display is in the middle will be 1,100 books per week. If there is no significant difference between the group means, your best forecast for each observation is simply the overall mean. Thus, in the second data set, you predict weekly sales of 1,117, independent of where the books are placed

You can also estimate the accuracy of your forecasts. The square root of the Within Groups MS (mean square) is the standard deviation of the forecasts from a one-way ANOVA. As shown in Fig. 10.48, the standard deviation of forecasts for the first data set is 156 (see the Signif worksheet again). By the rule of thumb, this means that you would expect, for example:

- During 68 percent of all the weeks in which books are placed at the front of the computer section, sales will be between $900-156=744$ and $900+156=1,056$ books.
- During 95 percent of all weeks in which books are placed at the front of the computer book section, sales will be between $900-2(156)=588$ books and $900+2(156)=1,212$ books

	C	D	E	F	G	H	I
11							
12	Anova: Single Factor						
13							
14	SUMMARY						
15	Groups	Count	Sum	Average	Variance		
16	Front	5	45	9	2.5		
17	Back	4	56	14	3.333333		
18	Middle	3	33	11	1		
19							
20							
21	ANOVA						
22	Source of Variation	SS	df	MS	F	P-value	F crit
23	Between Groups	55.66667	2	27.83333	11.38636	0.00343	4.25649
24	Within Groups	22	9	2.444444			
25							
26	Total	77.66667	11				
27							
28	est std error	1.563472					

Fig 10.48: Computation of forecast standard deviation.

Example:

A researcher wants to test the effectiveness of three different diets (A, B, and C) on weight loss. The weight loss of participants in each group is recorded. The null hypothesis (H_0) is that all diets result in the same mean weight loss, while the alternative hypothesis (H_1) is that at least one diet leads to different weight loss.

Using the ANOVA formula, we calculate the F-statistic:

$$F = \frac{\text{Between Group Variance}}{\text{Within Group Variance}}$$

If the F-value is greater than the critical value from the F-distribution, we reject H_0 , concluding that at least one group mean is significantly different.

These statistical tests are fundamental tools in hypothesis testing and are widely used in research to make data-driven decisions. Each test has its own specific application, depending on the type of data and the research question.

☞ Check Your Progress 1

1. What is Hypothesis Testing?

.....

.....

.....

.....

.....

2. Compare between t-test and Z-test.

.....

.....

.....

3. What are the common applications of the chi-square test?

.....

.....

.....

4. What does an F-test help in determining?

.....

.....

.....

5. How does ANOVA differ from t-tests in analyzing group differences?

.....

.....

.....

10.5 CORRELATION AND REGRESSION ANALYSIS

Correlation and regression analysis are two fundamental statistical techniques used to explore relationships between variables. They help to understand the strength, direction, and nature of the relationship between variables.

- **Correlation** measures the strength and direction of a linear relationship between two variables.
- **Regression** is used to predict the value of one variable based on the value of another (or others) by fitting a model (often a linear equation) to the data.

10.5.1 Univariate Analysis

Univariate analysis involves the analysis of one variable at a time. This analysis often focuses on describing the distribution, central tendency, and variability of the variable.

In the context of **regression analysis**, univariate refers to **simple linear regression**, where there is one independent variable (predictor) and one dependent variable (outcome). The goal is to model the relationship between these two variables.

Example: Simple Linear Regression

Consider a scenario where a company wants to understand the relationship between **advertising expenditure** and **sales revenue**. The company collects data for the past 10 months:

Table 10.2: Sample Data

Month	Advertising Expenditure (X)	Sales Revenue (Y)
1	20	30
2	25	35
3	30	40
4	35	50
5	40	55
6	45	60
7	50	65
8	55	70
9	60	80
10	65	85

The **simple linear regression model** has the form:

$$Y = \beta_0 + \beta_1 X$$

Where:

- Y is the dependent variable (sales revenue).
- X is the independent variable (advertising expenditure).
- β_0 is the intercept (the value of Y when X=0).
- β_1 is the slope (the change in Y for a one-unit increase in X).

Using the data, we can fit the regression line, which helps predict sales revenue based on advertising expenditure.

THE BASICS-REGRESSION ANALYSIS IN EXCEL

In statistical modeling, regression analysis is used to estimate relationships between two or more variables:

The dependent variable, also known as the criterion variable, is the primary factor you're trying to understand and predict.

Independent variables, also referred to as explanatory variables or predictors, are the factors that may influence the dependent variable.

Regression analysis helps determine how the dependent variable changes when an independent variable varies, allowing you to mathematically assess which variables have a significant impact.

Technically, a regression model is built on the sum of squares, a mathematical approach to measure data point dispersion. The objective is to minimize the sum of squares, creating a line that best fits the data.

In statistics, simple and multiple linear regressions are distinguished. Simple linear regression models the relationship between a dependent variable and

one independent variable using a linear function. When two or more independent variables are used to predict the dependent variable, it becomes multiple linear regression. If the data relationships are non-linear, nonlinear regression is applied. This tutorial focuses on simple linear regression.

For example, if you analyze sales data for umbrellas over the past 24 months and gather the average monthly rainfall for the same period, plotting the data would show the relationship between the independent variable (rainfall) and the dependent variable (umbrella sales). The regression line would illustrate this relationship.

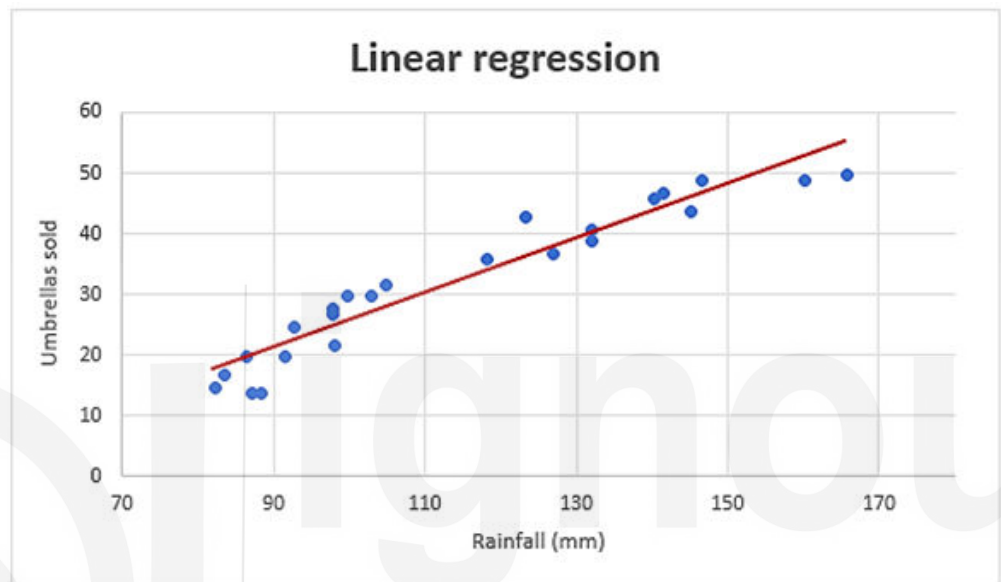


Fig. 10.49: Linear Regression

Linear regression equation

Mathematically, a linear regression is defined by this equation:

$$y = bx + a + \epsilon$$

Where:

- x is the independent variable.
- y is the dependent variable.
- a is the Y-intercept, representing the expected average value of y when all x variables are zero. On a regression graph, this is the point where the line intersects the Y-axis.
- b is the slope of the regression line, indicating how y changes as x varies.
- ϵ is the random error term, reflecting the difference between the actual value of the dependent variable and its predicted value.

The linear regression equation always includes an error term because, in reality, predictors are never perfectly accurate. However, some software programs, such as Excel, handle the error term calculations automatically. In Excel, linear regression is performed using the least squares method, aiming to find the coefficients a and b so that:

$$y = bx + a$$

In our example, the linear regression equation is as follows:

$$\text{Umbrellas sold} = b \times \text{rainfall} + a$$

There are several methods to calculate a and b. The three primary ways to perform linear regression analysis in Excel are:

- Using the Regression tool from the Analysis Toolpak
- Creating a scatter chart with a trendline
- Applying the linear regression formula

Detailed instructions for each method are provided below.

HOW TO PERFORM LINEAR REGRESSION IN EXCEL USING THE ANALYSIS TOOLPAK:

This example demonstrates how to execute regression analysis in Excel with the help of a specialized tool available through the Analysis ToolPak add-in.

Enable the Analysis ToolPak add-in

The Analysis ToolPak is included in all versions of Excel, from Excel 365 to 2003, but it isn't enabled by default. You'll need to activate it manually. Here's how:

1. In Excel, go to **File > Options**.
2. In the Excel Options window, select **Add-ins** from the sidebar on the left. Ensure **Excel Add-ins** is chosen in the **Manage** box, then click **Go**.

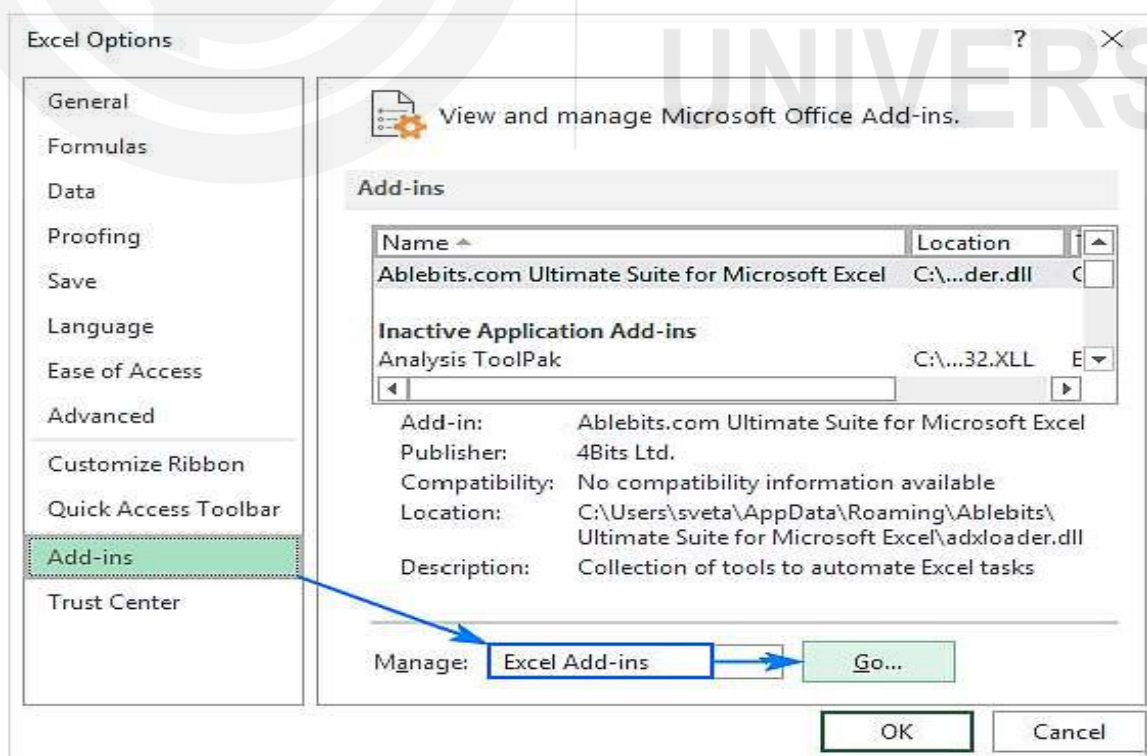


Fig. 10.50: Excel Add-Ins

3. In the Add-ins dialog box, check the box for **Analysis ToolPak** and click **OK**.

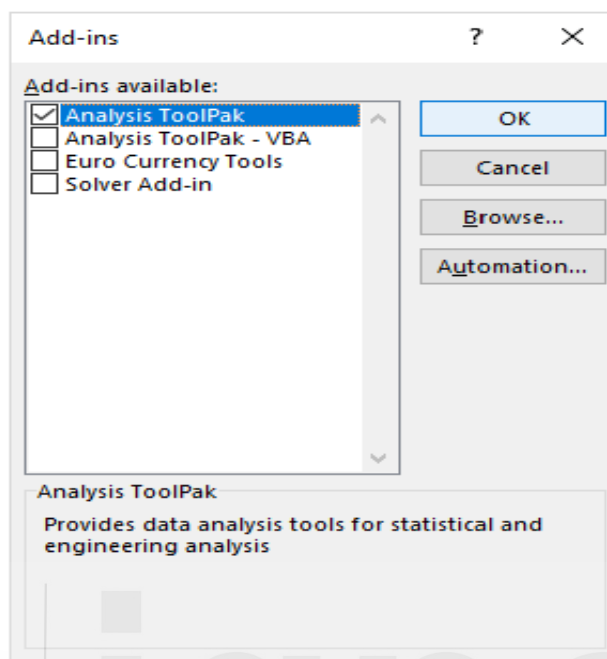


Fig. 10.51: Analysis ToolPak

This will add the **Data Analysis** tools to the **Data** tab in your Excel ribbon.

In this example, we will perform a simple linear regression in Excel. We have a list of average monthly rainfall for the past 24 months in column B, which serves as our independent variable (predictor), and the number of umbrellas sold in column C, which is the dependent variable. While other factors could influence sales, we will focus solely on these two variables for now.

	A	B	C
1	Month	Rainfall (mm)	Umbrellas sold
2	Jan	82	15
3	Feb	92.5	25
4	Mar	83.2	17
5	Apr	97.7	28
6	May	131.9	41
7	Jun	141.3	47
8	Jul	165.4	50
9	Aug	140	46
10	Sep	126.7	37

Fig. 10.52: Sample Data for Linear Regression

Once the Analysis ToolPak is enabled, follow these steps to perform regression analysis in Excel:

1. Go to the **Data** tab, and in the **Analysis** group, click on the **Data Analysis** button.

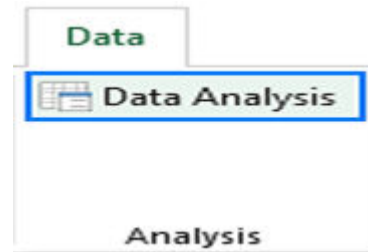


Fig. 10.53: Data Analysis Tab

2. Choose **Regression** from the list and click **OK**.

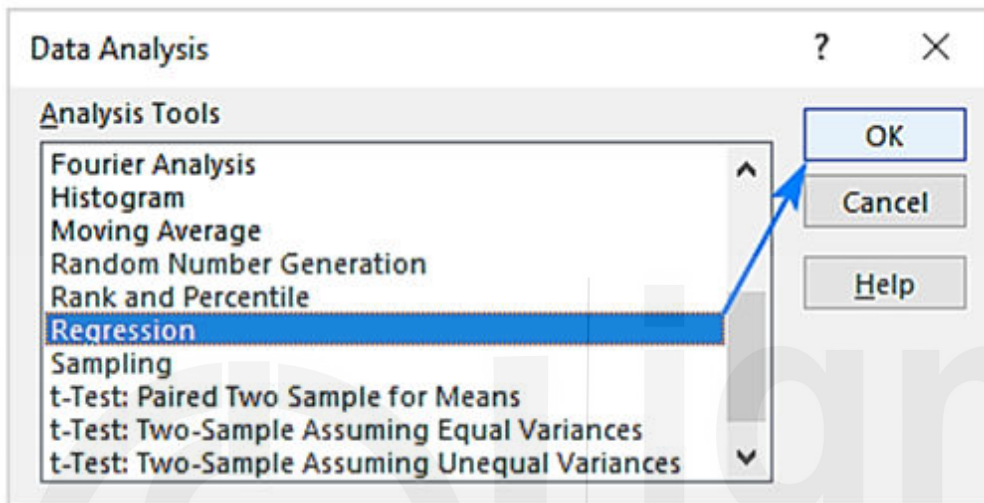


Fig. 10.54: Regression Tab

In the Regression dialog box, set the following options:

- Select the **Input Y Range**, which represents your dependent variable. In this example, it's umbrella sales (C1).
- Select the **Input X Range**, which is your independent variable. Here, it's the average monthly rainfall (B1).

If you are creating a multiple regression model, select two or more adjacent columns for different independent variables.

- Check the **Labels** box if your X and Y ranges include headers at the top.
- Choose your preferred **Output** option; in this case, we will use a new worksheet.
- Optionally, check the **Residuals** box to obtain the difference between the predicted and actual values.

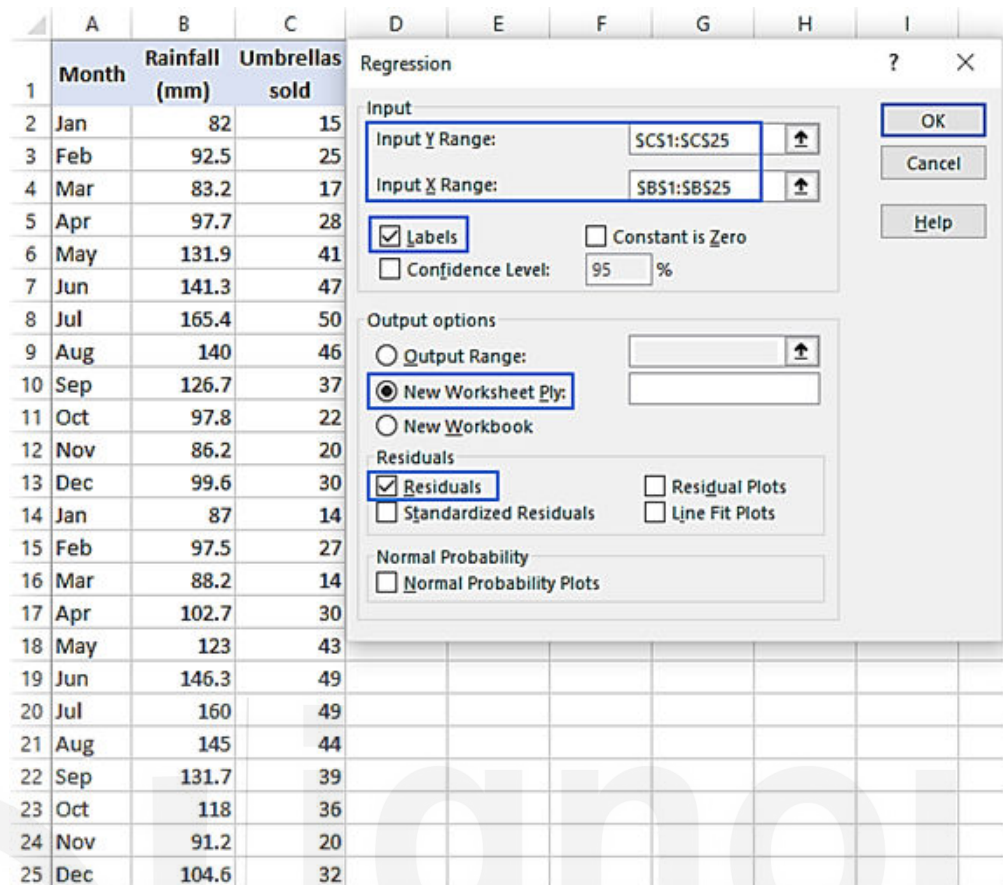


Fig. 10.55: Regression Dialog Box

- Click **OK** and review the regression analysis output generated by Excel.

Interpret regression analysis output

As you just witnessed, performing regression in Excel is straightforward since all calculations are done automatically. However, interpreting the results can be more challenging, as it requires understanding the significance of each number. Below, you'll find a breakdown of the four main components of the regression analysis output.

Summary Output: Regression analysis output

This section indicates how effectively the calculated linear regression equation aligns with your source data.

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.957666798
R Square	0.917125697
Adjusted R Square	0.913358683
Standard Error	3.58141382
Observations	24

Fig. 10.56: Output for Linear Regression

Here's the meaning of each piece of information:

Multiple R: This is the Correlation Coefficient, which measures the strength of the linear relationship between two variables. The correlation coefficient can range from -1 to 1, with its absolute value indicating the strength of the relationship. A larger absolute value signifies a stronger relationship:

- **1** indicates a strong positive relationship
- **-1** indicates a strong negative relationship
- **0** indicates no relationship at all

R Square: This is the Coefficient of Determination, which serves as an indicator of goodness of fit. It represents how well the data points align with the regression line. The R^2 value is derived from the total sum of squares, specifically as the sum of the squared deviations of the original data from the mean.

- In our example, R^2 is 0.91 (rounded to two decimal places), which is fairly good. This means that 91% of our values fit the regression analysis model, indicating that 91% of the dependent variables (y-values) are explained by the independent variables (x-values). Generally, an R^2 value of 95% or higher is considered a good fit.
- **Adjusted R Square:** This value adjusts the **R square** for the number of independent variables in the model. It is preferable to use this metric instead of R^2 for multiple regression analyses.
- **Standard Error:** This is another measure of goodness of fit that reflects the precision of your regression analysis—the smaller the value, the more confidence you can have in your regression equation. While R^2 indicates the percentage of variance in the dependent variables explained by the model, Standard Error provides an absolute measure of the average distance that the data points deviate from the regression line.

Observations: This simply refers to the number of observations included in your model.

ANOVA: Regression analysis output

The second component of the output is the Analysis of Variance (ANOVA):

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	3122.775	3122.775	243.4623	2.21604E-13
Residual	22	282.1835	12.82652		
Total	23	3404.958			

Fig. 10.57: ANOVA Regression Analysis

Essentially, it breaks down the sum of squares into individual components that provide insights into the variability levels within your regression model:

- **df:** This represents the degrees of freedom associated with the sources of variance.

- **SS:** This is the sum of squares. A smaller Residual SS compared to the Total SS indicates a better fit of your model to the data.
- **MS:** This stands for mean square.
- **F:** This is the F statistic, or F-test for the null hypothesis, used to assess the overall significance of the model.
- **Significance F:** This refers to the P-value of F.

The ANOVA section is seldom utilized in simple (LR) linear regression analyses in Excel, but it's important to pay close attention to the last component. The Significance F value provides an indication of how reliable (statistically significant) your results are. If Significance F is less than 0.05 (5%), your model is considered acceptable. Conversely, if it exceeds 0.05, you may want to consider selecting a different independent variable.

Coefficients: Regression analysis output

This section offers detailed information about the components of your analysis:

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-19.07410899	3.372182168	-5.656310378	1.09E-05	-26.06758677	-12.08063122
Rainfall	0.45000132	0.02884018	15.6032773	2.22E-13	0.390190448	0.509812192

Fig. 10.58: Coefficients: Regression Analysis Output

The most valuable component in this section is **Coefficients**, which allows you to construct a linear regression equation in Excel:

$$y = bx + a$$

For our dataset, where Y represents the number of umbrellas sold and x is the average monthly rainfall, our linear regression formula is as follows:

$$Y = \text{Rainfall Coefficient} \times x + \text{Intercept}$$

With the b and a value rounded to three decimal places, it becomes:

$$Y = 0.45 \times x - 19.074$$

For instance, if the monthly average rainfall is 82 mm, the umbrella sales would be approximately 17.8:

$$0.45 \times 82 - 19.074 = 17.8$$

In a comparable manner, you can determine the expected number of umbrellas sold for any specified average rainfall monthly (the variable x).

Output of regression analysis: residuals

When comparing the estimated and actual number of umbrellas sold for a monthly rainfall of 82 mm, you'll observe a small difference:

- **Estimated:** 17.8 (as shown in above calculations)
- **Actual:** 15 (from the source data row 2)

What accounts for this difference? Independent variables are not entirely accurate in predicting dependent variables. Residuals allow you to assess the degree to which the actual values differ from the predicted values.

RESIDUAL OUTPUT		
Observation	Predicted Umbrellas sold	Residuals
1	17.82599924	-2.825999237
2	22.5510131	2.448986904
3	18.36600082	-1.366000821
4	24.89101996	3.10898004
5	40.2810651	0.7189349
6	44.51107751	2.488922493
7	55.35610932	-5.356109317
8	43.92607579	2.073924208
9	37.94105824	-0.941058237
10	24.93602009	-2.936020092
11	19.71600478	0.283995219
12	25.74602247	4.253977532

Fig. 10.59: Residual Output

For the initial data point, which has a rainfall of 82 mm, the residual is roughly -2.8. Adding this residual to the predicted value gives us the actual value: $17.8 - 2.8 = 15$.

PRACTICE: HOW TO MAKE A LINEAR REGRESSION GRAPH IN EXCEL

If you want to quickly visualize the relationship between the two variables, you can create a linear regression chart. It's quite simple! Here's how:

1. Select the two columns containing your data, including the headers.
2. Go to the **Insert** tab, and in the **Charts** group, click on the **Scatter chart** icon, then select the first Scatter thumbnail.

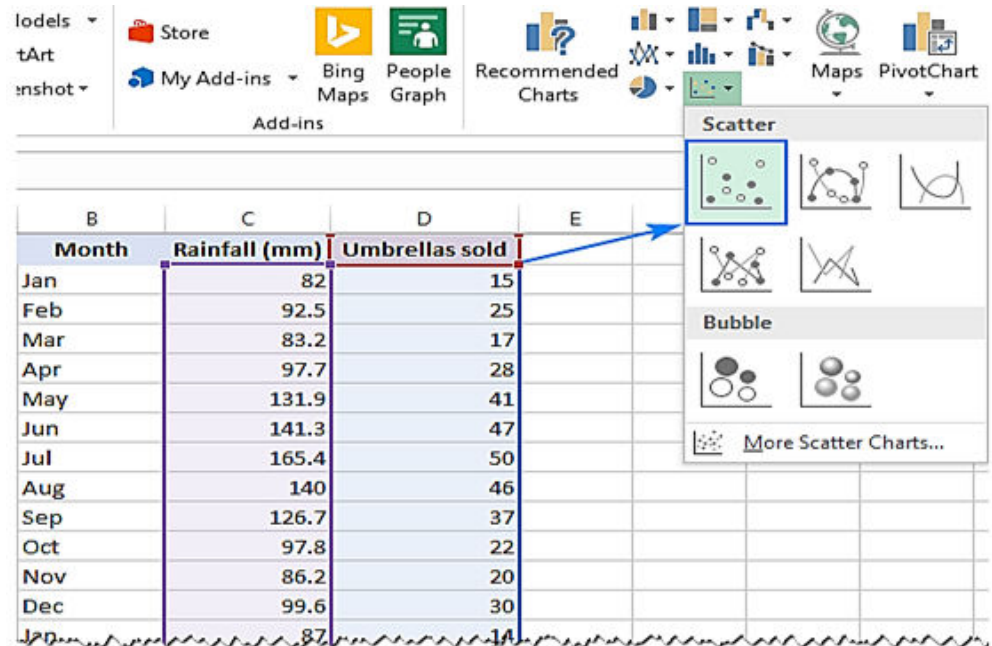


Fig. 10.60: Scatter Chart

This will create a scatter plot in your worksheet, which will look like this:

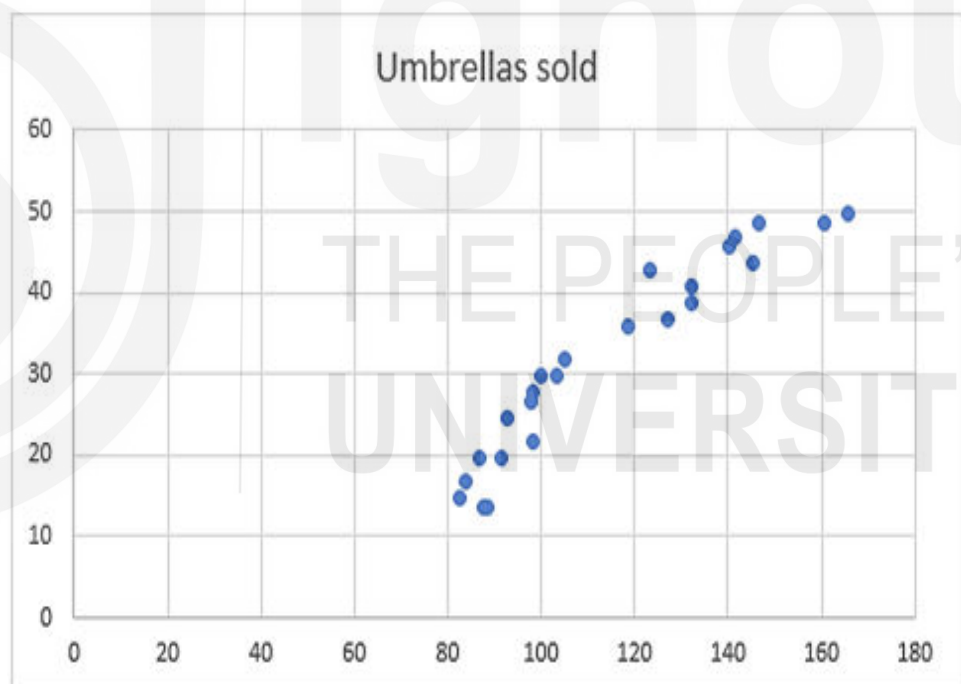


Fig. 10.61: Scatter Chart of Umbrella's Sold

- Next, we need to add the least squares regression line. To do this, right-click on any data point and select **Add Trendline...** from the context menu.

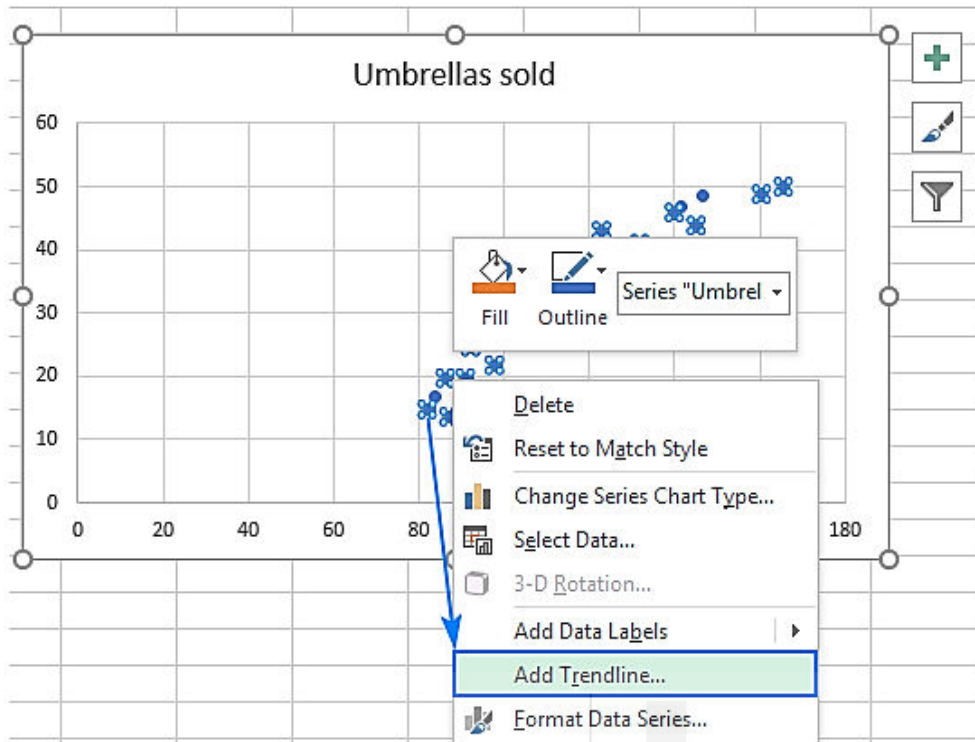


Fig. 10.62: Adding Trendline

4. In the right pane, choose the **Linear** trendline option and, if you wish, check the **Display Equation on Chart** box to show your regression formula:

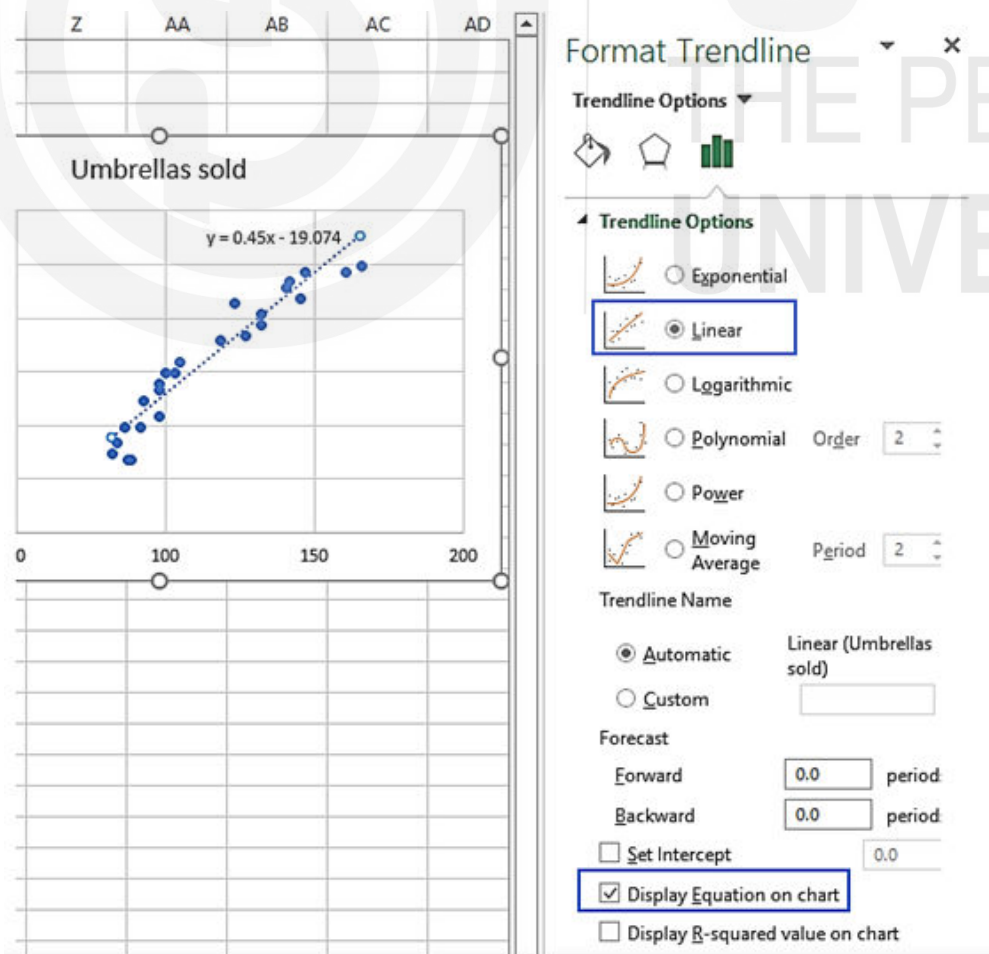


Fig. 10.63: Display Equation on Chart

As you can see, the regression equation produced by Excel matches the linear regression formula we derived from the Coefficients output.

Now, go to the **Fill & Line** tab to adjust the line settings to your liking. For example, you can change the line color and select a solid line instead of a dashed one by choosing **Solid line** in the **Dash type** box.

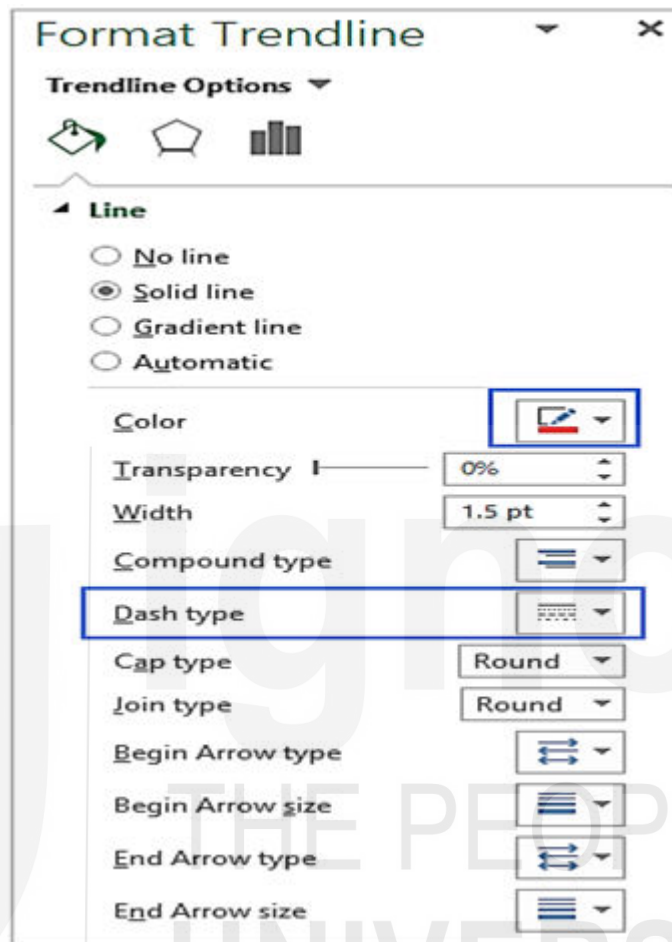


Fig. 10.64: Fill & Line Tab

At this stage, your chart should already resemble a satisfactory regression graph:

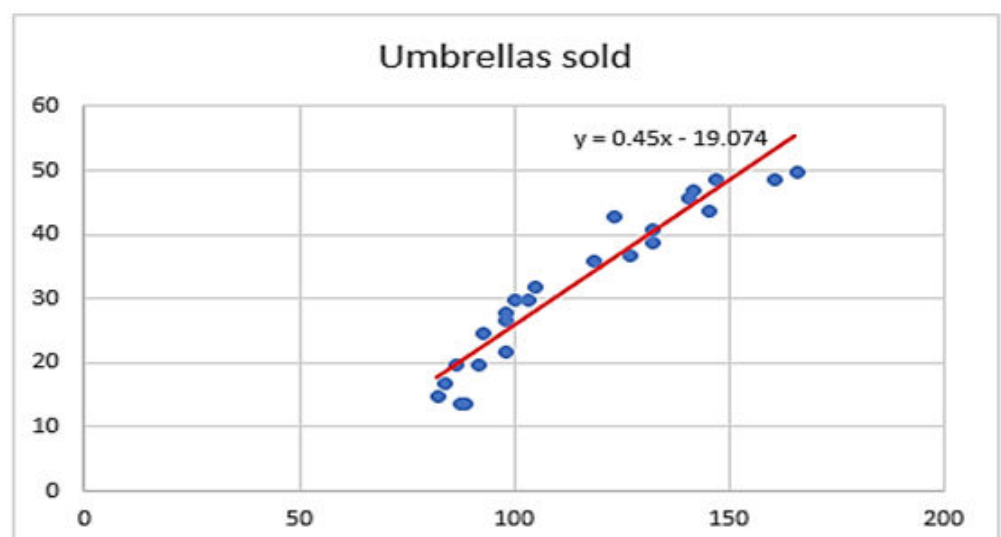


Fig. 10.65: Regression Graph

You might still want to make a few additional enhancements:

- Move the equation to a position that you find suitable.
- Add titles to the axes (using the **Chart Elements** button > **Axis Titles**).
- If your data points begin in the middle of the horizontal and/or vertical axes, as seen in this example, you may want to eliminate any unnecessary white space. The following tip explains how to do this: Scale the chart axes to minimize white space.

Here's how our improved regression graph appears:

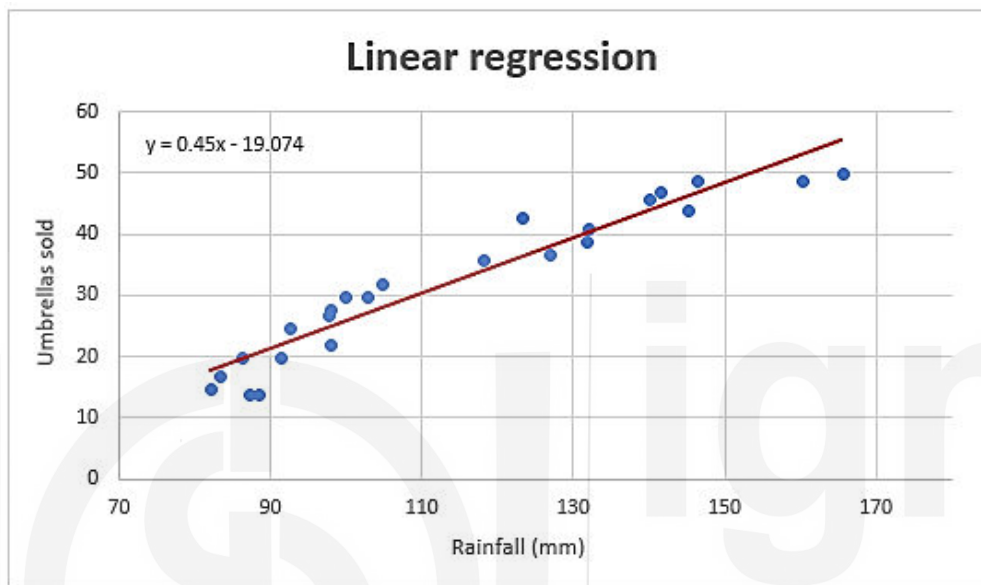


Fig. 10.66: Chart Elements

Note! In a regression graph, the independent variable should always be placed on the X-axis, while the dependent variable belongs on the Y-axis. If your graph is reversed, simply swap the columns in your worksheet and recreate the chart. If rearranging the source data isn't an option, you can switch the X and Y axes directly within the chart.

USING FORMULAS: HOW TO DO REGRESSION IN EXCEL?

Microsoft Excel offers several statistical functions to assist with linear regression analysis, including **CORREL**, **INTERCEPT**, **LINEST** and **SLOPE**.

The **LINEST** function applies the least squares regression method to find the straight line that most accurately represents the relationship between your variables, returning an array that outlines that line. A comprehensive explanation of the function's syntax is available in this tutorial. For now, let's create a formula for our sample dataset:"

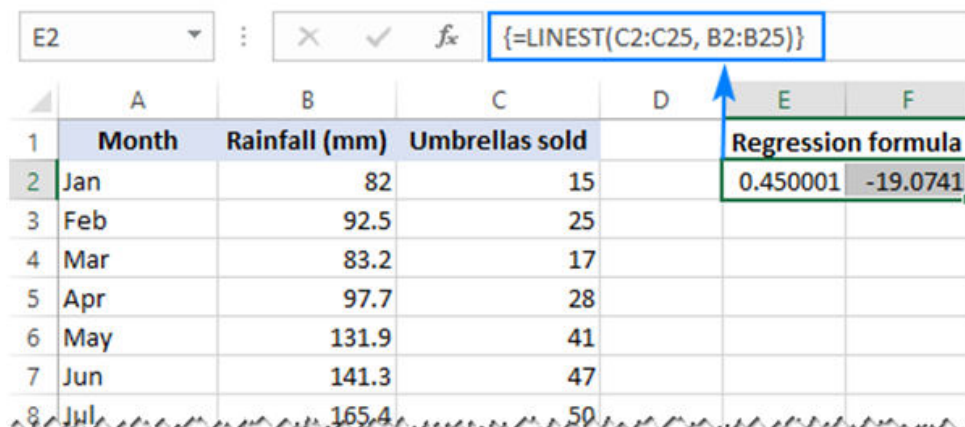
= LINEST(C2: C25, B2: B25)

Since the **LINEST** function generates an array of values, so it must be entered as an array formula. To do this, select two neighboring cells in the

same row (for example, E2), input the formula, and press **Ctrl + Shift + Enter** to run it.

The formula will yield the b coefficient (in cell E1) and the a constant (in cell F1) for the familiar linear regression equation:

$$y = bx + a$$



	A	B	C	D	E	F
1	Month	Rainfall (mm)	Umbrellas sold		Regression formula	
2	Jan	82	15		0.450001	-19.0741
3	Feb	92.5	25			
4	Mar	83.2	17			
5	Apr	97.7	28			
6	May	131.9	41			
7	Jun	141.3	47			
8	Jul	165.4	50			

Fig. 10.67: Regression Formula

If you prefer not to use formulas of an array in your worksheets, you can calculate a and b separately using standard formulas:

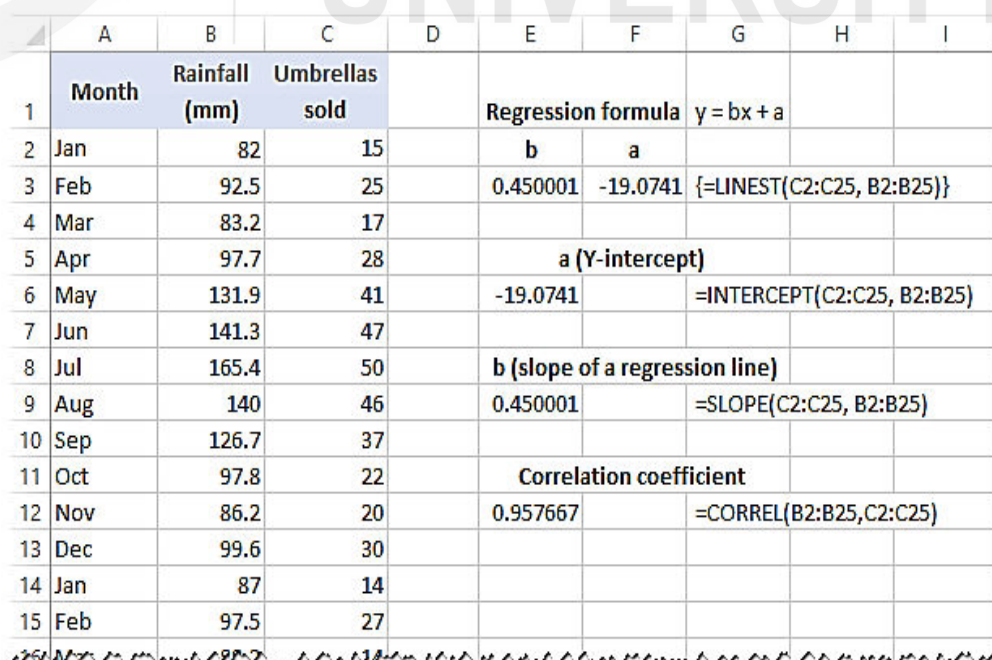
To obtain the Y-intercept (a): = **INTERCEPT(C2: C25, B2: B25)**

Get the slope (b): = **SLOPE(C2: C25, B2: B25)**

You can also calculate the correlation of coefficient (labeled as Multiple R in the regression analysis summary output), which indicates the strength of the relationship between the two variables:

$$= \text{CORREL}(B2: B25, C2: C25)$$

The following screenshot shows all these Excel regression formulas in action:



	A	B	C	D	E	F	G	H	I
1	Month	Rainfall (mm)	Umbrellas sold		Regression formula	y = bx + a			
2	Jan	82	15		b	a			
3	Feb	92.5	25		0.450001	-19.0741	{=LINEST(C2:C25, B2:B25)}		
4	Mar	83.2	17						
5	Apr	97.7	28		a (Y-intercept)				
6	May	131.9	41		-19.0741		=INTERCEPT(C2:C25, B2:B25)		
7	Jun	141.3	47						
8	Jul	165.4	50		b (slope of a regression line)				
9	Aug	140	46		0.450001		=SLOPE(C2:C25, B2:B25)		
10	Sep	126.7	37						
11	Oct	97.8	22		Correlation coefficient				
12	Nov	86.2	20		0.957667		=CORREL(B2:B25, C2:C25)		
13	Dec	99.6	30						
14	Jan	87	14						
15	Feb	97.5	27						

Fig. 10.68: Details of Regression Formula

💡 **Tip:** To get additional statistics for your regression analysis, use the **LINEST** function with the stats parameter set to TRUE, as shown in this example.

This is how you perform linear regression in Excel. However, please remember that Microsoft Excel is not primarily a statistical program. If you need to conduct regression analysis at a professional level, consider using specialized software like XLSTAT, RegressIt, and others.

Correlation:

In addition to regression, we can calculate the **correlation coefficient (r)** to measure the direction and strength of the linear relationship between the two variables. The correlation coefficient ranges from -1 to 1:

- Perfect positive linear relationship is $r=1$
- Perfect negative linear relationship is $r=-1$.
- No linear relationship $r=0$.

For the above data, the correlation between advertising and sales revenue is expected to be positive ($r > 0$), indicating that as advertising increases, sales also tend to increase.

Let us start by describing the following simple example:

In any computer system, there are various components like Memory, Processor, Motherboard etc, and say you want to study how the performance of any computer system varies with different permutation and combination of some constituent components. For example, you may want to know “is there any relationship between the size of Random Access Memory - RAM and the performance of Computer system” OR “is there any relationship between Hard disk storage capacity and the performance of Computer system” etc. in order to address such queries, you are required to conduct a statistical experiment that entails in the collection and tabulate the data as discussed earlier. So, collect and tabulate data under heading say “RAM-Size” and “Performance-Status” OR “Hard Disk-Capacity” and “Performance-Status”.

Thus, it is required to carry out a statistical investigation of the fact that a Computer System’s performance improves on increasing the RAM size. Since it is known that there is non-linear relationship between the two variables, a linear regression (which involves linear relationship between variables), will “not be a good approximation” in the study of relationship between system performance and RAM size. Consequently, a non-linear relationship of appropriate type, between computer system performance and RAM size will only give a better approximation (read chapter 8, Book 3). Since, Non Linear regression is not within the scope of this course, we will restrict our discussion only to Correlation and Linear Regression. Recall that a linear relationship between variables x (independent variable) and y (dependent variable) can be put in the form

$$y = a + b x.$$

The objective of our study is to investigate the question

“Is there any relation between the size of Random Access Memory (RAM) and the Performance of Computer System?”

Steps:

- From the objective of the problem, identify the two variables, which in this case are “size of Random Access Memory (RAM)” and the “performance of Computer system”.
- In order to proceed with our study, we are required to collect data, for which we are required to have RAM of different sizes. Further, for the sake of simplicity, let us restrict the study to the processor existing in the system. It is an exhaustive exercise, in which we have to mount different RAM on the Motherboard slot repeatedly, restart the system and monitor the system performance parameters. Say, we record the Percent (%) variation in RAM size and the percent (%) variation in the System performance in the Excel spread sheet.
- Since we are investigating the effect on System performance, the influence of variation in RAM size, i.e., the % variation in RAM is the independent variable x and % variation in system performance y is the dependent variable.
- Notice that the importance of identifying the two variables rests on the fact that there are *two different lines of regression* viz., y on x and x on y .

Now to calculate the correlation between the two continuous variables in excel, tabulate the recorded observations into the excel worksheet. Excel enables us to investigate the question stated above in three different ways, viz., through

1. **Excel Formula**
2. **Standard formula - Correl(x,y)**
3. **Data Analysis ToolPak**

As an exercise you are required to observe the consistency of the results obtained through all three methods. This will help you to identify the mistakes, which might have occurred while writing your own formula.

In the screen shot given below you can identify that all three options leads to same result.

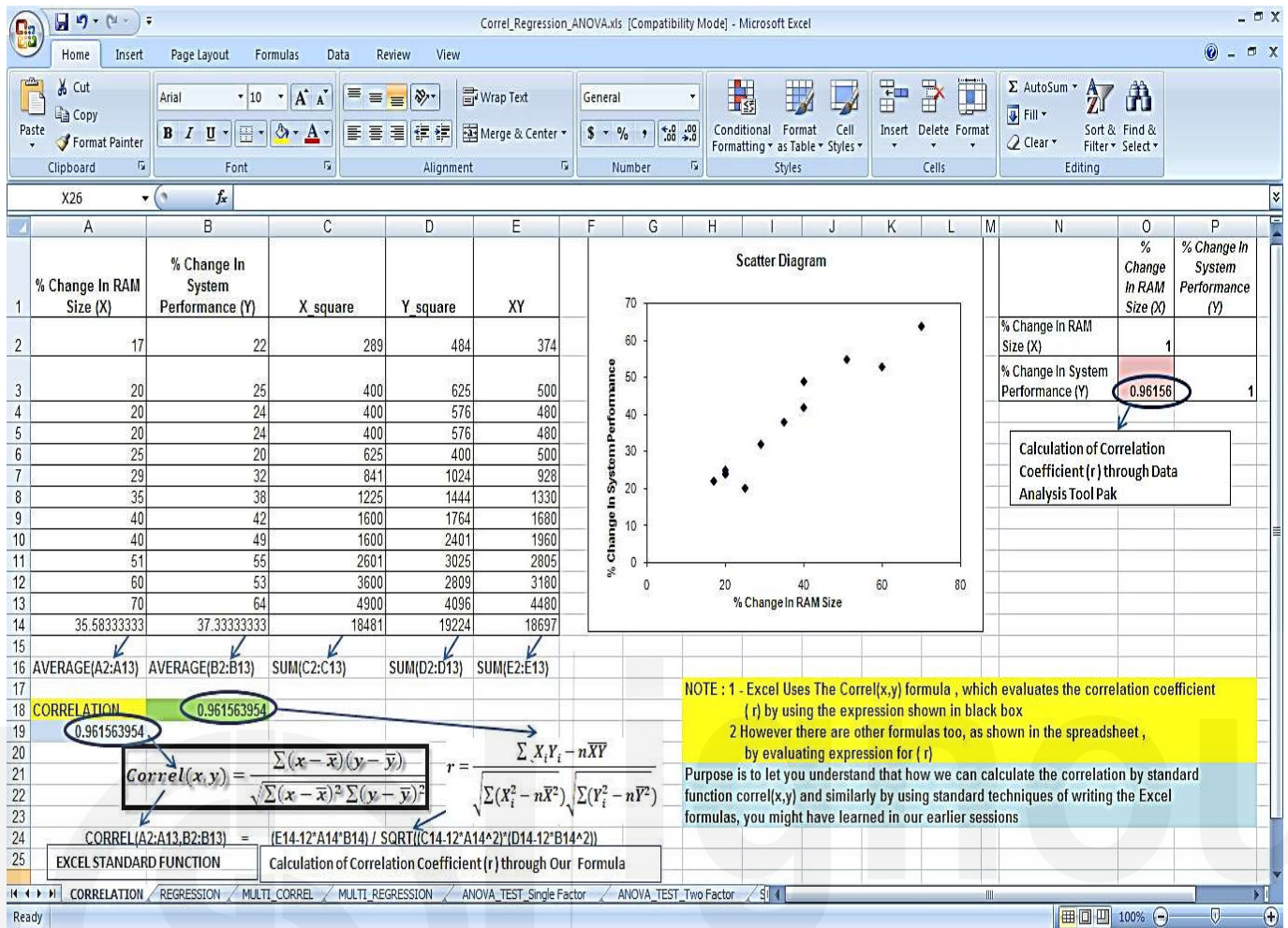


Figure 10.69: Correlation

Now, let us explore each of the option to calculate the correlation coefficient (r) i.e., using own Excel Formula, Standard formula - **correl(x,y)** and Data Analysis ToolPak

1. **Own Excel Formula:** There are different forms of the formula to calculate correlation coefficient (r). Standard function described in excel i.e. **correl(x,y)** uses the formula in the form

$$r = r(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

However, we will calculate the same using the form

$$r = \frac{\sum X_i Y_i - n \bar{X} \bar{Y}}{\sqrt{\sum X_i^2 - n \bar{X}^2} \sqrt{\sum Y_i^2 - n \bar{Y}^2}},$$

in our excel implementation.

To implement it, tabulate the data as shown in the above screen shot and write the formula **(E14-12*A14*B14)/SQRT((C14-12*A14^2)*(D14-12*B14^2))** to calculate correlation coefficient (r). You might have learned the method of formula writing in excel in earlier courses. Another form useful in computations is

$$r = \frac{n \sum X_i Y_i - (\sum X_i)(\sum Y_i)}{\sqrt{n \sum X_i^2 - (\sum X_i)^2} \sqrt{n \sum Y_i^2 - (\sum Y_i)^2}}$$

2. Standard formula - $\text{correl}(x,y)$:

- Tabulate the collected data in column A & B as shown in above screen shot
- Select any cell in which you wish to have result of correlation coefficient (r)
- In that cell write $\text{CORREL}(A2:A13,B2:B13)$, where A2:A13 and B2:B13 are the columns related to the tabulated/recorded data, i.e., the values of variables x and y .

*You can also use Excel Help by pressing function F1 key and type the function or formula you want to understand. Say, press F1 and type **correl** in search option you will get all related details of that formula. $\text{CORREL}(X,Y)$ is the Excel formula to calculate correlation coefficient (r).*

Here, $\text{Correl}(x,y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$ where $\bar{x}$ & $\bar{y}$ are average values of variable x & y respectively.

- Using Data Analysis ToolPak** : You may calculate the correlation coefficient (r) by using the Data Analysis Toolpak utility of Excel, which provides a set the tools necessary for statistical Data Analysis.

Steps to find Correlation coefficient (r) using Data Analysis ToolPak:

- Click the Data tab



- Click the Data Analysis tab



- Select correlation option and click OK



- Select the input range,
 - Which in our case is A1 to B13, relating to % variation in RAM and % variation in system performance.
 - Choose the columns option as the data is tabulated in the columnar manner.
 - Check the Option Labels in first row, as in A1 and B1 we are having

headings for respective columns.

- iv. Finally, choose the cell location where you want to have the output result.
- v. press OK



e. Output:

N	O	P
	% Change In RAM Size (X)	% Change In System Performance (Y)
% Change In RAM Size (X)	1	
% Change In System Performance (Y)	0.96156	1

Calculation of Correlation Coefficient (r) through Data Analysis Tool Pak

NOTE: We will prefer to use Data Analysis Toolpak for other applications subsequently and would like to put more emphasis on data interpretation.

Screen Shot of Final Outcome

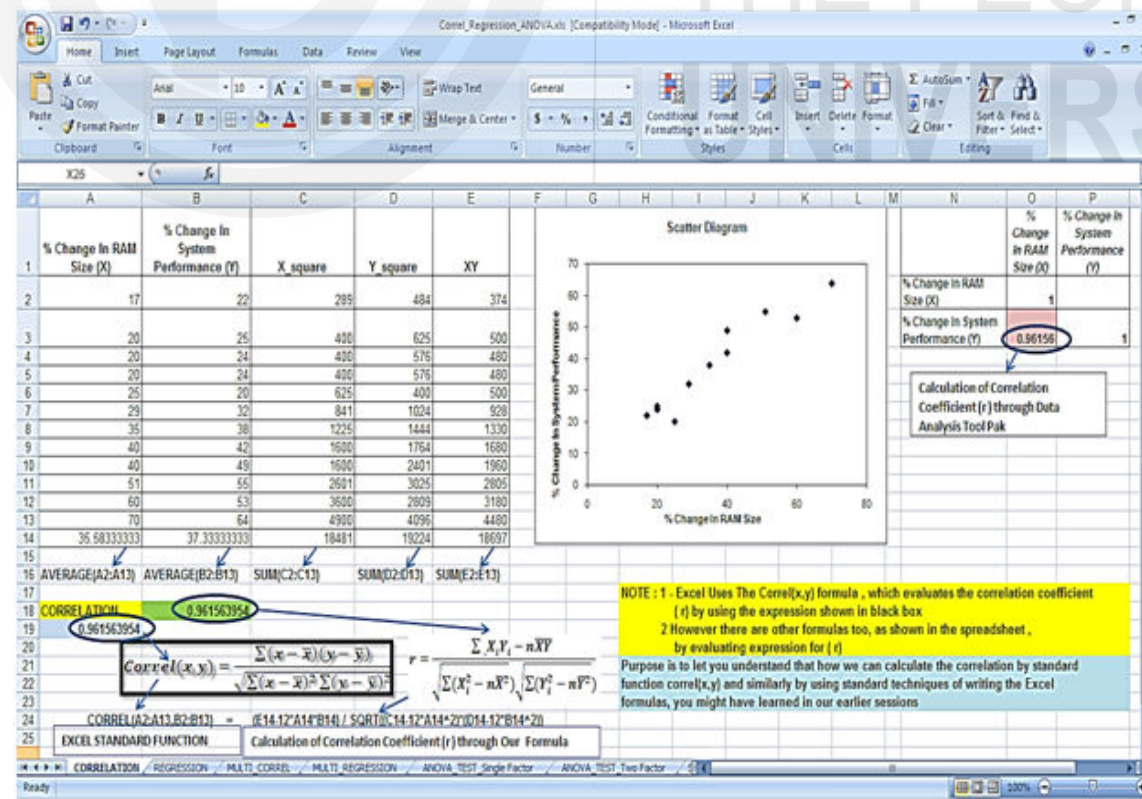


Figure 10.70 : Correlation final outcome

Remark on Data Interpretation

- **Which curve to fit?**
 - Between two continuous variables the relationship can be ascertained by first plotting a scatter diagram.
 - Then, simply looking at the trend of the resultant plot, it could be linear or curvilinear.
 - You are required to apply your understanding of our earlier sessions to plot scatter diagram etc. in Excel.
- **Why r ?**
 - If the resulting scatter diagram exhibits a linear relationship between x and y , the extent can be quantified using correlation coefficient (r), which “measures the extent of linearity” present in the data.
 - Value of r always lies between -1 and $+1$.
 - Closer the value of r towards ± 1 , the stronger is the linear relationship between the variables.
 - You can verify in the subsequent section on simple linear regression that depending on the value of r being $+ve$ or $-ve$, the slope of the line of regression will also be $+ve$ or $-ve$ respectively. In other words, when r is $+ve$ or $-ve$, an increase in the value of “independent variable (x)” will result in an increase or decrease in the value of the “dependent variable (y)”.
- If the relationship is found to be significantly strong, the line of “best fit” passing through the bi-variate data can be obtained and it is discussed in section 2.4.

10.5.2 Multivariate Analysis

Multivariate analysis involves the simultaneous analysis of more than one dependent or independent variable. In the context of regression, multivariate refers to **multiple regression analysis**, where multiple independent variables (predictors) are used to predict the outcome of a single dependent variable.

Correlation in Multivariate Analysis:

Now we extend our discussion to the case of coefficient of multiple correlation, which measures the extent to which a dependent variable can be predicted using the *linear function of a set of independent variables*. To explain the concept let us take another example from the IT sector, companies generally have a monitoring index called the technical index (TECH INDEX). The monitoring of this index goes on monthly basis, and the indexing is performed to identify growth of an independent company with respect to the industry requirements. The data related to the variation between industry parameter called TECH INDEX and the monthly rate of return is gathered for few companies viz. Google, Yahoo, Microsoft, Apple; the same is to be analyzed. Since variation of multiple companies in coordination with one parameter is to be analyzed, we take recourse to the concept of Multiple Correlation. Denoting the companies by x_1, x_2, x_3, x_4 and

industry index by y , a multiple regression model that describes this relationship can be put in the form

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \epsilon$$

You are required to perform the following:

1. Plot the graph for all collected data.

This will help you to visually identify the variation in variable synchronization.

2. Run Correlation analysis on all the variables simultaneously.

Given below is the data on various IT sector companies stated above:

Monthly Rate of Returns for it Sector Companies					
		Company			
Date	Tech Index	Google	Yahoo	Microsoft	Apple
1-Apr-11	0.8799	0.7541	2.1407	-4.6296	-18.8406
1-May-11	7.5187	14.9701	-2.5948	18.986	6.6964
1-Jun-11	5.558	11.9792	7.7869	-1.7226	-3.3473
1-Jul-11	1.3716	7.907	-8.5551	-0.5535	5.8442
1-Aug-11	-1.6289	-5.1724	1.2474	6.679	1.9427
1-Sep-11	2.4171	3.4091	0.8214	1.8261	2.1063

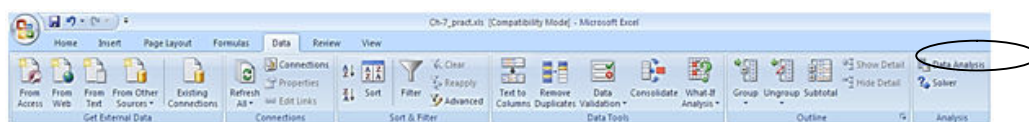
As the first step of analysis, we are interested in visualizing the data graphically and to this end, obtaining a scatter plot is simple as well as most appropriate. Notice that in the case of multiple correlation and regression with only *two independent variables*, scatter diagram can be plotted in the three dimensions. Thus, for the sake of completeness, we can generate a plot with date along the x -axis and the series values along the same y -axis to get an insight into the given data. (see Output Screen Shot given below)

Steps for Multiple Correlation using Data Analysis ToolPak

1. Click the Data tab



2. Click the Data Analysis tab



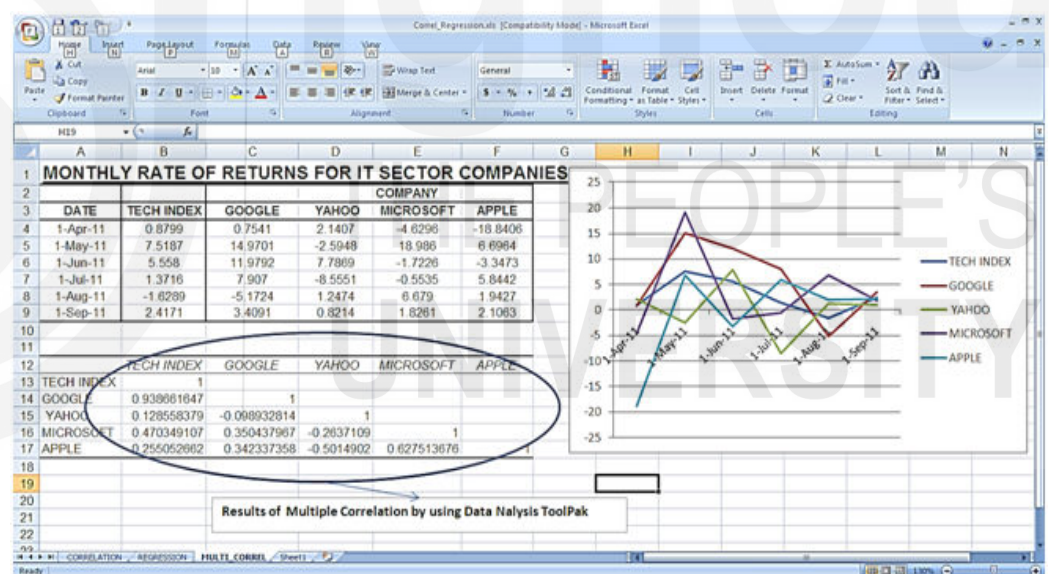
3. Select correlation option and click OK



4. Select the input range,
 - a. Which in our case is \$B\$3:\$F\$9, related to TECH Index and Companies data understudy.
 - b. Choose the columns option as the data is tabulated in the columnar manner.
 - c. Check the Option Labels in first row, as from B3 to F3 we are heaving headings for respective columns.
 - d. Finally, opt for the cell location where you want to have the output result
 - e. press OK



5. Output



DATA INTERPRETATION

	TECH INDEX	GOOGLE	YAHOO	MICROSOFT	APPLE
TECH INDEX	1				
GOOGLE	0.938661647	1			
YAHOO	0.128558379	-0.098932814	1		
MICROSOFT	0.470349107	0.350437967	-0.2637109	1	
APPLE	0.255052662	0.342337358	-0.5014902	0.627513676	1

The table shown above contains correlation coefficients discussed in the previous section, computed between pairs of variables for the data under study. Notice that it is in the form of a matrix, which is known as *matrix of*

correlation coefficients and it is symmetric about the diagonal (why?). Clearly, the elements along the diagonal are correlation coefficients computed between values of a variable with itself, which is *unity* (refer to section 2.2 and state why?). Further, our study requires us to analyze the relation between industry parameter TECH INDEX and Monthly rate of return of various companies. Clearly,

1. There is high correlation (0.938661647) between Google's monthly rate of return data and the industry TECH INDEX, which is on the expected lines.
2. Similarly, for the Microsoft's monthly rate of return data the correlation coefficient is 0.470349107 and it is reasonably high.
3. In contrast to these, the respective correlation coefficients in the case of Yahoo and Apple are 0.128558379 and 0.255052662 respectively, which is low.

Thus, we learned how to interpret the results obtained by using the correlation coefficient in Data analysis ToolPak.

Example of Multiple Linear Regression

From the correlation analysis done above we found that Google and Microsoft are in high correlation, thus the data on these two companies is considered for multiple regression analysis, which is listed below:

		COMPANY	
DATE	TECH INDEX	GOOGLE	MICROSOFT
1-Apr-11	0.8799	0.7541	-4.6296
1-May-11	7.5187	14.9701	18.986
1-Jun-11	5.558	11.9792	-1.7226
1-Jul-11	1.3716	7.907	-0.5535
1-Aug-11	-1.6289	-5.1724	6.679
1-Sep-11	2.4171	3.4091	1.8261

Screen shot is as follows

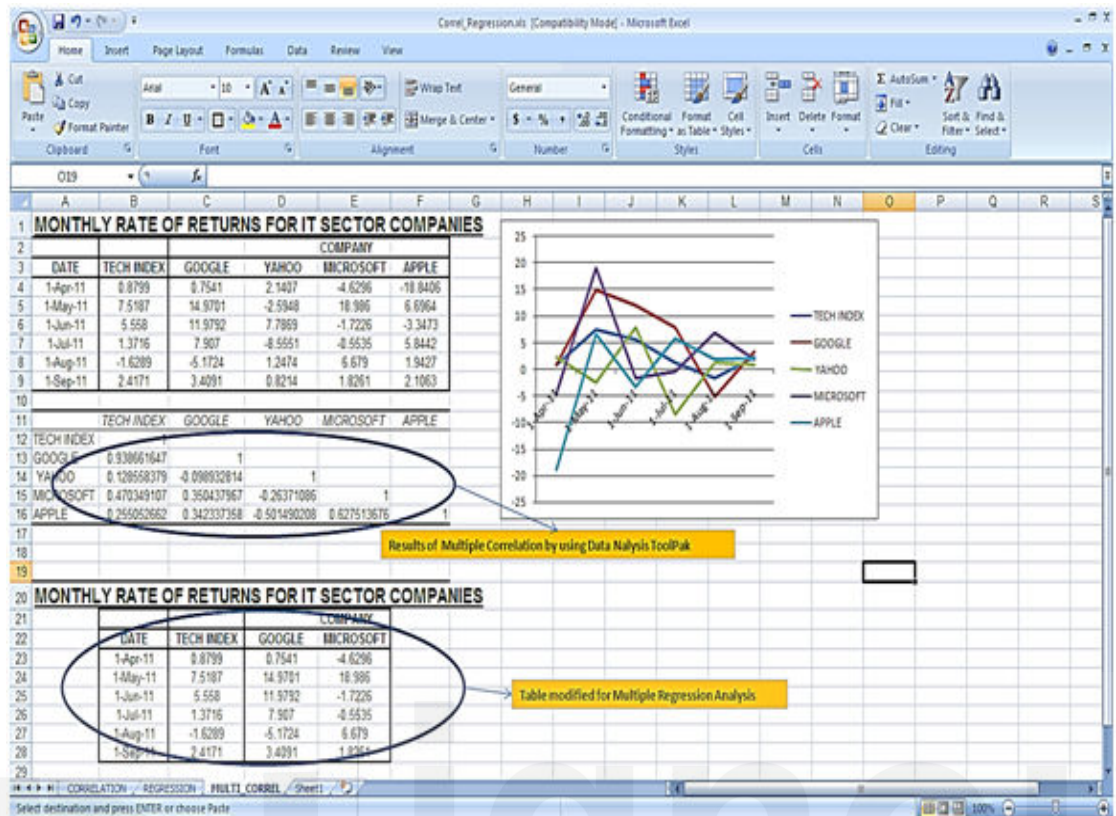


Figure 10.71: Multiple Regression

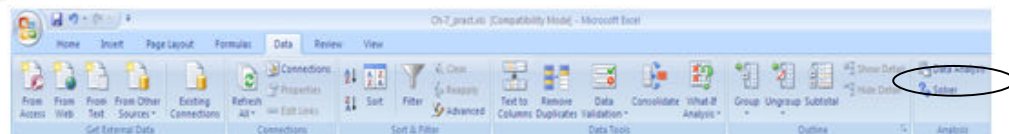
Now let us use the tool i.e., Data Analysis ToolPak's Regression facility to perform multiple regression analysis:

Steps for Multiple Regression using Data Analysis Tool Pak:

1. Click the Data tab



2. Click the Data Analysis tab

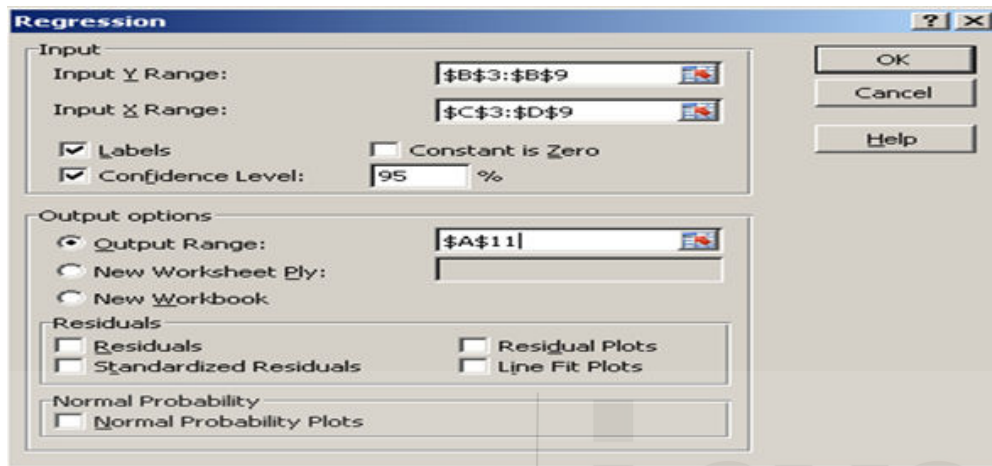


3. Select Regression option and click OK



4. Select the respective data range,
 - a. Which in our case is \$B\$3:\$B\$9 as Input Y- Range, related to TECH Index and \$C\$3:\$D\$9 as Input X-Range, related to shortlisted Companies viz. Google and Microsoft.

- Choose the Output range, we opted for cell location \$A\$11
- Check the Option Labels and confidence level at which we wish to fix and this in our case is 95%.
- Finally, opt for the cell location where you want to have the output result
- Press OK.



5. Output

File

Home

Insert

Page Layout

Formulas

Data

Review

View

From Access

From Web

From Text

From Other Sources

Existing Connections

Get External Data

Refresh All

Properties

Edit Links

Connections

Sort

Filter

Clear

Advanced

Text to Columns

Remove Duplicates

Data Validation

Data Tools

Consolidate

What-If Analysis

Group

Ungroup

Subtotal

Outline

Show Detail

Hide Detail

Data Analysis

Solver

Analysis

L10

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	MONTHLY RATE OF RETURNS FOR IT SECTOR COMPANIES														
2			COMPANY												
3	DATE	TECH INDEX	GOOGLE	MICROSOFT											
4	1-Apr-11	0.8799	0.7541	-4.6296											
5	1-May-11	7.5187	14.9701	18.986											
6	1-Jun-11	5.568	11.9792	-1.7226											
7	1-Jul-11	1.3716	7.907	-0.5535											
8	1-Aug-11	-1.6289	-5.1724	6.679											
9	1-Sep-11	2.4171	3.4091	1.6261											
10															
11	SUMMARY OUTPUT														
12															
13	Regression Statistics														
14	Multiple R	0.958726434													
15	R Square	0.903800967													
16	Adjusted R Square	0.839801446													
17	Standard Error	1.330891938													
18	Observations	6													
19															
20	ANOVA														
21		df	SS	MS	F	Significance F									
22	Regression	2	49.965867	24.9829335	14.10563395	0.029799897									
23	Residual	3	5.31382005	1.77127335											
24	Total	5	55.28368705												
25															
26		Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%						
27	Intercept	0.251395692	0.711022732	0.353699008	0.747049498	-2.011399064	2.514187227	-2.011399064	2.514187227						
28	GOOGLE	0.39330088	0.085205836	4.615891325	0.019133411	0.122137882	0.66446388	0.122137882	0.664463879						
29	MICROSOFT	0.062953947	0.074535551	0.843484722	0.462900097	-0.174569686	0.30047758	-0.174569686	0.30047758						

CORRELATION

REGRESSION

MULTI CORREL

MULTI REGRESSION

The Data Analysis ToolPak, gave exhaustive summary for the Data, out of which we will focus on the Regression statistics and subsequent sessions, we will extend our discussion to ANOVA.

Interpretation of Results

- Following is the summary output obtained using Data analysis ToolPak, which we seek to interpret.

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.950726494
R Square	0.903880867
Adjusted R Square	0.839801446
Standard Error	1.330891938
Observations	6

- The Multiple R is the multiple correlation coefficient and it measures the strength of the association. The range of Multiple R is -1 to +1, which in the present case yields to the value $R = 0.950726494$. This implies that the dependent variable Y which in our case is TECH INDEX has a high positive correlation with the combined effect of Google (X_1) and Microsoft (X_2).
- The R Square value (R^2) is the square of the multiple correlation coefficient R . It measures the strength of the regression prediction compared with predicting solely by the response mean. Alternatively, the value of $R^2 = 0.903880867$, can also be interpreted as that 90.3% of the variation in Y can be explained by variables X_1 & X_2 . It may further be noted the closer value of R to 1, the stronger is the regression prediction.
- Defining multiple correlation as $R^2 = \frac{\text{Explained Variation}}{\text{Total Variation}}$ which is in terms of sums of squares (SS) can be expressed as

$$R^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}} = \frac{49.969867}{55.28368705} = 0.90388086733083 \text{ (for values see screen shot above).}$$
- Discuss with other students or counselors that in general, the value of R^2 will always increase when an additional regressor / independent variable is added (say YAHOO). Details of which is beyond the scope of current content.
- We present below the Regression coefficients from the screen shot along with summary statistics.

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	0.251395602	0.711022732	0.353569008	0.747049498
GOOGLE	0.39330088	0.085205836	4.615891325	0.019133411
MICROSOFT	0.062953947	0.074635551	0.843484722	0.460900097

The column of coefficients contain the values of intercept (c) and the regression coefficients m_1 and m_2 of the regression equation

$$Y = c + m_1X_1 + m_2X_2,$$

which in the present case is

$$Y = 0.251395602 + 0.39330088X_1 + 0.062953947X_2$$

The value X_1 gives us how much will Y change for each unit change in X_1 , and X_2 is held constant. Similarly for X_2 . To understand this, consider the regression equation $Y = -39.62 - 0.5403X_1 + 13.4852X_2$. Here since the coefficient of X_1 is negative, keeping X_2 constant, an unit change in X_1 leads to a decrease in Y by 0.5403. Again, since the value of regression coefficient X_2 is 13.4852, keeping X_1 constant, an unit change in X_2 leads to an increase in Y by 13.4852.

Another Example: A company might want to predict sales revenue not only based on **advertising expenditure** but also on **social media marketing** and **product price**. The data might look like this:

Table 4: Sample Data on Social Media Marketing

Month	Advertising Expenditure (X ₁)	Social Media Marketing (X ₂)	Product Price (X ₃)	Sales Revenue (Y)
1	20	5	100	30
2	25	6	98	35
3	30	7	97	40
4	35	8	95	50
5	40	9	93	55
6	45	10	90	60
7	50	11	88	65
8	55	12	85	70
9	60	13	83	80
10	65	14	80	85

The structure of **multiple regression model** is:

$$Y = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3$$

Where:

- Dependent variable is Y (sales revenue).
- X_1, X_2 and X_3 are the independent variables (advertising expenditure, social media marketing, and product price)
- β_0 is the intercept.
- β_1, β_2 and β_3 are the coefficients for the independent variables?

This model allows the company to understand how each of these variables affects sales revenue and predict future sales by inputting values for the independent variables.

In multivariate analysis, we can also compute **partial correlations** to understand the relationship between each independent variable and the dependent variable while holding other variables constant. This allows the

impact of one predictor to be measured while considering the effects of others.

Interpretation:

By performing multivariate regression, the company can answer questions such as:

- How much will sales increase if both advertising expenditure and social media marketing increase by a certain amount?
- What is the effect of product price on sales when controlling for advertising and social media marketing?

In summary, **univariate analysis** (simple regression) focuses on one predictor, while **multivariate analysis** (multiple regression) uses more than one predictor to model the relationship with a dependent variable. Both are essential tools in understanding and predicting relationships in real-world data.

☞ Check Your Progress 2

1. What is the difference between regression and correlation analysis?

.....

.....

.....

2. Explain the concept of Univariate Regression.

.....

.....

.....

3. List some common uses of Multivariate Regression Analysis.

.....

.....

.....

4. What is the role of the correlation coefficient in regression analysis?

.....

.....

.....

5. How can multiple independent variables be utilized to predict a dependent variable?

.....

.....

10.6 TIME SERIES AND FORECASTING USING EXCEL

Time series analysis focuses on examining data points gathered or observed at regular time intervals. The goal of time series analysis is to understand the underlying patterns such as trends, seasonality, or cyclical behaviors to make forecasts. **Forecasting** leverages this analysis to estimate future values using past data."

In **Excel**, time series forecasting is facilitated using functions like **LINEST**, **FORECAST**, or the **Data Analysis Toolpak**.

Steps for Time Series Forecasting in Excel:

1. **Collect the data:** Time-based data such as monthly sales, stock prices, or temperature readings.
2. **Plot the data:** Use a line chart to visualize trends, seasonality, or irregular patterns.
3. **Apply Moving Averages:** This smooths out the data to reveal the underlying trend.

Note: Also, Try to Use the **Moving Average** option in the Data Analysis Toolpak.

4. **Trendline Forecasting:**
 - Select your data and add a **Trendline** using Excel's chart options.
 - Choose options like **Linear** or **Exponential** depending on the pattern in your data.
 - Excel will generate a line of best fit, which can be extended to forecast future values.
5. **Using Excel's FORECAST function:** Excel's built-in **FORECAST.ETS** function allows for more sophisticated time series forecasting, including support for seasonal data.

Example: Suppose you have monthly sales data for a product and you want to forecast sales for the next three months. Input your sales data in Excel, then use the Forecast Sheet (found in the Data tab) to generate predictions for future months.

1. Enter your data in two columns (Time and Sales).
2. Highlight the data and go to the **Data tab > Forecast Sheet**.
3. Excel will automatically generate a forecast, showing a chart and predicted values for future periods.

Here is a stepwise example to elaborate, how to use data analysis Toolpak for determining moving averages

Let's go step by step with an example. First create a simple dataset, then apply **Moving Average**.

Example Dataset (Monthly Sales Data)

Month	Sales
Jan	120
Feb	135
Mar	128
Apr	150
May	160
Jun	145
Jul	170
Aug	175
Sep	168
Oct	180
Nov	190
Dec	200

Steps to Apply Moving Average in Excel

1. **Enable the ToolPak (if not already active):**
 - Go to **File > Options > Add-ins**.
 - At the bottom, select **Excel Add-ins > Go...**
 - Check **Analysis ToolPak** and click **OK**.
2. **Open the Moving Average Tool:**
 - Go to **Data tab > Data Analysis** (at the far right).
 - From the list, select **Moving Average > Click OK**.
3. **Set Input Range:**
 - Select your data range for Sales (e.g., **B2:B13**).
4. **Specify Interval:**
 - For example, if you want a **3-month moving average**, type 3 in the interval box.
 - This means the average of every 3 consecutive months will be calculated.
5. **Output Range:**
 - Select a blank column where you want results to appear (say, **C2**).
6. **Chart Output (Optional):**
 - You can check the **Chart Output** option to generate a graph of actual vs. moving average.
7. **Click OK.**

Result (3-Month Moving Average)

Month	Sales	3-Month Moving Avg
Jan	120	—
Feb	135	—
Mar	128	127.67
Apr	150	137.67
May	160	146.00
Jun	145	151.67
Jul	170	158.33
Aug	175	163.33
Sep	168	171.00
Oct	180	174.33
Nov	190	179.33
Dec	200	190.00

Notice that the first two months don't have averages (because 3 months are needed). From March onward, each value is the **average of the last 3 months**.

The utility of determining moving averages of the data are as follows:

- Helps identify **trends** in data (e.g., sales growing steadily).
- Reduces the impact of **random fluctuations**.
- Useful in **forecasting** and business decision-making.

Let's go step by step on **Time Series Data Analysis and Forecasting in Excel** with an example.

1. Example Dataset (Monthly Sales Data)

Suppose you have the following sales data for one year:

Month	Sales
Jan	120
Feb	135
Mar	128
Apr	150
May	160
Jun	145
Jul	170
Aug	175
Sep	168
Oct	180
Nov	190
Dec	200

2. Plot Time Series

1. Select your **Month** and **Sales** columns.
2. Go to **Insert > Line Chart** → choose a simple line chart.
 - This gives you a visual trend of sales over time.
 - You'll notice sales generally **increase**, but with some ups and downs.

3. Time Series Forecasting Options in Excel

You have two main approaches in Excel:

(A) Using Data Analysis ToolPak → Moving Average or Regression

- Moving Average (explained above) smooths fluctuations.
- Regression can fit a **trend line** (linear, exponential, etc.) to forecast future values.

Approach – A : We had already discussed the usage of Data analysis toolpak for Moving averages

B) Using Forecast Sheet (Recommended)

Excel 2016+ has a built-in **Forecast Sheet** option that automatically generates forecasts using **Exponential Smoothing (ETS)**.

Approach-B : Below are the steps of Using Forecast Sheet, for the purpose of forecasting

4. Forecasting with Forecast Sheet (Step-by-Step)

1. Select the **Month** and **Sales** columns.
2. Go to **Data > Forecast Sheet**.
3. A window will open showing a chart with your historical data and forecasted future values.
4. Set forecast **end date** (e.g., extend forecast for 6 months).
5. Click **Create**.

Excel generates:

- A forecast chart (with confidence intervals).
- A new table with actual data + predicted future sales.

5. Example Output (Forecast for Next 3 Months)

Month	Sales	Forecast
Dec	200	—
Jan (Next Yr)	—	210.5
Feb (Next Yr)	—	218.7
Mar (Next Yr)	—	227.3

□ These numbers depend on Excel's **ETS algorithm**, which adapts to trends and seasonal patterns.

6. Adding Trendline Forecast in Charts (Quick Method)

If you just want a **trend-based forecast**:

1. Insert a line chart for your data.
2. Right-click on the data line → **Add Trendline**.
3. Choose **Linear, Exponential, or Polynomial**.
4. Check **Display Equation on Chart** and **Forecast Forward (e.g., 3 months)**.

Example: If Excel fits a linear trend like

$$y = 7.9x + 112$$

you can use this equation to predict future sales.

☞ Check Your Progress 3

1. What is a time series, and why is it important in forecasting?

.....

.....

.....

2. List the steps involved in time series forecasting using Excel.

.....

.....

.....

3. Describe the use of the moving average method in time series analysis.

.....

.....

.....

4. How does a trend differ from seasonality in time series data?

.....

.....

.....

5. How can Excel's FORECAST function be applied for future predictions?

.....

.....

.....

10.7 SUMMARY

This Unit titled “**Excel for Inferential Data Analysis**” explains how Microsoft Excel can be used as a tool to perform statistical tests and draw conclusions about populations from sample data. It begins with an introduction to inferential statistics, highlighting how it goes beyond

descriptive statistics by enabling hypothesis testing, predictions, and deeper insights. The objectives focus on helping learners understand statistical concepts, conduct hypothesis testing, perform regression and correlation analysis, apply ANOVA, and interpret results effectively.

The text then discusses **sampling techniques** such as random, systematic, stratified, cluster, and bootstrapping, showing how these can be implemented using Excel's Analysis ToolPak. It emphasizes the efficiency, cost-effectiveness, and accuracy that sampling provides in data analysis.

The section on **hypothesis testing** covers several statistical methods:

- **t-tests** (one-sample, paired, and independent samples) to compare means.
- **Chi-square tests** for categorical data and association in contingency tables.
- **F-tests** to compare variances.
- **Z-tests** for large samples with known population variance.
- **ANOVA** to compare means across three or more groups, with examples illustrating significance and forecasting applications.

Moving further, the unit explains **correlation and regression analysis**. Correlation measures the strength and direction of relationships, while regression is used to build predictive models. It demonstrates both **univariate regression** (simple linear regression with one predictor) and **multivariate regression** (multiple predictors), supported with practical IT and business-related examples. Learners are guided through using Excel functions and the Analysis ToolPak to compute regression outputs, coefficients, significance values, and graphical visualizations such as scatter plots with trendlines.

Finally, the unit encourages self-learning by providing examples, exercises, and interpretations, stressing the importance of correct model selection, proper data handling, and critical analysis of statistical results. Overall, it equips learners with practical skills to conduct inferential data analysis in Excel, making it a hands-on guide for data-driven decision-making.

Frequently Asked Questions:

1. **What is the purpose of statistical analysis in research?** To collect, analyze, interpret, and present data to draw meaningful conclusions and support decision-making.
2. **How do descriptive statistics differ from inferential statistics?** Descriptive statistics summarize data, while inferential statistics make predictions or inferences about a population based on a sample.
3. **What role does data play in statistical analysis?** Data serves as the foundation for analysis, enabling researchers to identify patterns, trends, and relationships.
4. **What are the primary objectives of statistical analysis?** To describe data, test hypotheses, make predictions, and inform decision-making.
5. **How can clear objectives improve the quality of research?** They

provide focus, guide methodology, and help in evaluating the effectiveness of the research.

6. **What are some specific objectives that researchers might aim to achieve with statistical methods?** Identifying relationships, measuring effects, predicting outcomes, and validating theories.
7. **What is sampling, and why is it important in research?** Sampling is selecting a subset of a population to represent the whole; it's crucial for making inferences without surveying the entire population.
8. **Differentiate between probability sampling and non-probability sampling.**
Probability sampling involves random selection, ensuring every member has a chance to be included, while non-probability sampling does not ensure equal chances.
9. **What factors should be considered when determining the sample size for a study?** Population size, desired confidence level, margin of error, and variability within the population.
10. **Why is hypothesis testing critical in statistical analysis?** It helps validate or reject assumptions about a population based on sample data, guiding research conclusions.
11. **What are the steps involved in conducting a hypothesis test?** Define hypotheses, select significance level, choose the appropriate test, calculate test statistic, and make a decision based on p-value.
12. **Explain the concept of Type I and Type II errors in hypothesis testing.**
Type I error occurs when a true null hypothesis is rejected, while Type II error happens when a false null hypothesis is not rejected.
13. **In what scenarios would you use a t-test instead of a z-test?**
When the sample size is small (typically <30) or when the population standard deviation is unknown.
14. **What are the assumptions that need to be met when conducting a t-test?** Normality of data, independence of observations, and homogeneity of variances.
15. **How would you interpret the results of a t-test?** By comparing the p-value to the significance level; if $p < \alpha$, reject the null hypothesis.
16. **What types of data are suitable for chi-square tests?** Categorical data, often in frequency counts, where the relationship between variables is assessed.
17. **What are the key assumptions underlying the chi-square test?** Observations should be independent, categories must be mutually exclusive, and expected frequencies should be sufficiently large (usually > 5).
18. **How would you report the results of a chi-square test?** Report the chi-square statistic, degrees of freedom, and p-value, indicating whether the null hypothesis is rejected.

19. **When is the F-test used in hypothesis testing?** To compare variances between two or more groups, often as part of ANOVA.
20. **What is the relationship between the F-test and ANOVA?** ANOVA uses the F-test to determine if there are significant differences among group means.
21. **What assumptions must be met for the F-test to be valid?** Independence of observations, normal distribution of groups, and homogeneity of variances.
22. **What are the main differences between a one-sample z-test and a two-sample z-test?** A one-sample z-test compares the sample mean to a known population mean, while a two-sample z-test compares means from two different samples.
23. **Under what conditions can a z-test be performed?** When the sample size is large ($n \geq 30$) and the population standard deviation is known.
24. **How is the z-test statistic calculated?** $z = \frac{(\bar{x} - \mu)}{(\sigma / \sqrt{n})}$, where $\bar{x}$ is the sample mean, μ is the population mean, σ is the population standard deviation, and n is the sample size.
25. **What is the purpose of ANOVA in statistical analysis?** To determine if there are statistically significant differences among the means of three or more groups.
26. **What are the different types of ANOVA, and when would you use each?**
27. One-Way ANOVA: compares means across one independent variable.
28. Two-Way ANOVA: examines two independent variables and their interaction.
29. Repeated Measures ANOVA: analyzes the same subjects across multiple conditions.
30. **How do you interpret the results of an ANOVA test?** If the p-value is less than the significance level, reject the null hypothesis, indicating at least one group mean is different.
31. **What is the significance of correlation analysis in research?** It assesses the strength and direction of the relationship between two variables, helping to identify potential associations.
32. **How do outliers affect correlation coefficients?** Outliers can skew correlation coefficients, often leading to misleading results by inflating or deflating the perceived strength of a relationship.
33. **What is the difference between simple linear regression and multiple regression?** Simple linear regression uses one independent variable to predict a dependent variable, while multiple regression uses two or more independent variables.
34. **What is univariate analysis, and why is it useful?** Univariate analysis examines one variable at a time, providing insights into its distribution, central tendency, and dispersion.

35. **What types of visualizations are commonly used in univariate analysis?** Histograms, bar charts, and box plots are commonly used to visualize univariate data.
36. **How can univariate regression be applied in real-world scenarios?** To predict outcomes based on one predictor, such as forecasting sales based on advertising spend.
37. **What are the advantages of using multivariate regression over univariate regression?** Multivariate regression accounts for multiple predictors, providing a more comprehensive understanding of relationships and improving prediction accuracy.
38. **Explain how multicollinearity can affect multivariate regression analysis.** Multicollinearity inflates standard errors, making it difficult to determine the individual effect of predictors on the dependent variable.
39. **What methods can be used to assess the goodness of fit in multivariate regression?** R-squared, adjusted R-squared, and residual analysis are commonly used to evaluate model fit.
40. **What are the key components of a time series dataset?** Trend, seasonality, cyclic patterns, and irregular fluctuations are the primary components.
41. **How does Excel facilitate time series analysis compared to other software?** Excel offers built-in functions, easy visualization tools, and user-friendly interfaces for basic time series analysis.
42. **What are some common forecasting methods available in Excel?** Moving averages, exponential smoothing, and linear regression are common forecasting methods in Excel.

10.10 SOLUTIONS/ANSWERS

☞ Check Your Progress 1

1. **What is Hypothesis Testing?** Hypothesis testing is a statistical method used to make decisions or inferences about population parameters based on sample data. It involves formulating a null hypothesis (H_0) and an alternative hypothesis (H_1), then using a test statistic to determine whether to reject or fail to reject the null hypothesis.
2. **Compare between t-test and Z-test.**
 - **t-test:** Used when the sample size is small ($n < 30$) and the population standard deviation is unknown. It is applicable for comparing means of two groups.
 - **Z-test:** Used for large sample sizes ($n > 30$) when the population standard deviation is known. It also compares means but is generally more suited to larger datasets.
3. **What are the common applications of the chi-square test?**
 - **Goodness of fit:** To determine if a sample matches a population.
 - **Test of independence:** To assess whether two categorical variables are related.

- **Homogeneity:** To compare the distribution of categories across different populations.

4. **What does an F-test help in determining?** The F-test helps in comparing the variances of two or more groups to determine if they are significantly different. It is often used in ANOVA to assess whether multiple group means are equal or whether at least one differs significantly.

5. **How does ANOVA differ from t-tests in analyzing group differences?**

ANOVA (Analysis of Variance) is used to compare means across three or more groups simultaneously, while t-tests are used to compare the means of two groups. ANOVA helps identify whether there is a statistically significant difference between group means but does not specify which groups differ from each other.

☞ Check Your Progress 2

1. **What is the difference between correlation and regression analysis?**

Correlation analysis measures the strength and direction of a linear relationship between two variables, producing a correlation coefficient (r) that ranges from -1 to 1. In contrast, regression analysis quantifies the relationship between a dependent variable and one or more independent variables, allowing predictions of the dependent variable based on the independent variables. While correlation does not imply causation, regression can provide insight into causal relationships.

2. **Explain the concept of Univariate Regression.** Univariate regression, also known as simple linear regression, involves modeling the relationship between a single independent variable (predictor) and a dependent variable (outcome). The relationship is represented by the equation $Y = \beta_0 + \beta_1 X$, where Y is the predicted value, β_0 is the intercept, β_1 is the slope, and X is the independent variable. This method is used to assess how changes in the independent variable affect the dependent variable.

3. **List some common uses of Multivariate Regression Analysis.**

- **Marketing Analysis:** To evaluate the impact of various marketing strategies (e.g., advertising spend, social media presence) on sales.
- **Economics:** To predict economic indicators (e.g., GDP) based on multiple factors like unemployment rates, inflation, and consumer spending.
- **Healthcare:** To analyze the effects of different treatments and patient demographics on health outcomes.
- **Real Estate:** To determine the value of properties based on various factors such as location, size, and amenities.

4. **What is the role of the correlation coefficient in regression analysis?**

The correlation coefficient indicates the strength and direction of the linear relationship between the independent and dependent variables. It

helps to assess how well the independent variable explains the variation in the dependent variable. A high absolute value of the correlation coefficient (close to 1 or -1) suggests a strong relationship, while a value close to 0 indicates a weak relationship. It is essential for evaluating the appropriateness of using regression analysis.

- 5. How can multiple independent variables be used to predict a dependent variable?** Multiple independent variables can be included in a regression model to explain the variation in a dependent variable through a multiple regression equation of the form $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$. Each coefficient (β) represents the expected change in the dependent variable for a one-unit increase in the respective independent variable while holding all other variables constant. This approach allows for a more comprehensive understanding of the factors influencing the dependent variable.

🔍 Check Your Progress 3

1. **What is a time series, and why is it important in forecasting?** A time series is a sequence of data points recorded at specific intervals over time, often representing trends, cycles, or seasonal variations in a dataset. It is important in forecasting because it allows analysts to identify patterns and behaviors in historical data, which can be used to make informed predictions about future events. Understanding time series helps businesses and researchers anticipate changes, allocate resources effectively, and improve decision-making.
2. **List the steps involved in time series forecasting using Excel.**
 - **Collect Data:** Gather historical data points at consistent intervals.
 - **Visualize Data:** Create a line chart to observe trends and patterns visually.
 - **Determine the Forecasting Method:** Choose between methods like moving averages, exponential smoothing, or regression based on the data characteristics.
 - **Apply Moving Averages or Trendlines:** Use Excel's functions to calculate moving averages or add trendlines to the chart for better insight.
 - **Use the Forecast Function:** Implement Excel's FORECAST function or the Forecast Sheet feature to generate predictions for future time points.
 - **Evaluate Results:** Compare the forecasted values with actual data to assess accuracy and adjust models as needed.
3. **Explain how the moving average method is used in time series analysis.** The moving average method smooths out short-term fluctuations in time series data to highlight longer-term trends. It calculates the average of a specific number of past data points (the "window") and updates this average as new data becomes available. For

example, a 3-month moving average takes the average of the last three months of data to forecast the next month. This method helps reduce noise in the data, making trends and cycles easier to identify.

4. **What is the difference between a trend and seasonality in time series data?**

- **Trend:** A trend represents the long-term movement or direction in the data over time, showing whether it is generally increasing, decreasing, or remaining stable. Trends can last for several years and indicate the overall pattern in the dataset.
- **Seasonality:** Seasonality refers to regular, repeating patterns or fluctuations that occur at specific intervals within the data, often linked to the calendar year, such as increased retail sales during the holiday season. Seasonal effects are typically short-term and predictable based on the time of year.

5. **How can Excel's FORECAST function be applied for future predictions?** Excel's FORECAST function predicts a future value based on existing values. It takes historical data points and a specific future point (x) as inputs to estimate the corresponding y value. The function uses linear regression to calculate the predicted value, considering the relationship between the dependent and independent variables. For example, if you have monthly sales data, you can use the FORECAST function to predict sales for the next month by inputting the appropriate historical data range and the next month's index. This allows businesses to make data-driven decisions based on expected future performance.

10.9 REFERENCES

- Keller, G. (2012). **Statistics for Management and Economics**. South-Western Cengage Learning.
- Anderson, D., Sweeney, D., & Williams, T. (2016). **Essentials of Modern Business Statistics with Microsoft Excel**. Cengage Learning.
- Mendenhall, W., Beaver, R., & Beaver, B. (2013). **Introduction to Probability and Statistics**. Brooks/Cole.
- Jaggia, S., & Kelly, A. (2020). **Business Statistics: Communicating with Numbers**. McGraw-Hill Education.
- Microsoft Support. (n.d.). **Time Series Forecasting**. Retrieved from <https://support.microsoft.com/en-us/excel>

10.12 FURTHER READINGS

1. Machine learning an algorithm perspective, Stephen Marsland, 2nd Edition, CRC Press,, 2015.
2. Machine Learning, Tom Mitchell, 1st Edition, McGraw- Hill, 1997.
3. Machine Learning: The Art and Science of Algorithms that Make Sense of Data, Peter Flach, 1st Edition, Cambridge University Press, 2012.